

Submission of project

Title: Digital Forensics and Steganography: Recovering Hidden Data from a JPG Image

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Objective

The objective of this project is to demonstrate how text can be hidden inside a JPG image using steganography techniques and then stored on a simulated 2-disk storage system. The project further aims to show how forensic tools such as FTK Imager and Autopsy can be used to recover a deleted JPG file and extract the hidden message, simulating a real-world digital forensics investigation.

Solution / Implementation

1. Creating the Environment & Preparing the Disk

A virtual forensic environment was set up using **Oracle VirtualBox**.

Two virtual disks were added:

- **Disk 1 (2 GB)** – Used to store the steganographic JPG file
- **Disk 2 (20 GB)** – Used as an Evidence drive for storing recovered data

Both disks were initialized and formatted using **Disk Management**, assigning:

- Volume Label: **Data** (2 GB Disk)
- Volume Label: **Evidence** (20 GB Disk)

This created a controlled environment to simulate data storage and data loss.

2. Embedding Hidden Text into the JPG Image

1. A sample JPG image (**01.jpg**) was downloaded.
2. **SilentEye**, an open-source steganography tool, was installed.
3. The image was opened in SilentEye, and a text message was encoded using:
 - No passphrase
 - Compression disabled
4. The encoded JPG file was saved into the **Data (F:) > StegFiles** folder.

This step simulated a real scenario where a user hides confidential information inside an image.

3. Deleting the File to Simulate Data Loss

The steganographic file was:

- Deleted manually
- Then removed from the Recycle Bin

The file became inaccessible and appeared permanently deleted—setting up the scenario for forensic data recovery.

4. Recovering the Deleted File Using FTK Imager

1. **FTK Imager** was installed and launched.
2. A **physical disk image** of the 2 GB disk was created.
3. The disk image was saved into the **Evidence** drive under *StegFilesRecovery*.
4. The disk image (.001 format) served as evidence for further forensic examination.

This demonstrated how forensic investigators capture exact replicas of storage devices for safe analysis.

5. Forensic Recovery Using Autopsy

1. **Autopsy**, a digital forensics suite, was installed.
2. A new case (“Case 1”) was created.
3. The FTK-created disk image (**StegContent.001**) was added as a data source.
4. Autopsy automatically scanned the disk image.
5. Under **Deleted Files**, Autopsy successfully identified the deleted *01.jpg*.

6. The file was extracted to the Evidence drive.

This step demonstrated how forensic software can restore deleted data even after it is removed from the recycle bin.

6. Extracting the Hidden Text from the Recovered Image

1. The recovered JPG file was opened again in **SilentEye**.
2. The **Decode** function was used.
3. The hidden message was successfully retrieved.

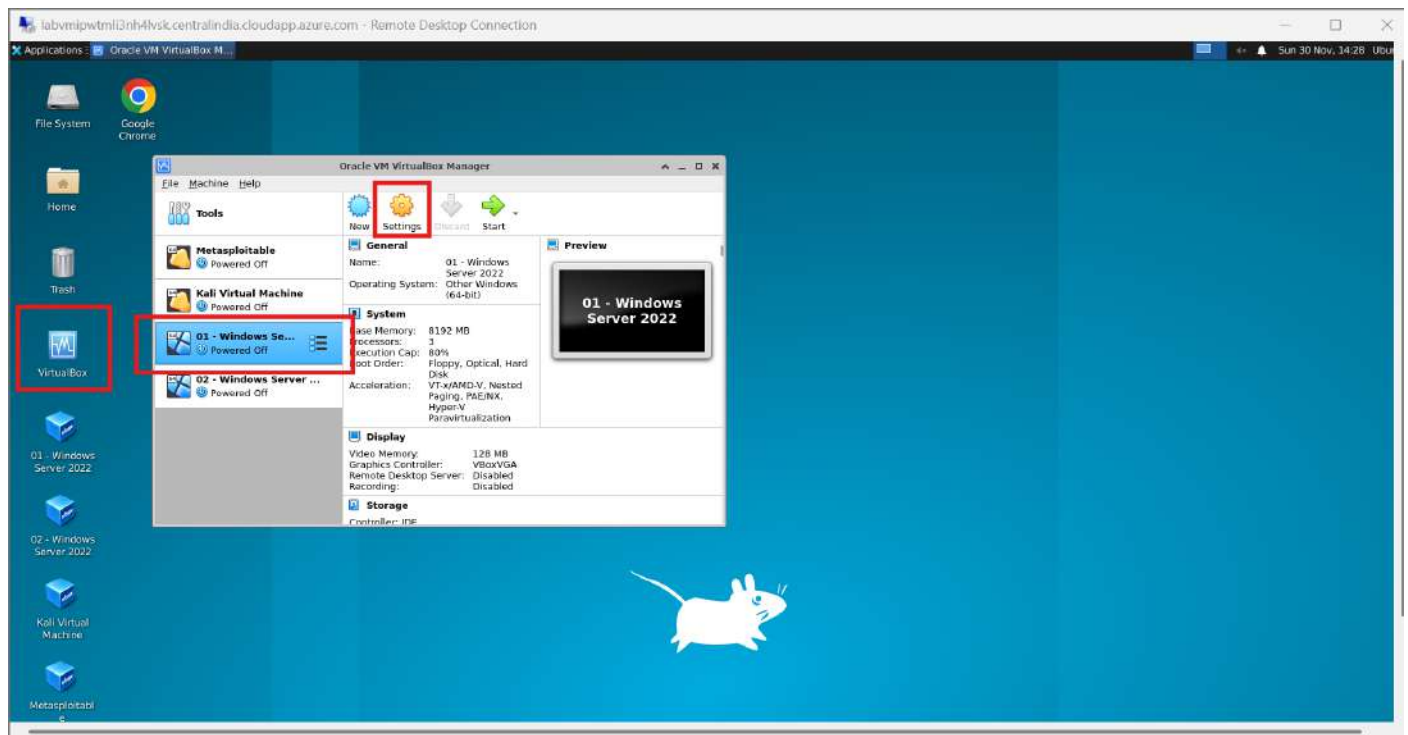
This confirmed that:

- The file was recovered properly
- The concealed information was fully intact
- Steganography artifacts persist even after deletion

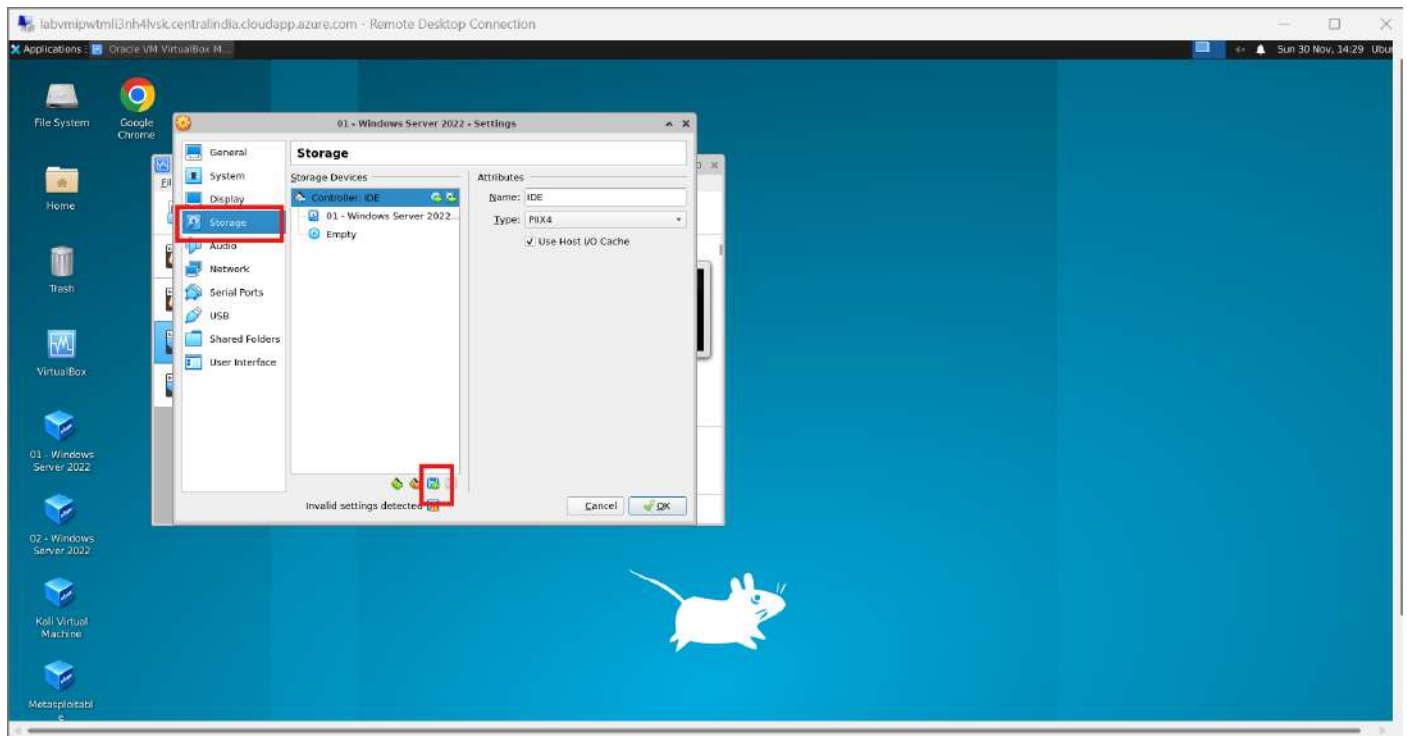
Task 1: Setting up Disk and Creating a JPG Image and Hiding Text Using Steganography:

Step 1: Setup the Disk

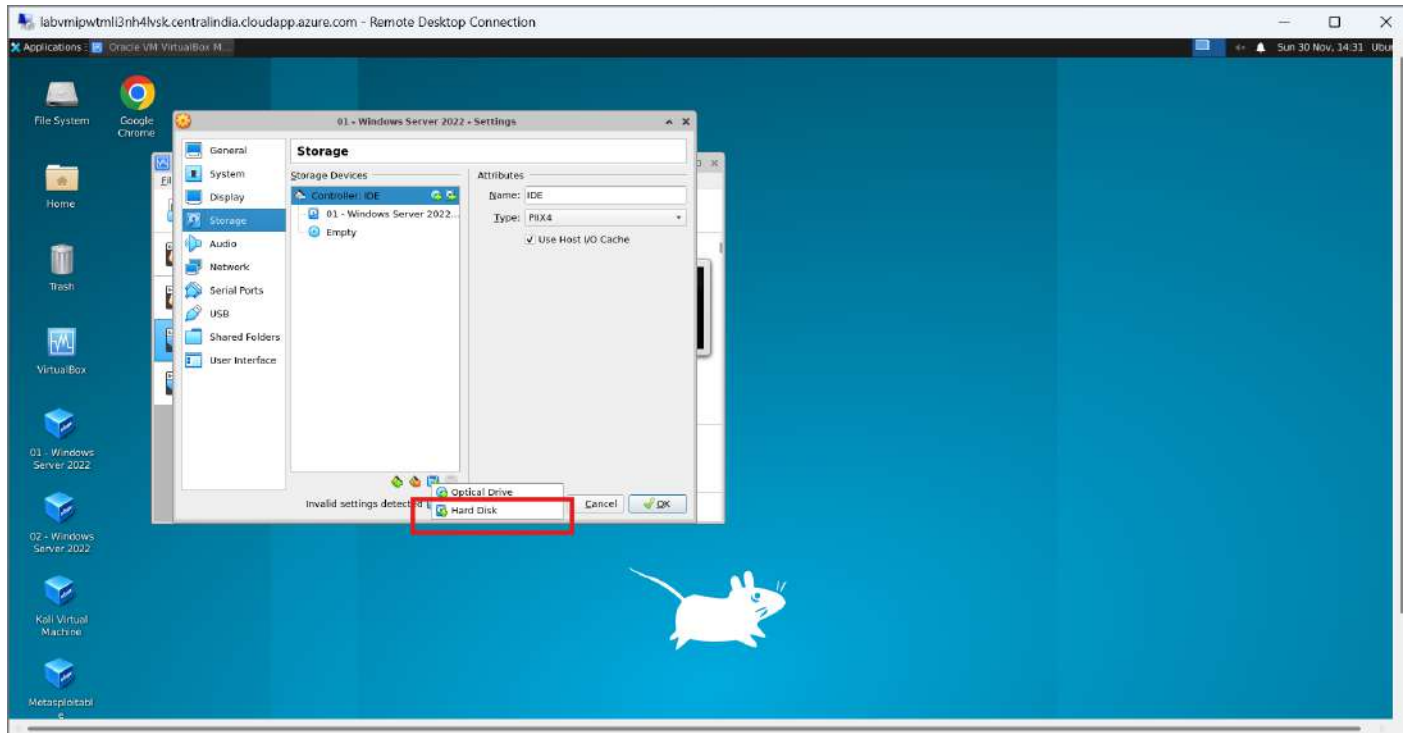
1.1 Double clicked on the **Oracle virtual box** and selected the “**01-Windows Server 2022**” then clicked on **Settings**.



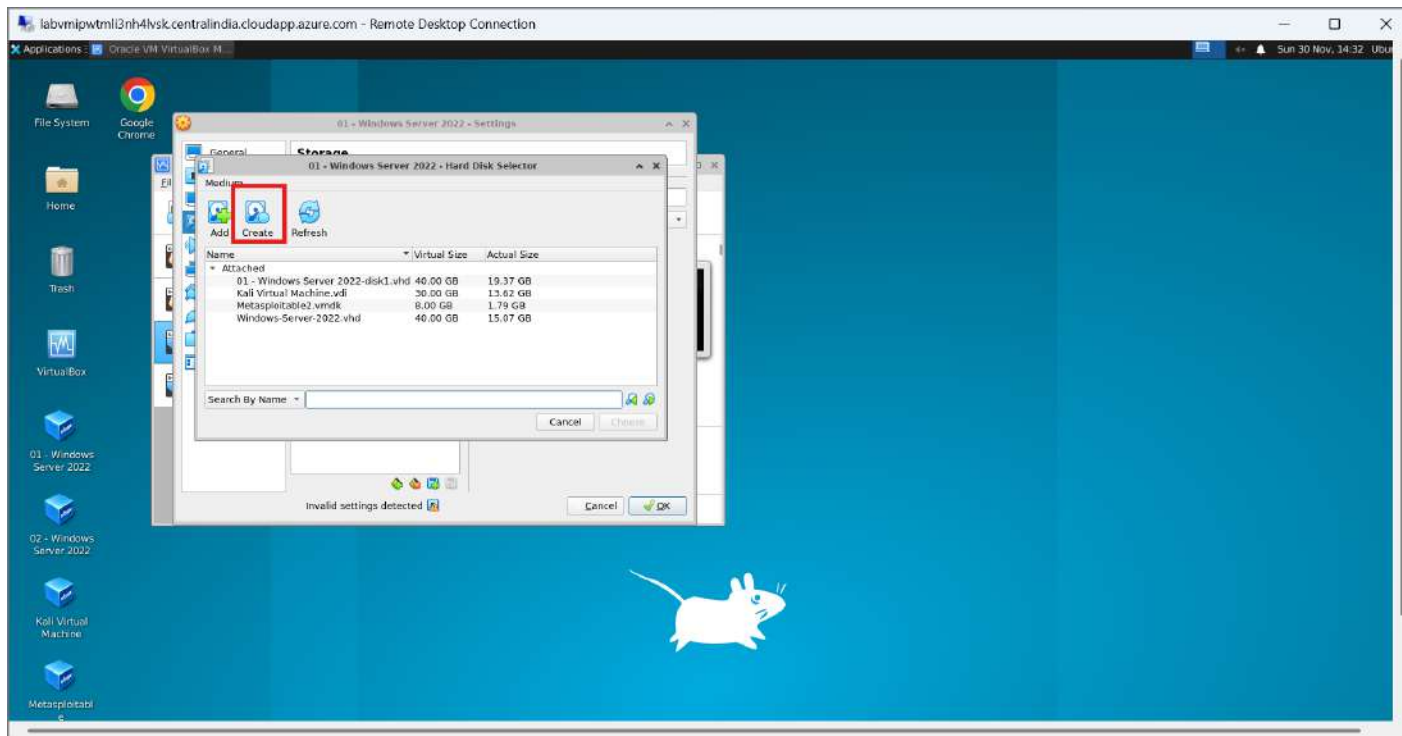
1.2 Selected the **Storage** and clicked on **Add attachment**.



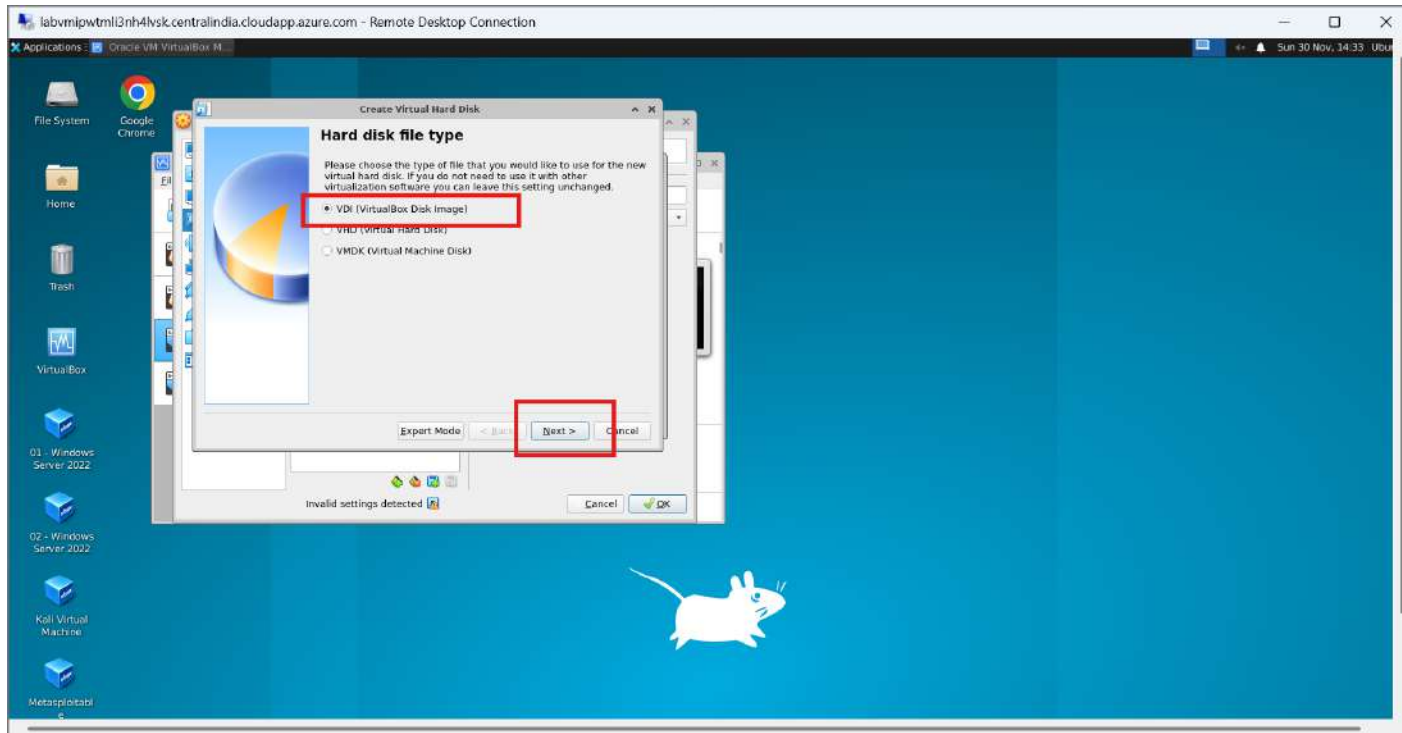
1.3 Clicked on **Hard Disk**.



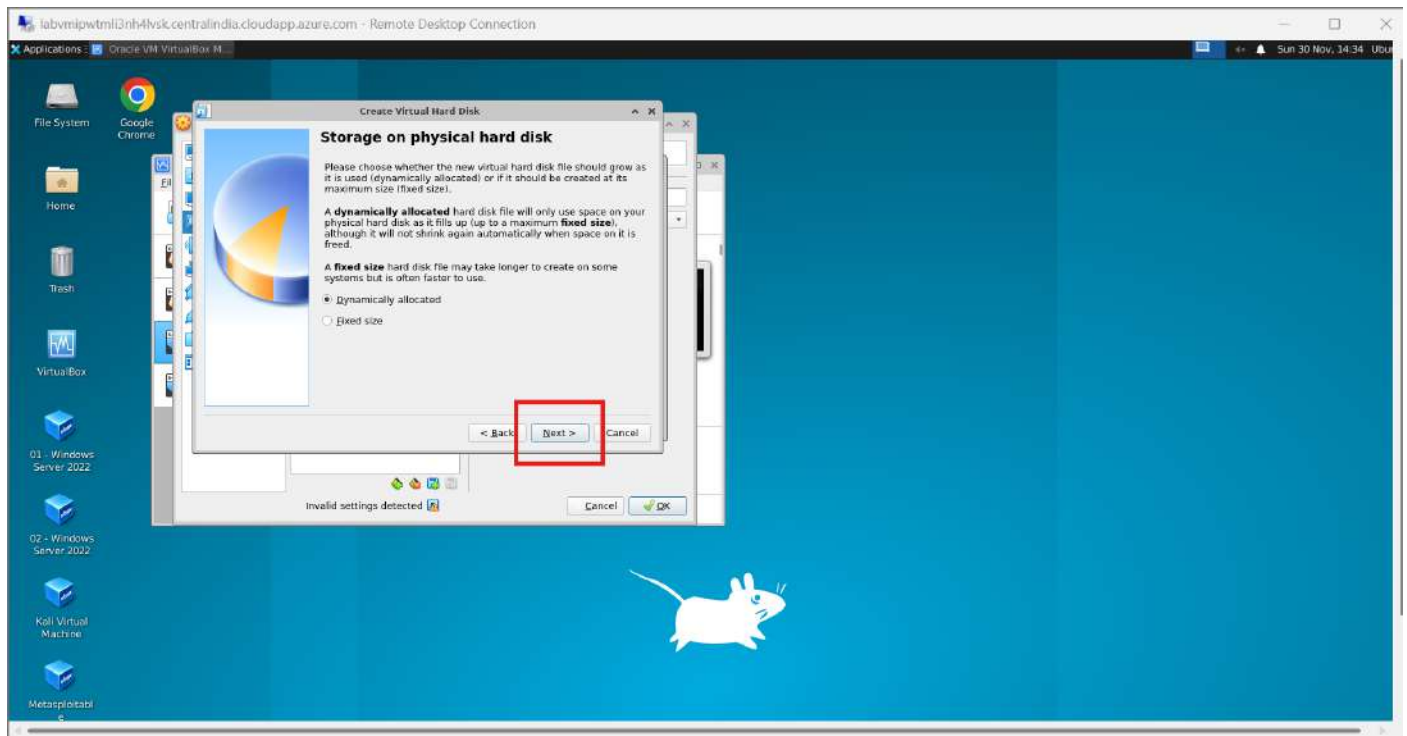
1.4 Clicked on **Create**.



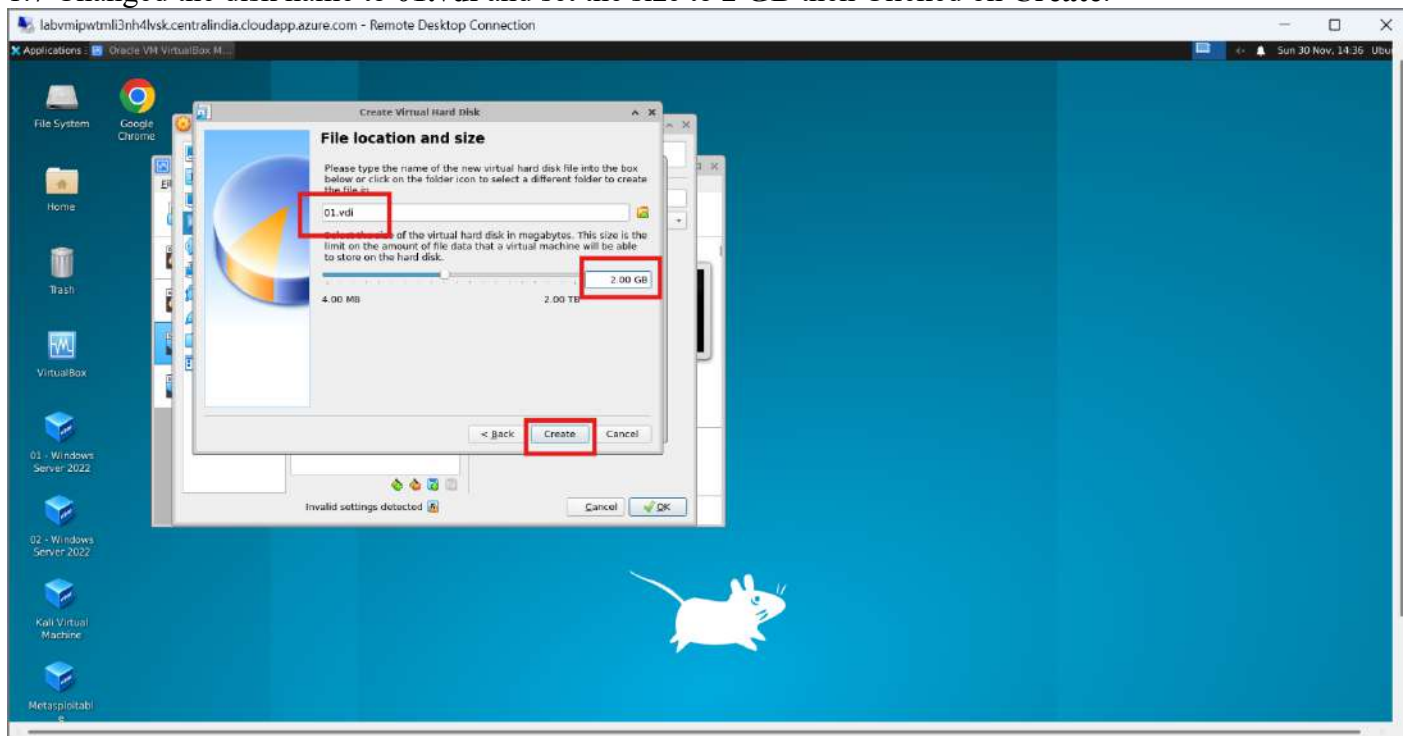
1.5 Checked the radio button of **VHD** and clicked on **Next**.



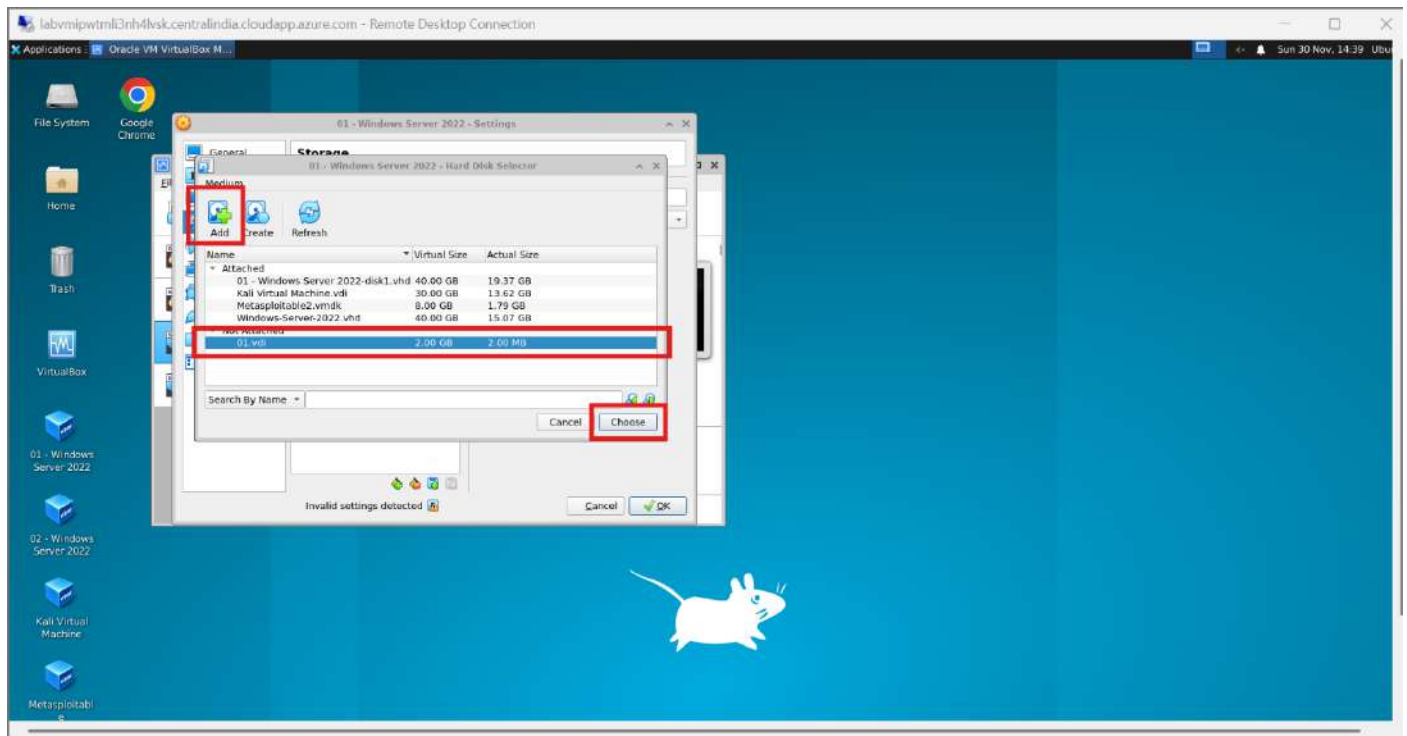
1.6 Clicked on **Next**.



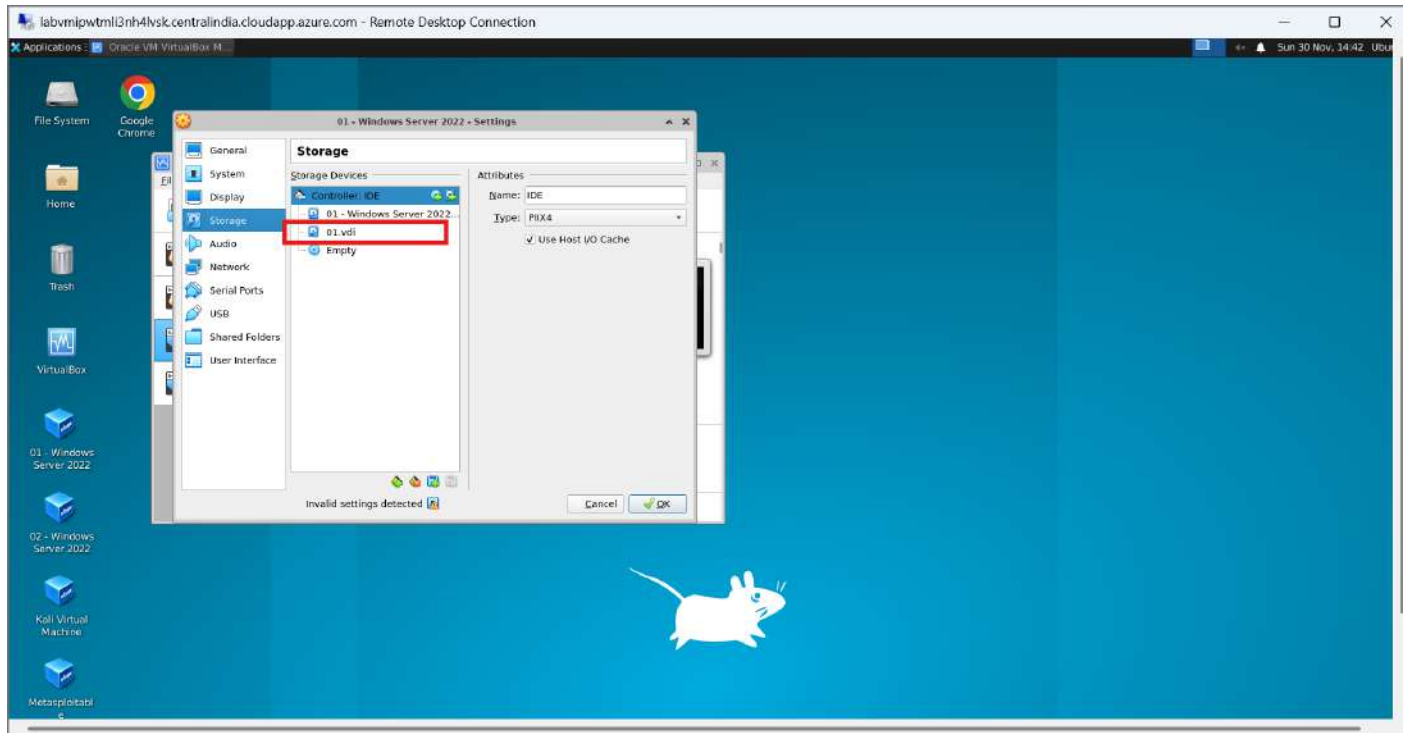
1.7 Changed the disk name to **01.vdi** and set the size to **2 GB** then Clicked on **Create**.



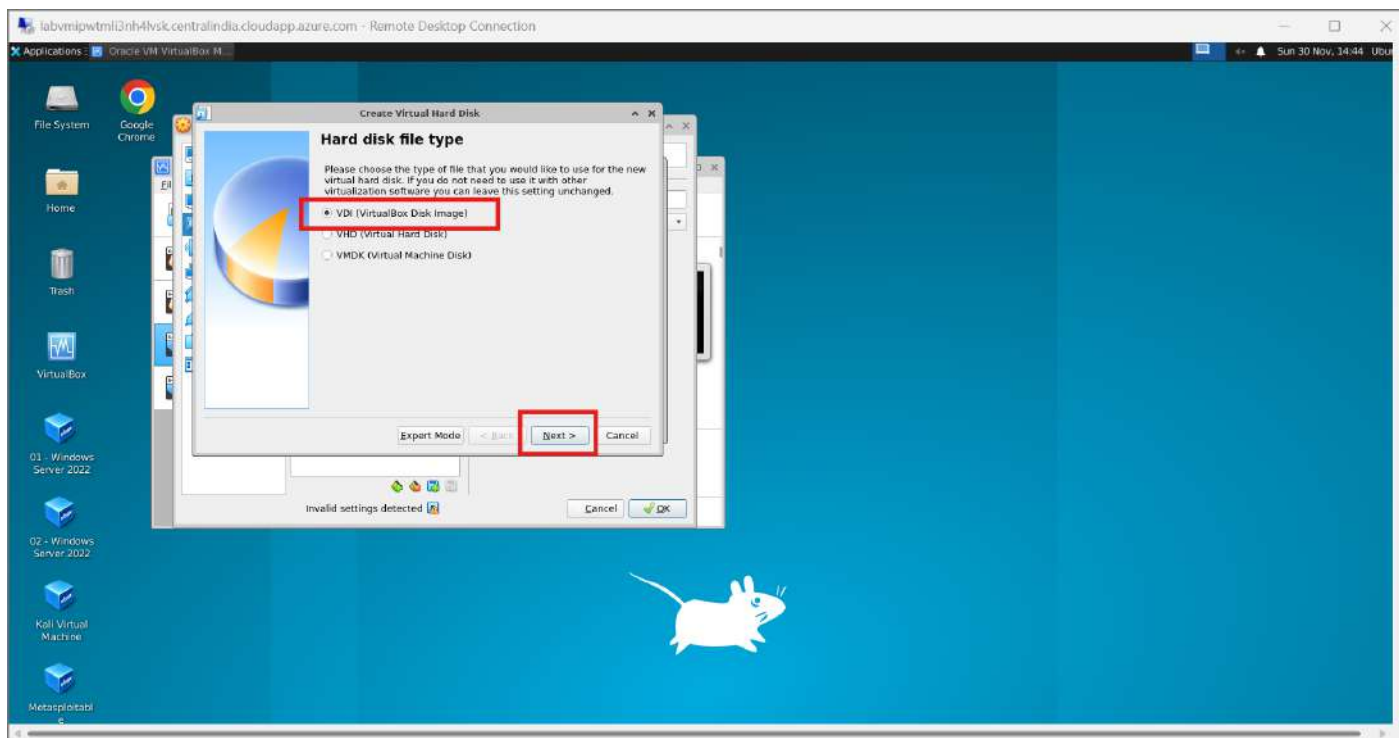
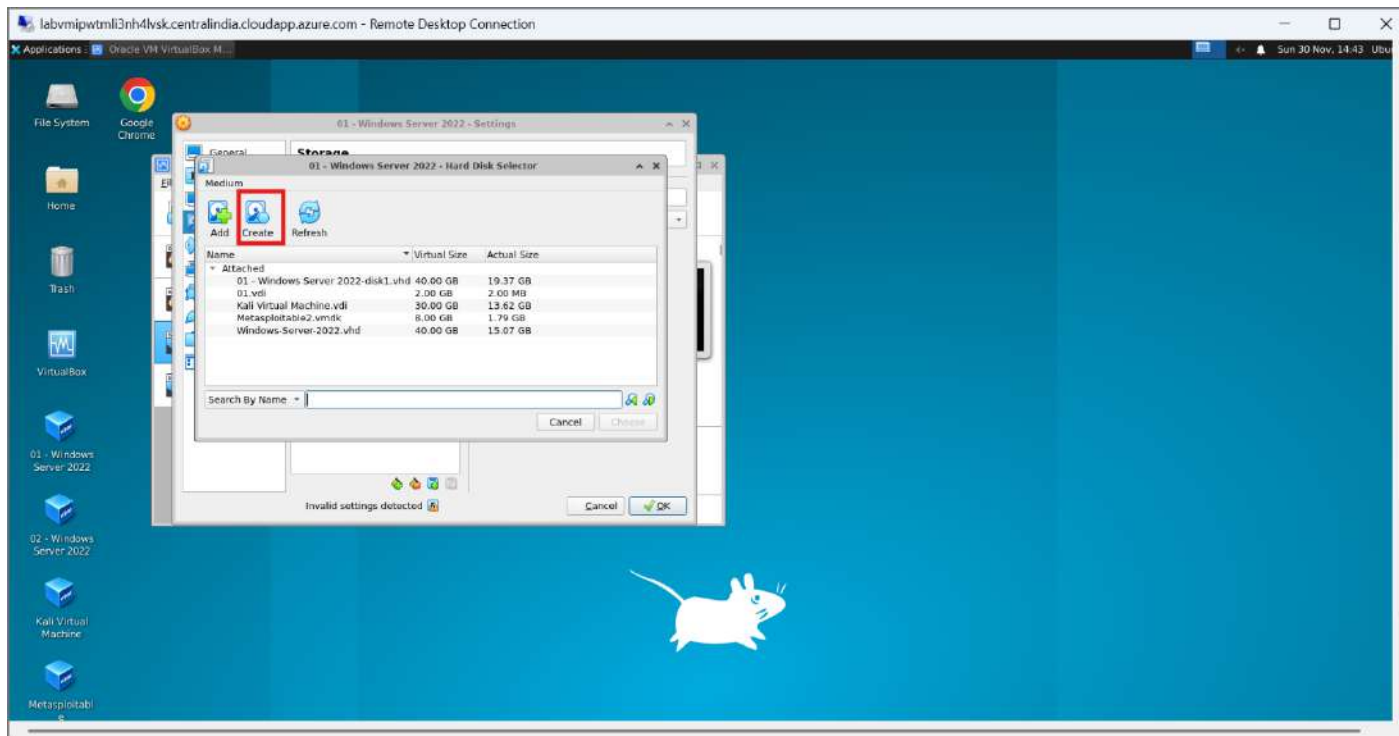
1.8 Clicked on **Add**, selected the **01.vhd** and Clicked on **Choose**.

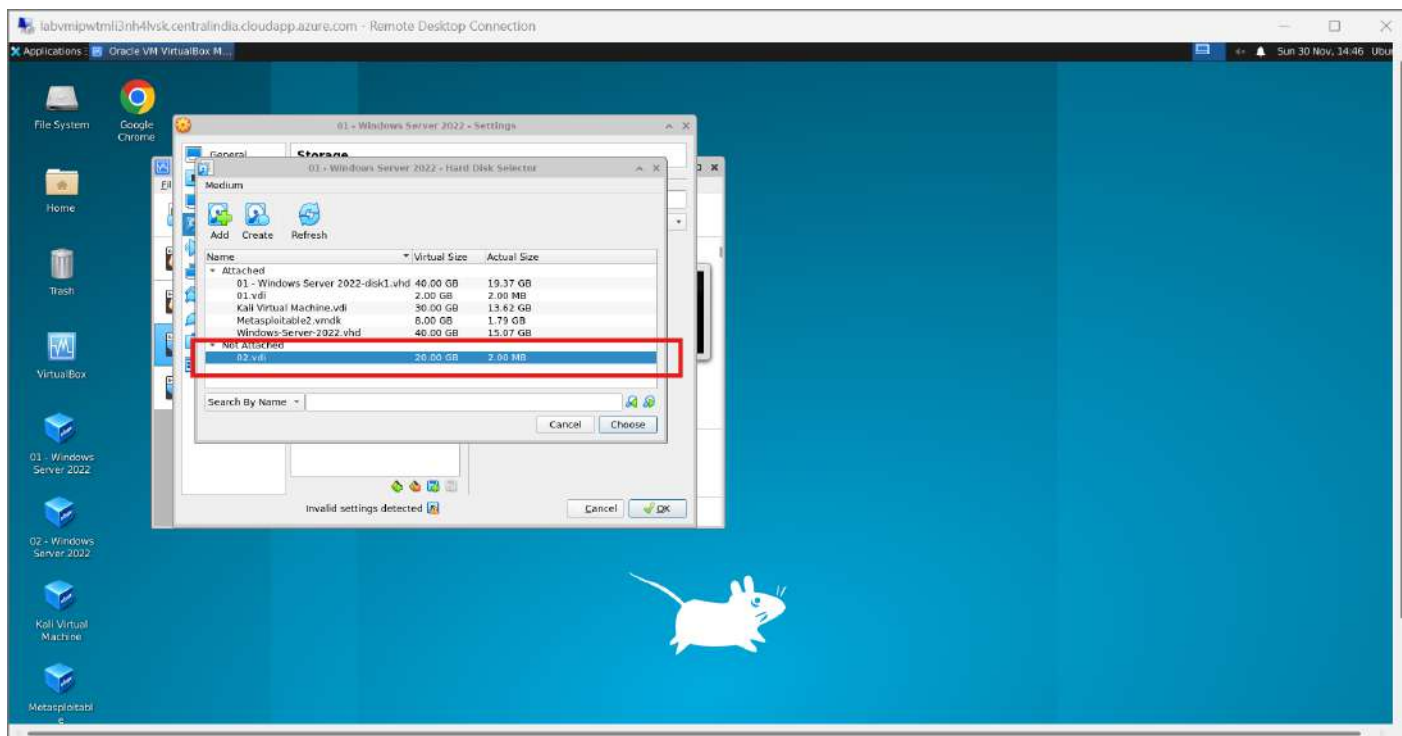
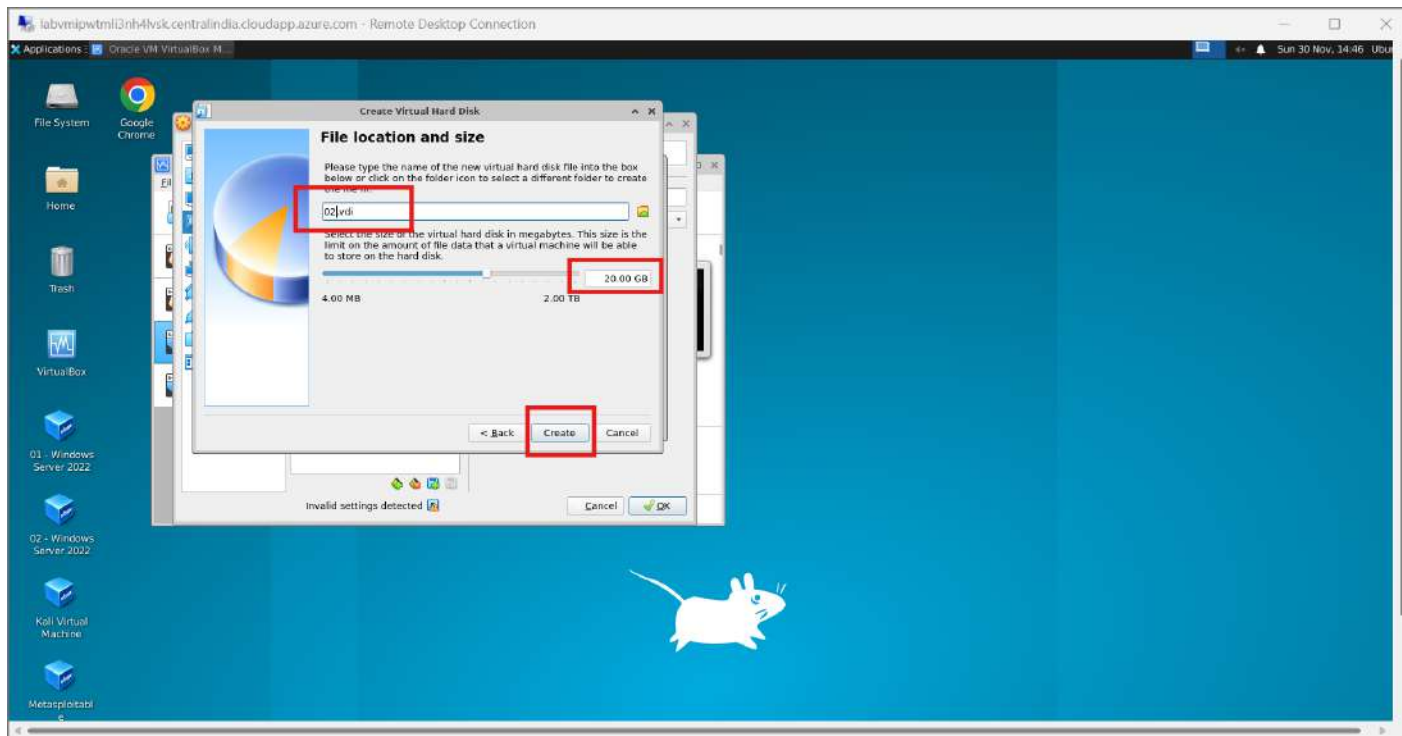


1.9 Disk is added

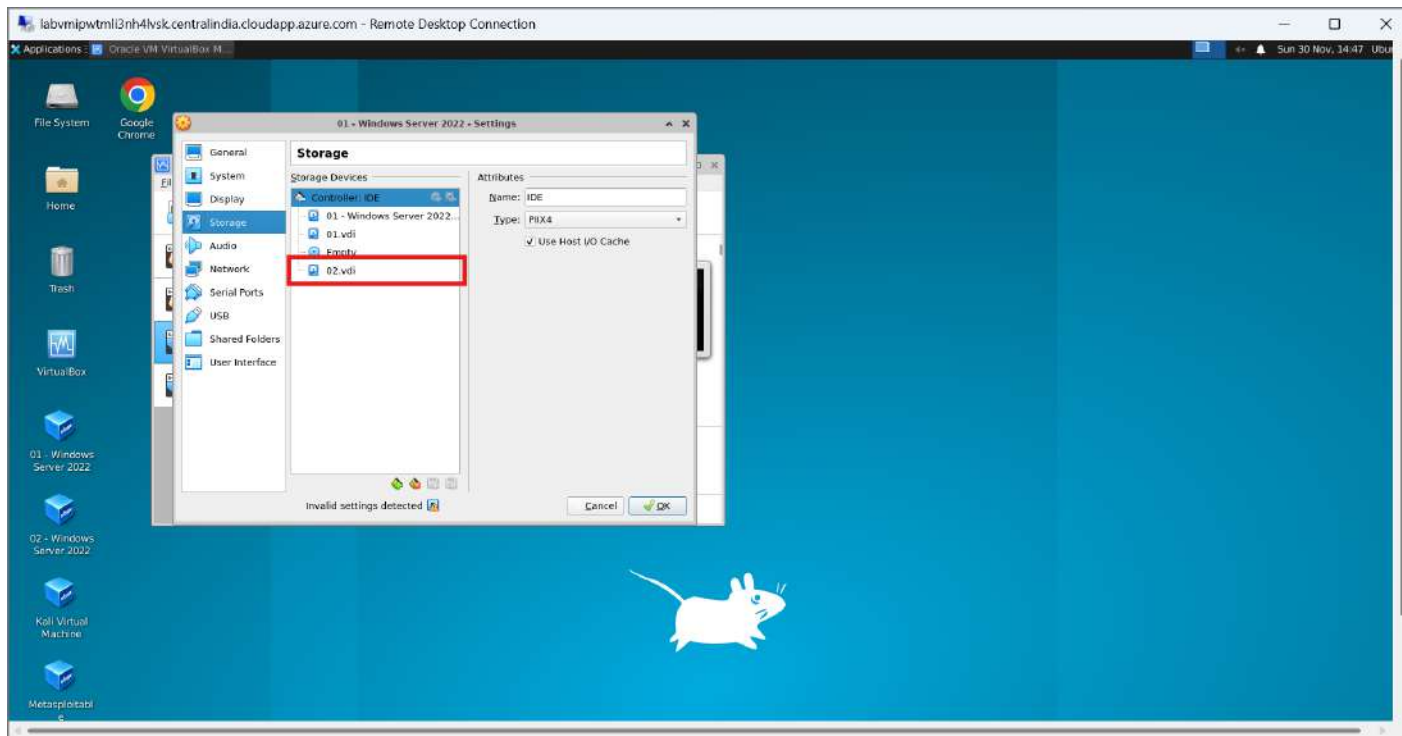


1.10 Now again doing same steps to create a **20 GB** disk.

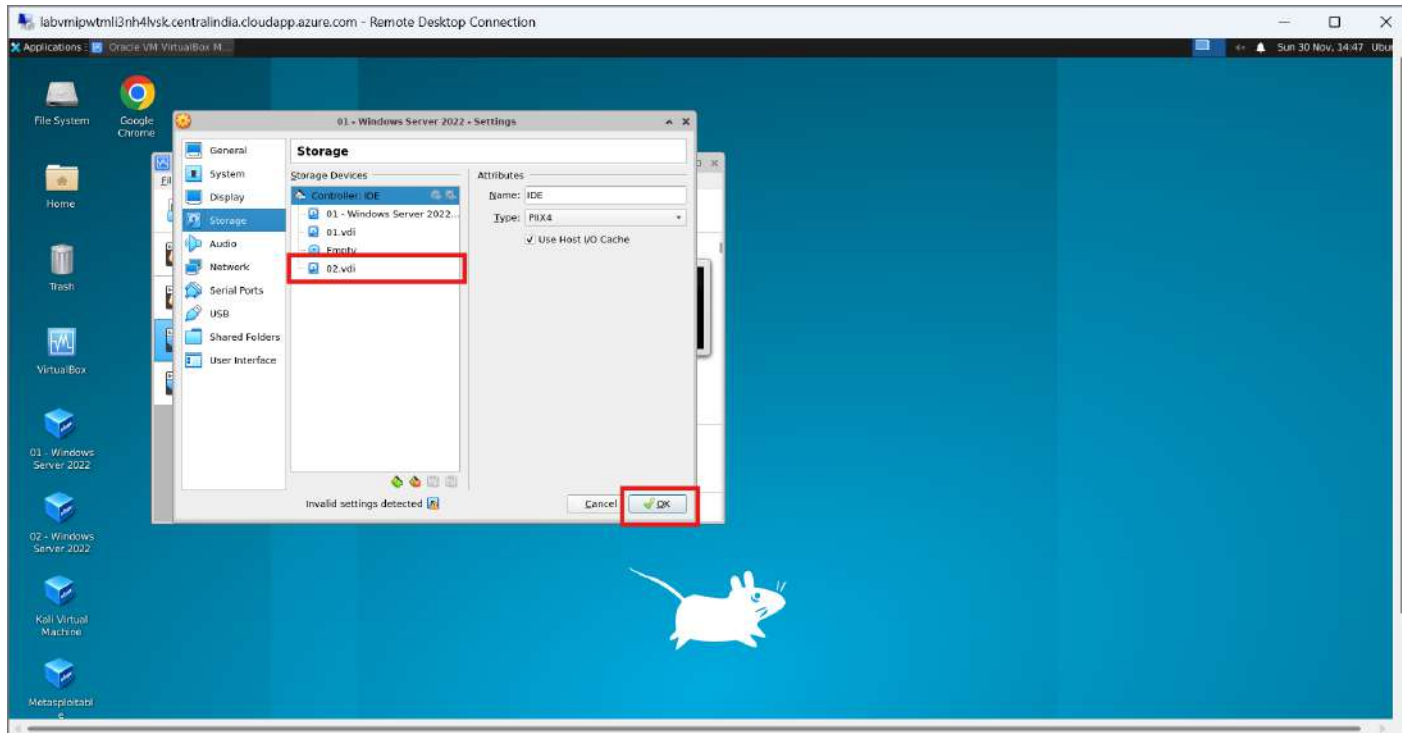




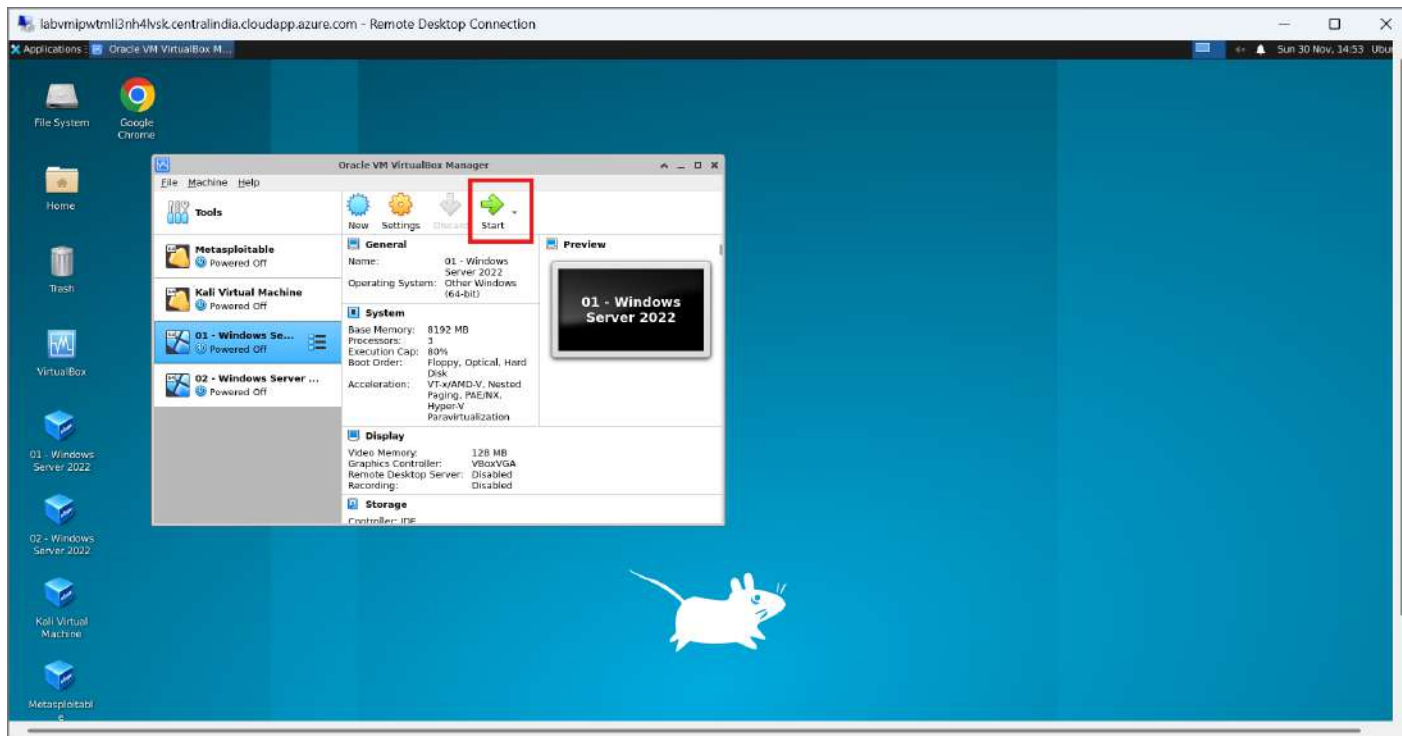
1.11 Disk is added



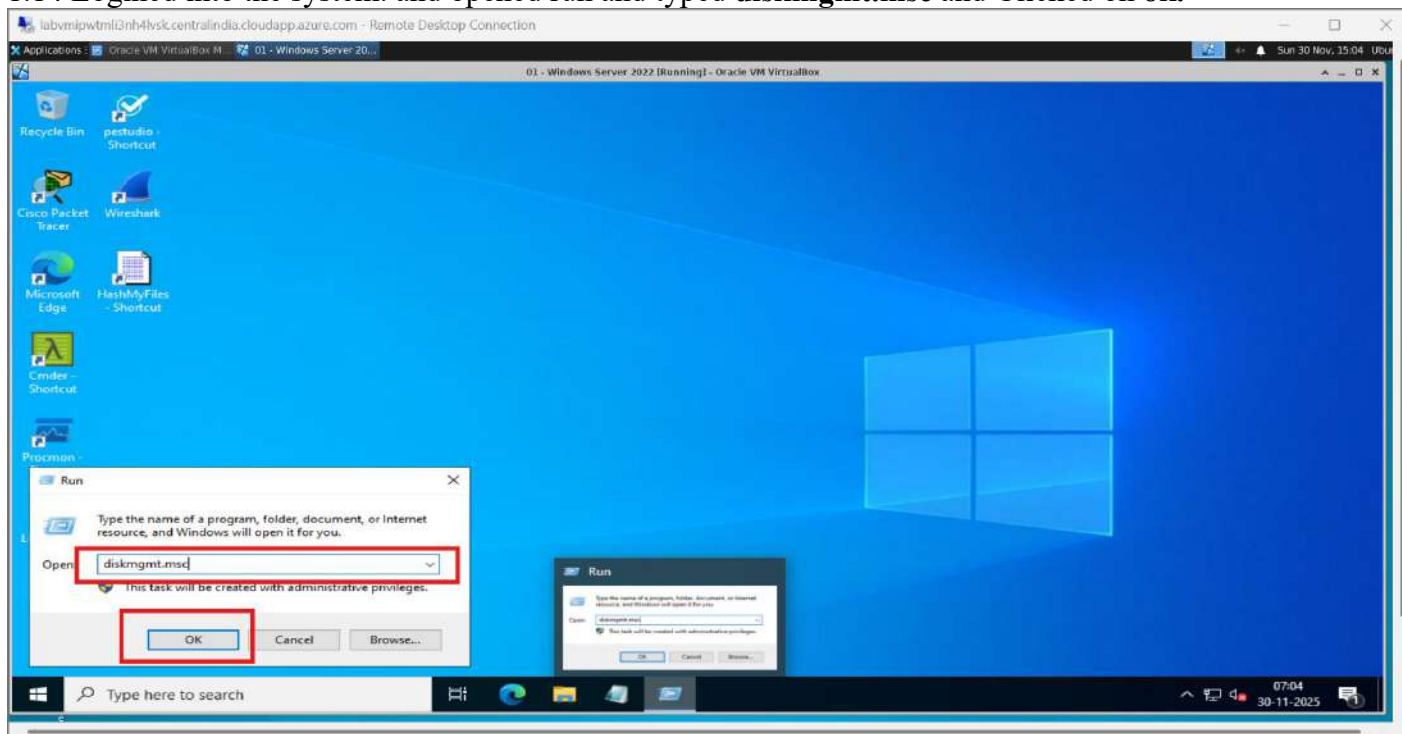
1.12 Now Clicked on **Ok**.



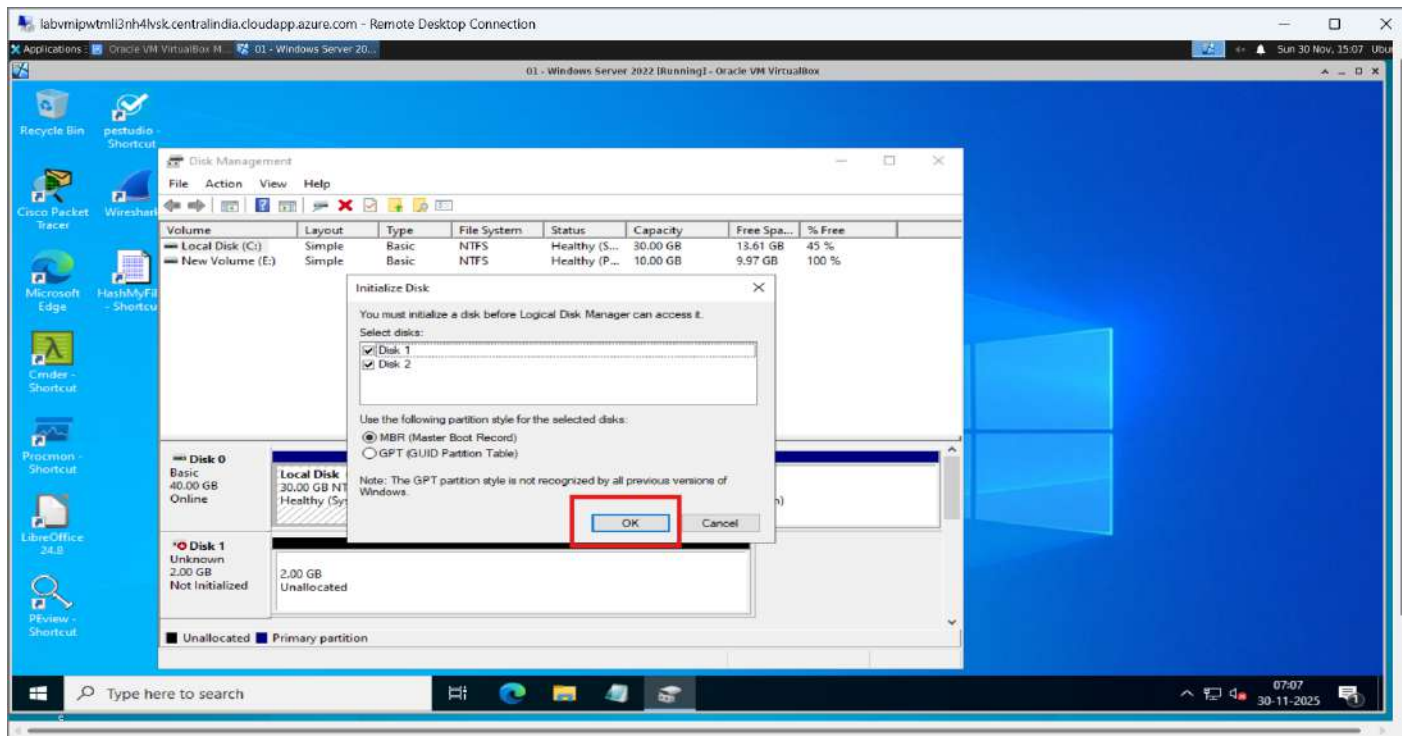
1.13 Start the **Virtual machine**.



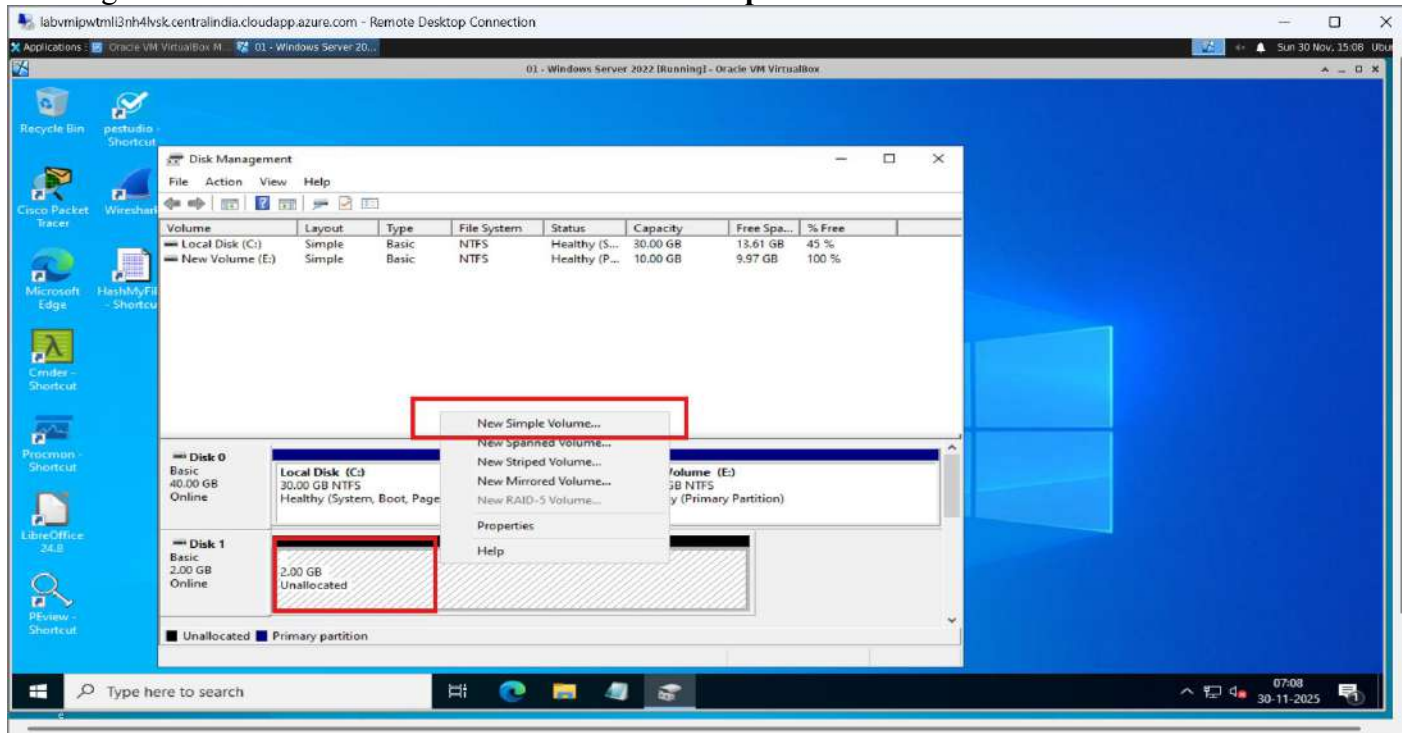
1.14 Logged into the system. and opened run and typed **diskmgmt.msc** and Clicked on **ok**.



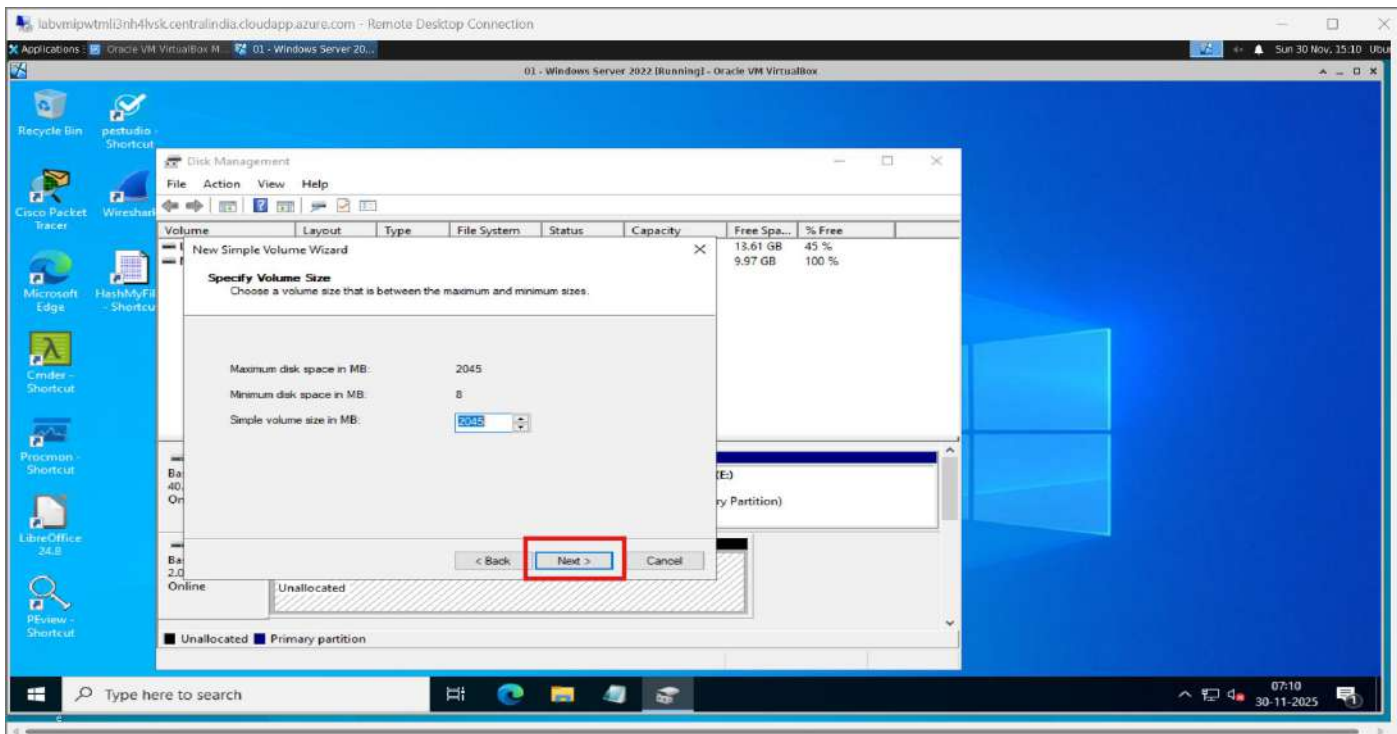
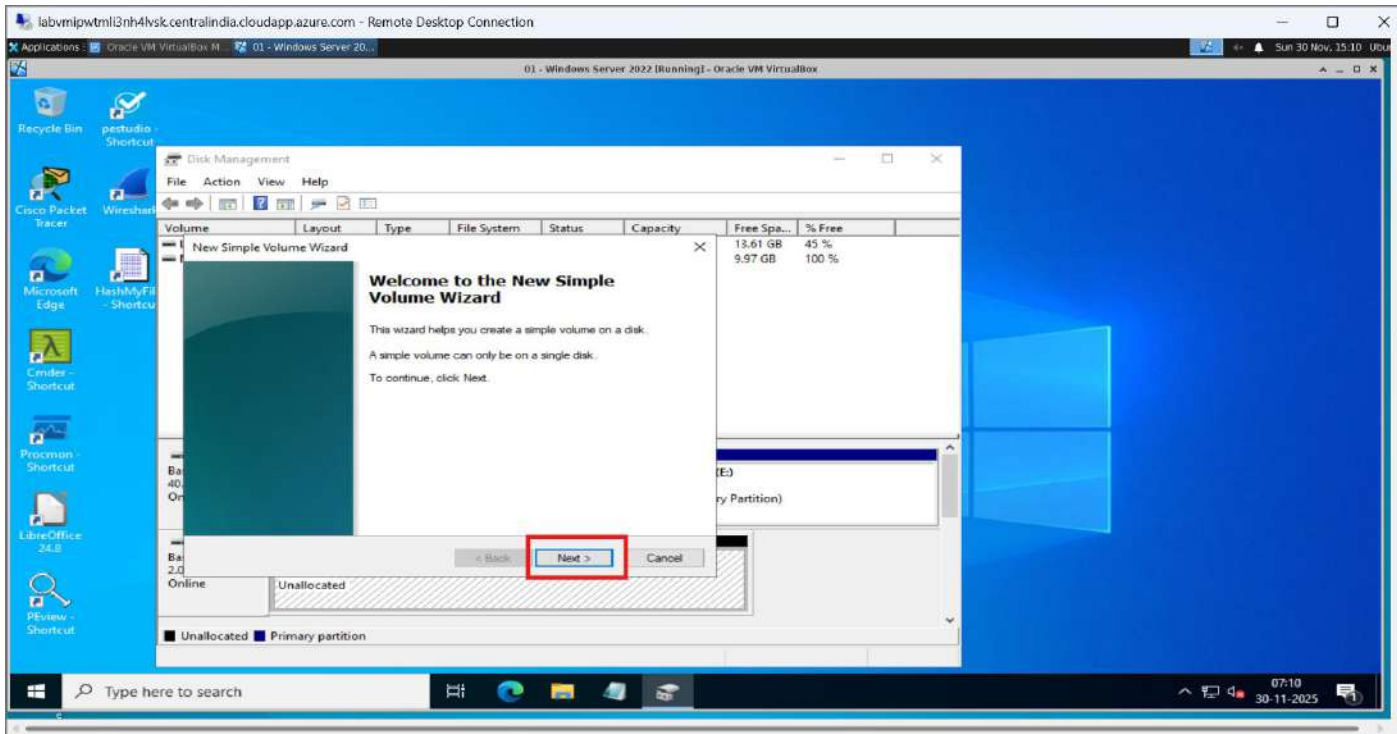
1.15 Clicked on **Ok**.

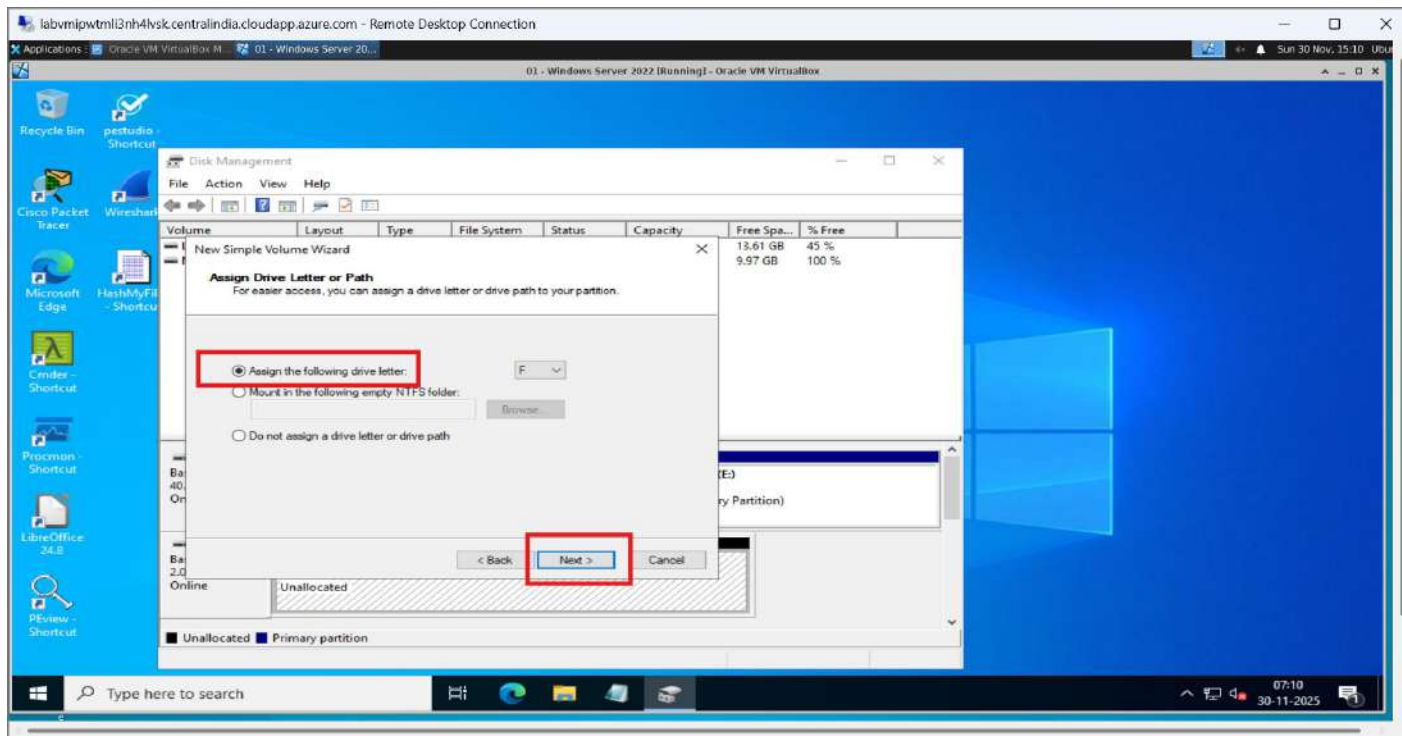


1.16 Right-Clicked on **Disk 1** and selected the “New Simple Volume”

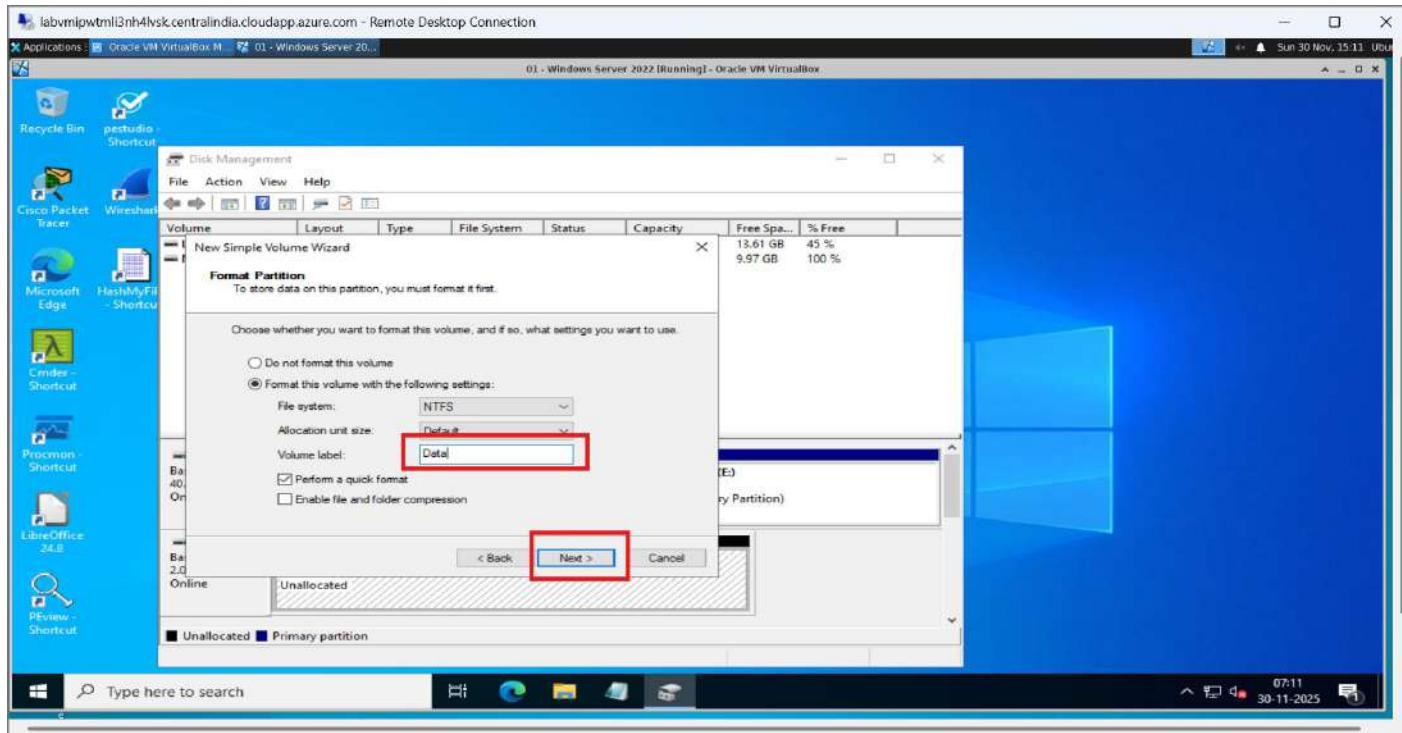


1.17 Clicked on **Next** until volume labels comes.

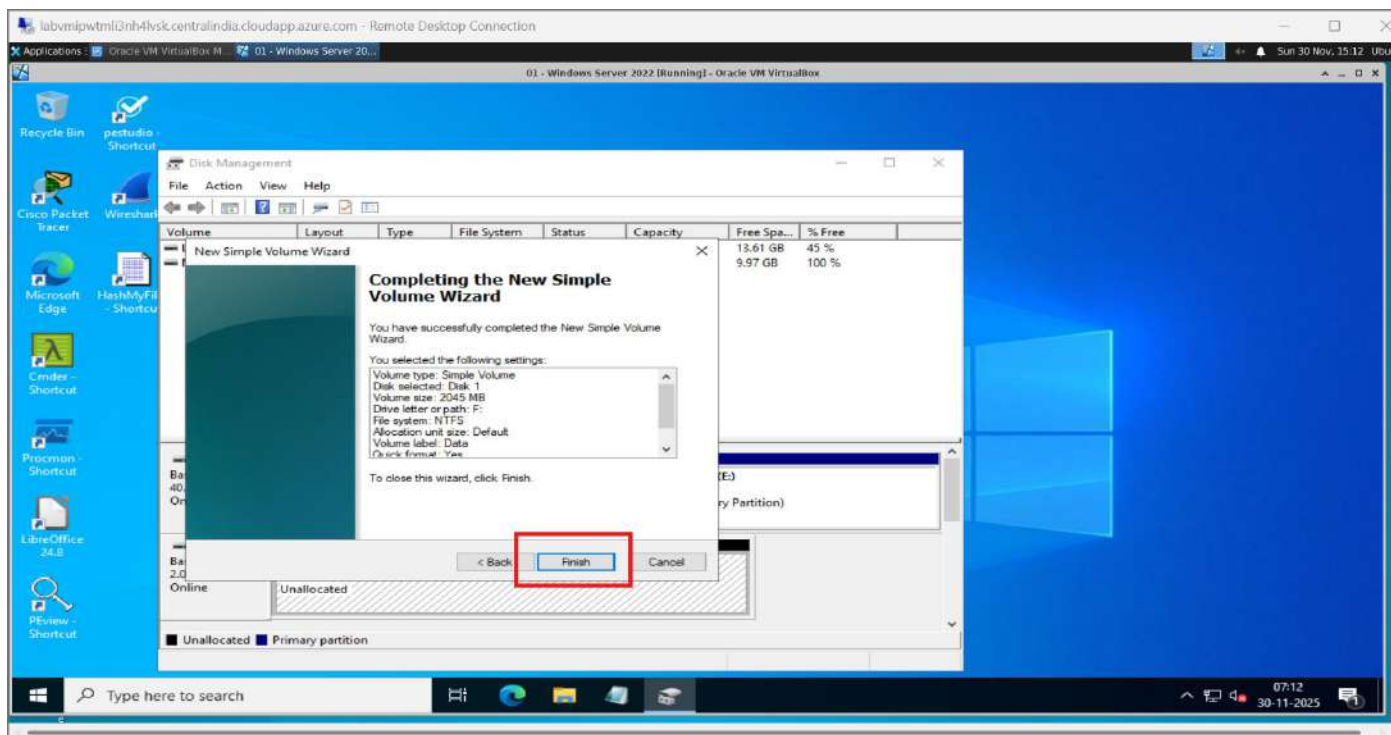




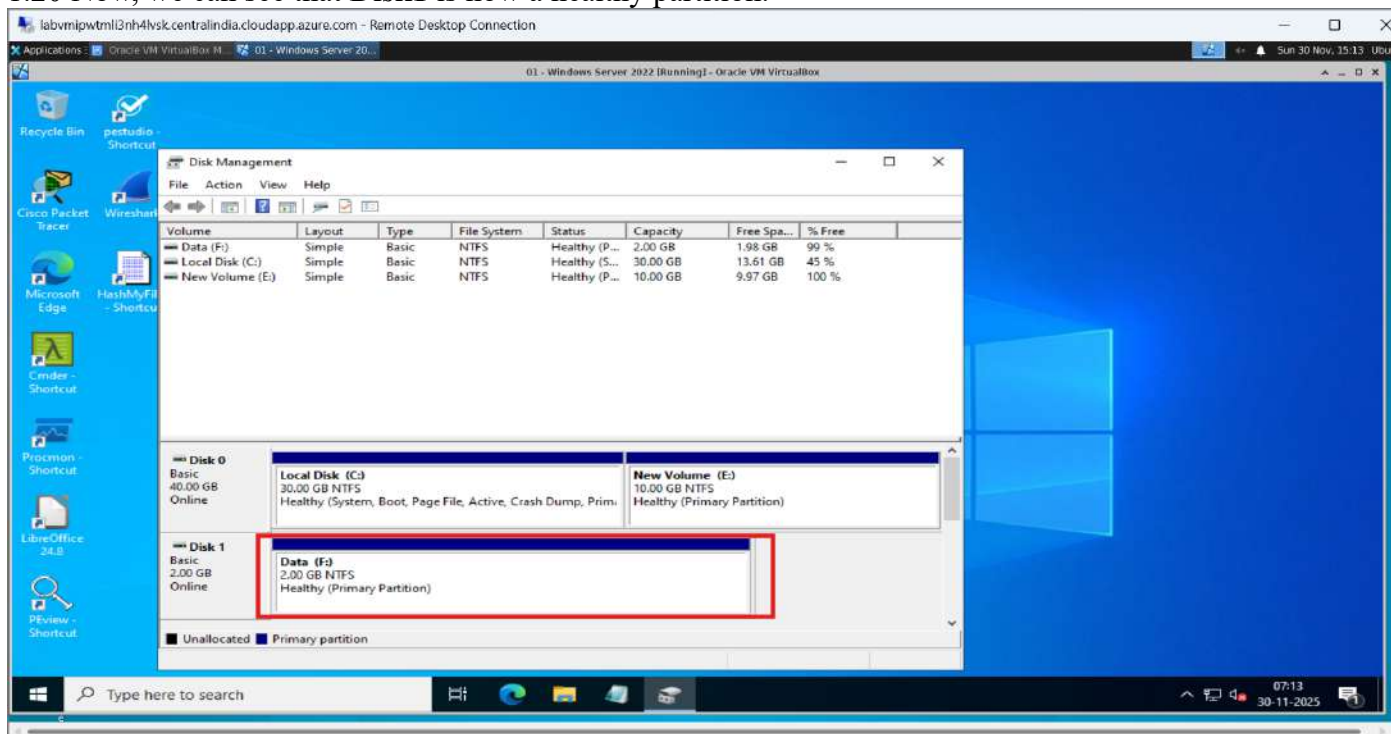
1.18 Write the Volume label as **Data** and Clicked on **Next**.



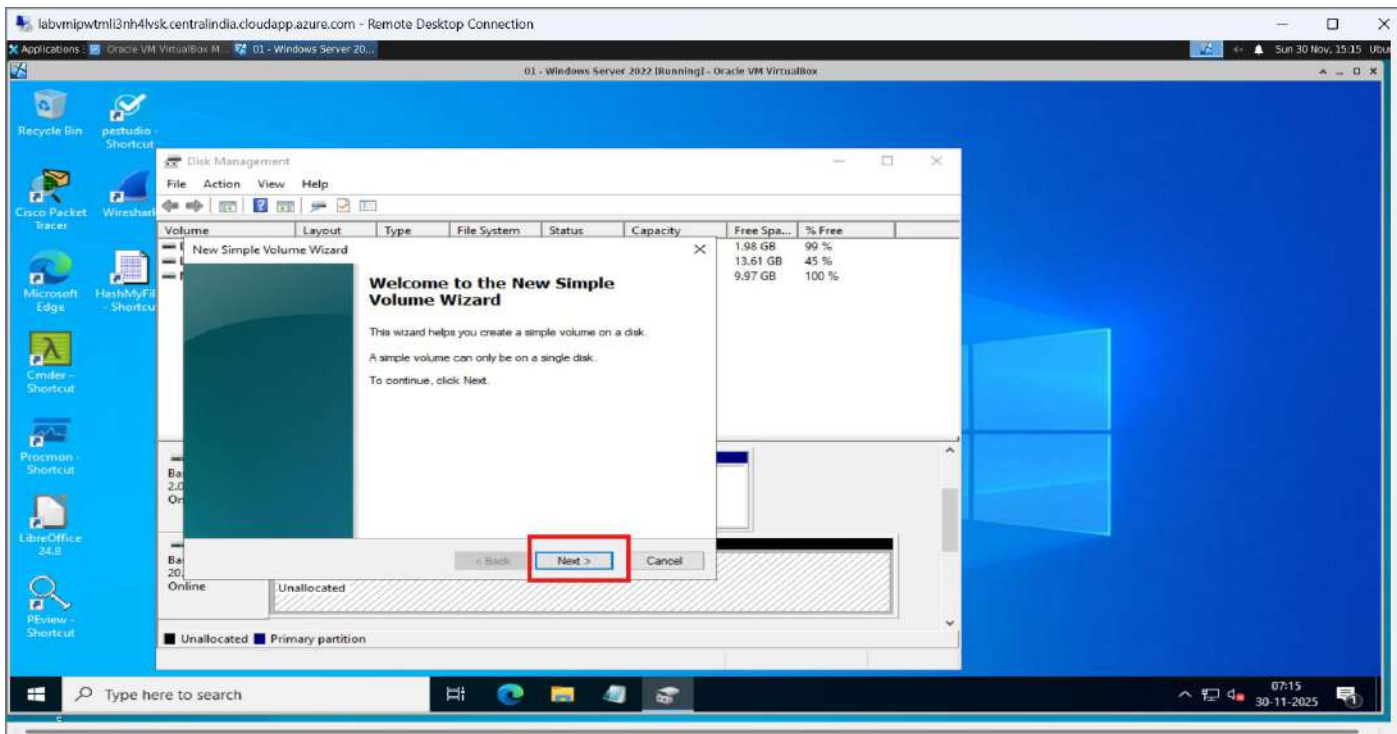
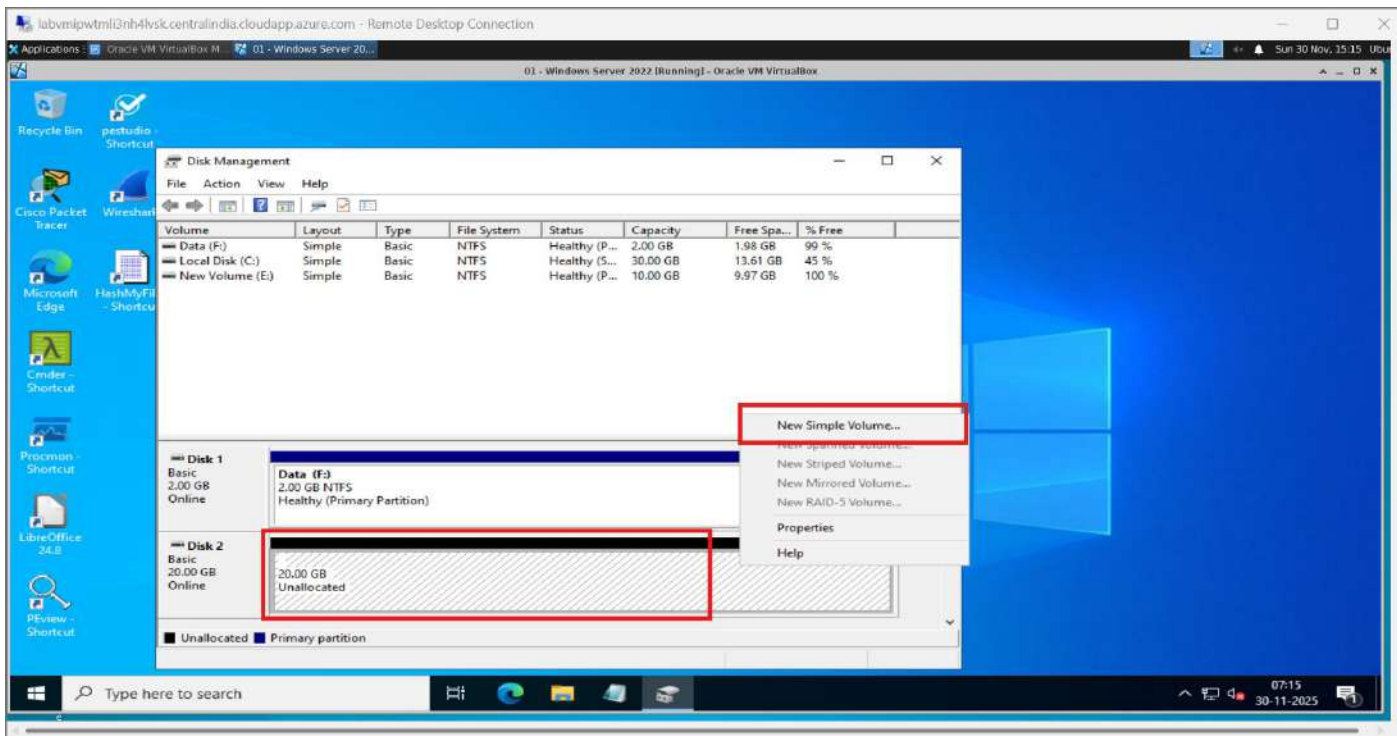
1.19 Clicked on **Finish**.

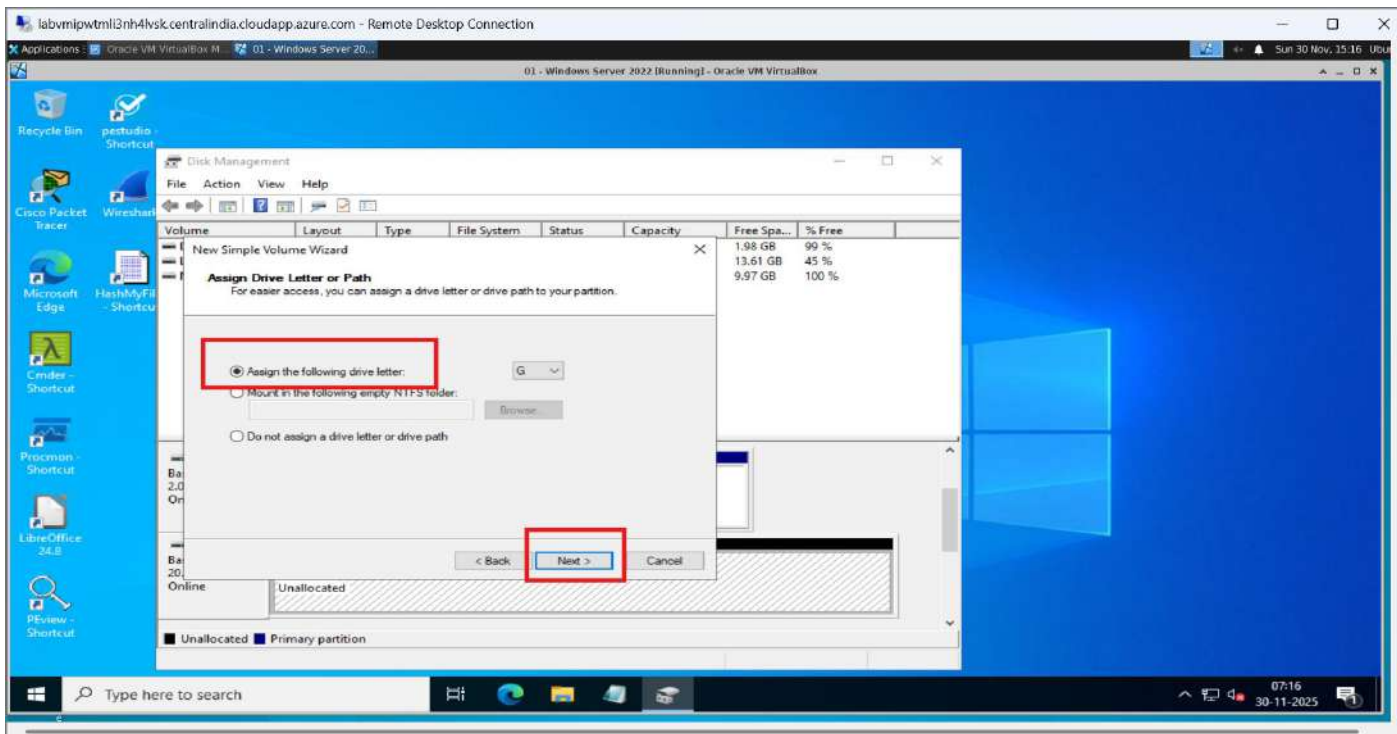
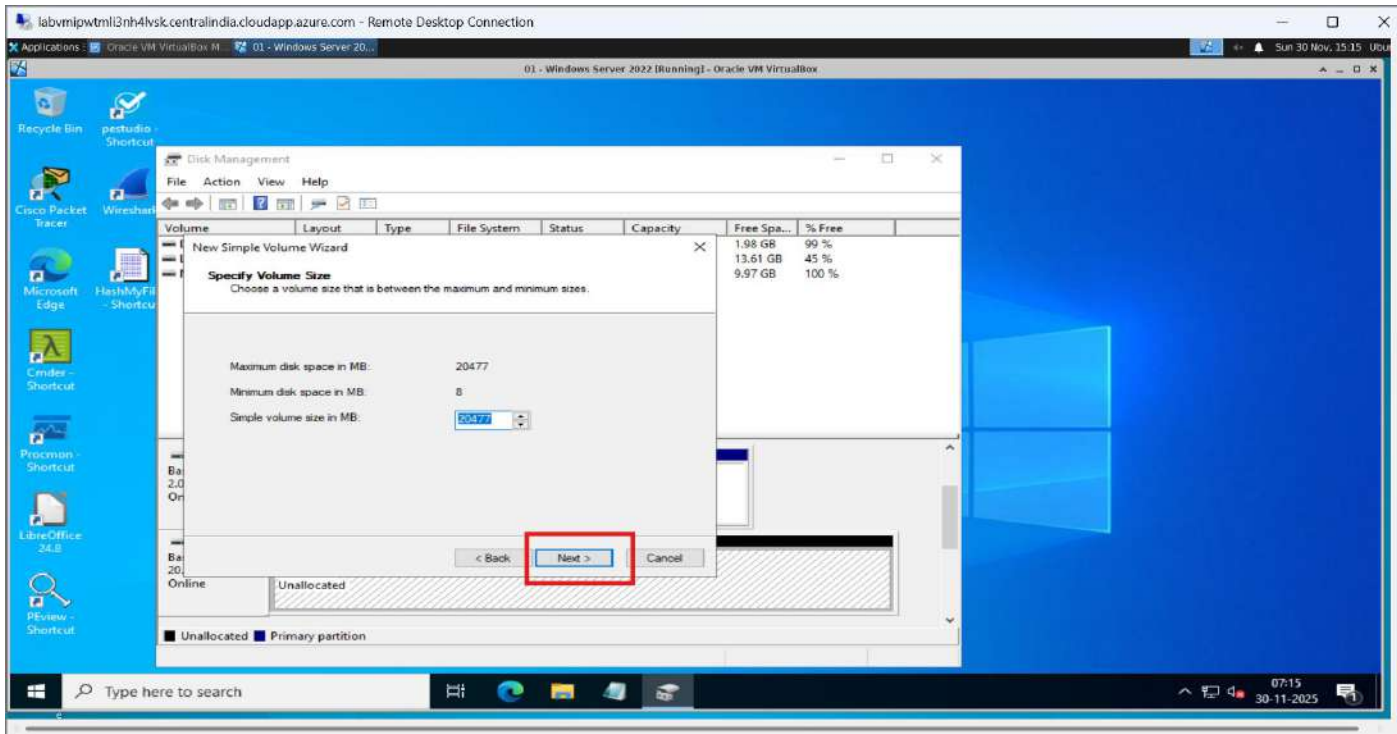


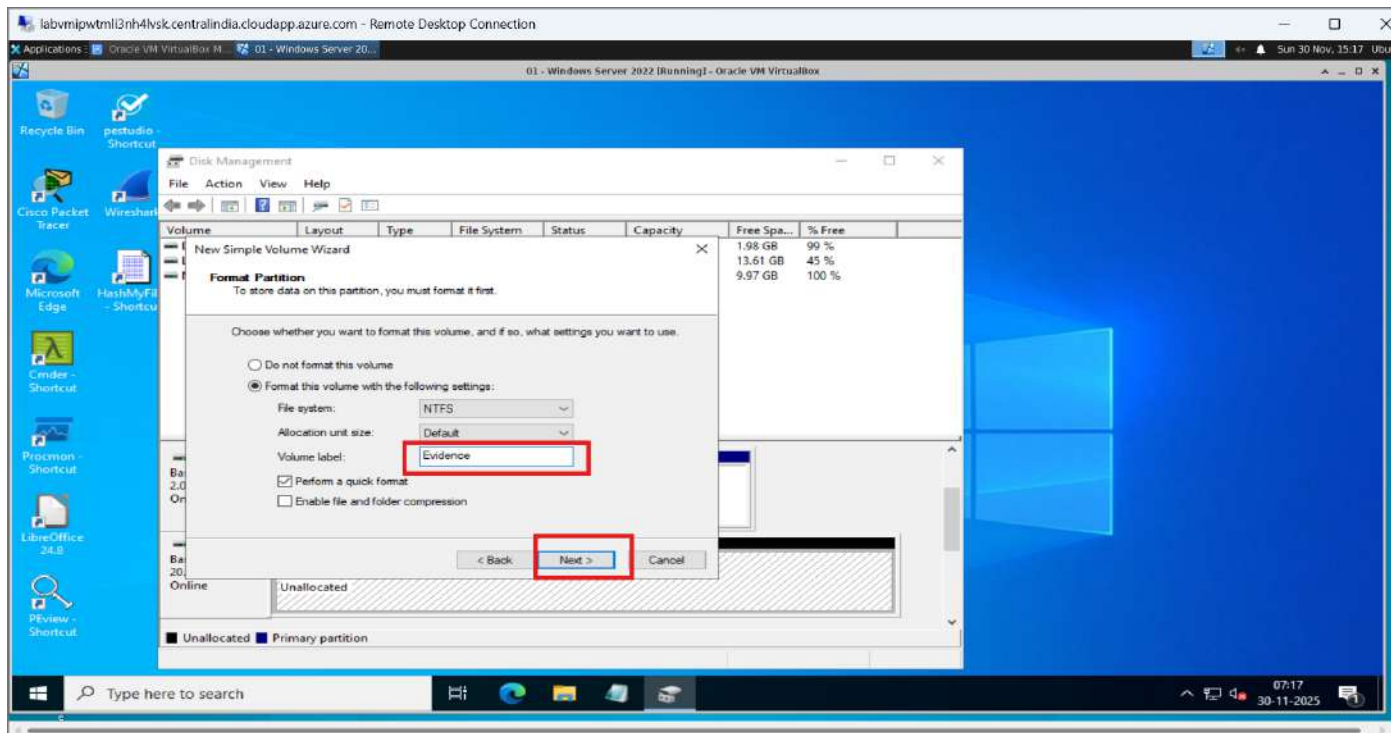
1.20 Now, we can see that **Disk1** is now a healthy partition.



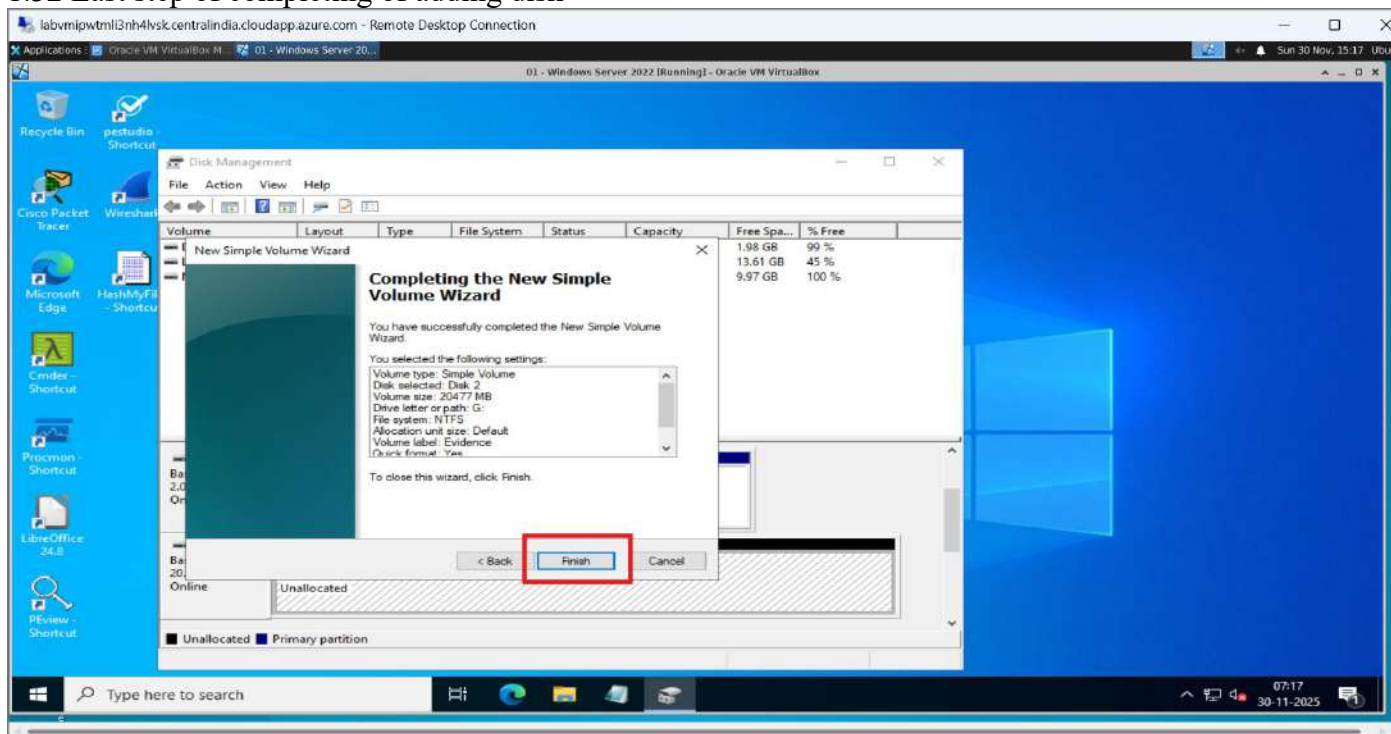
1.21 Similarly, creating a new partition of 20 GB and labeling it as Evidence.

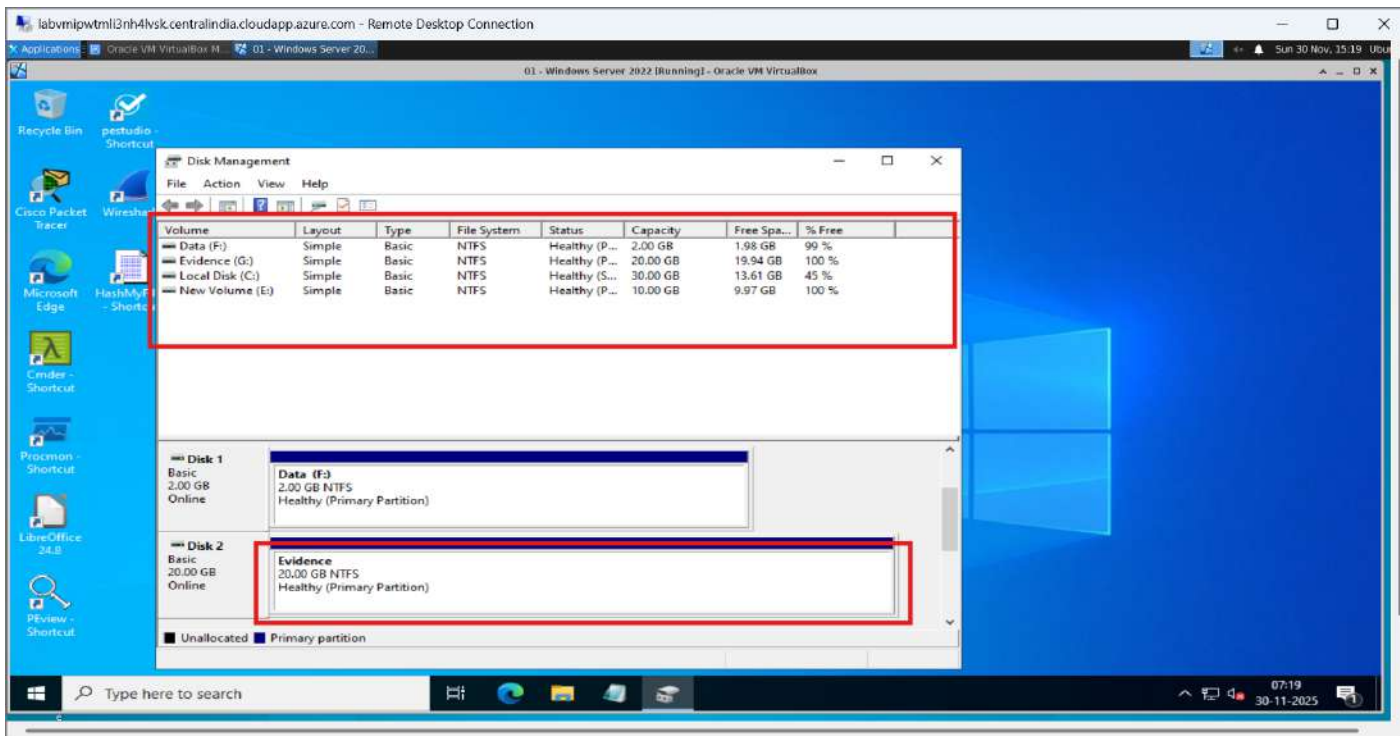






1.32 Last step of completing of adding disk

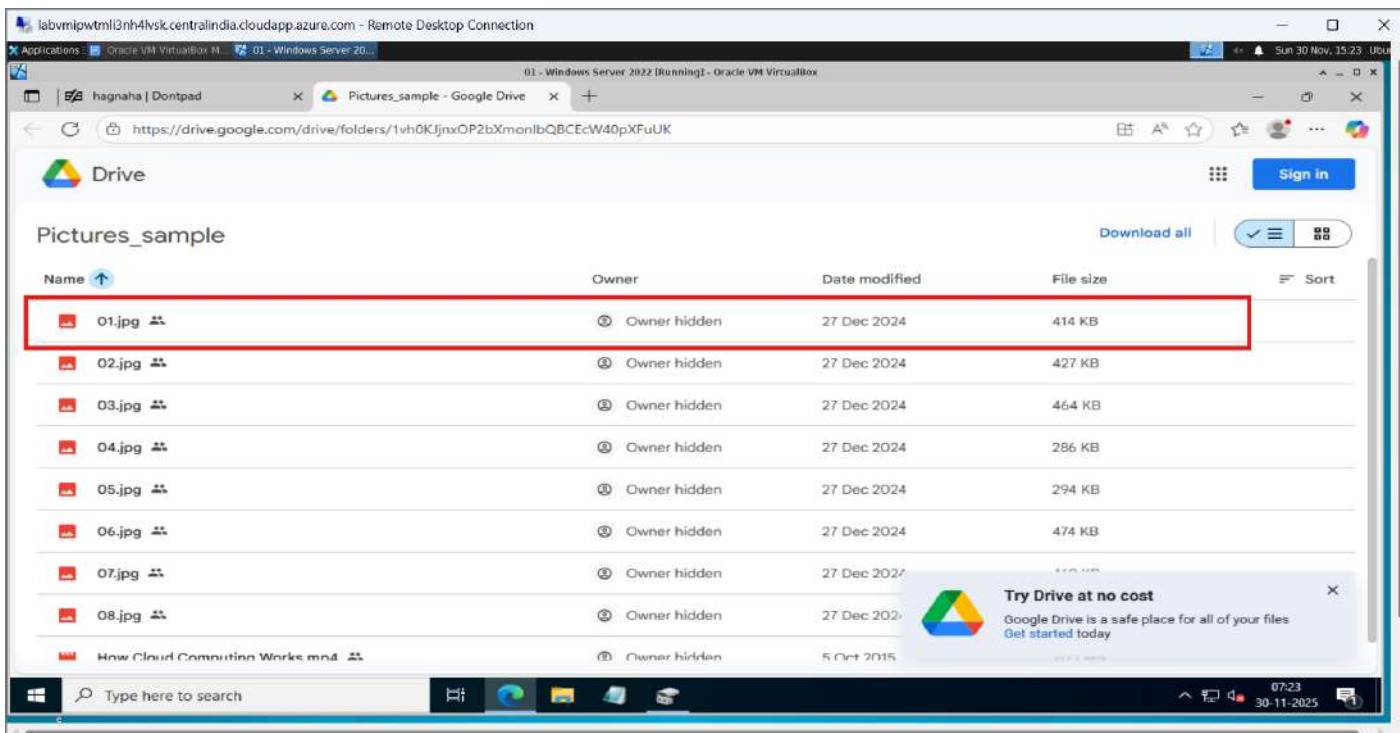




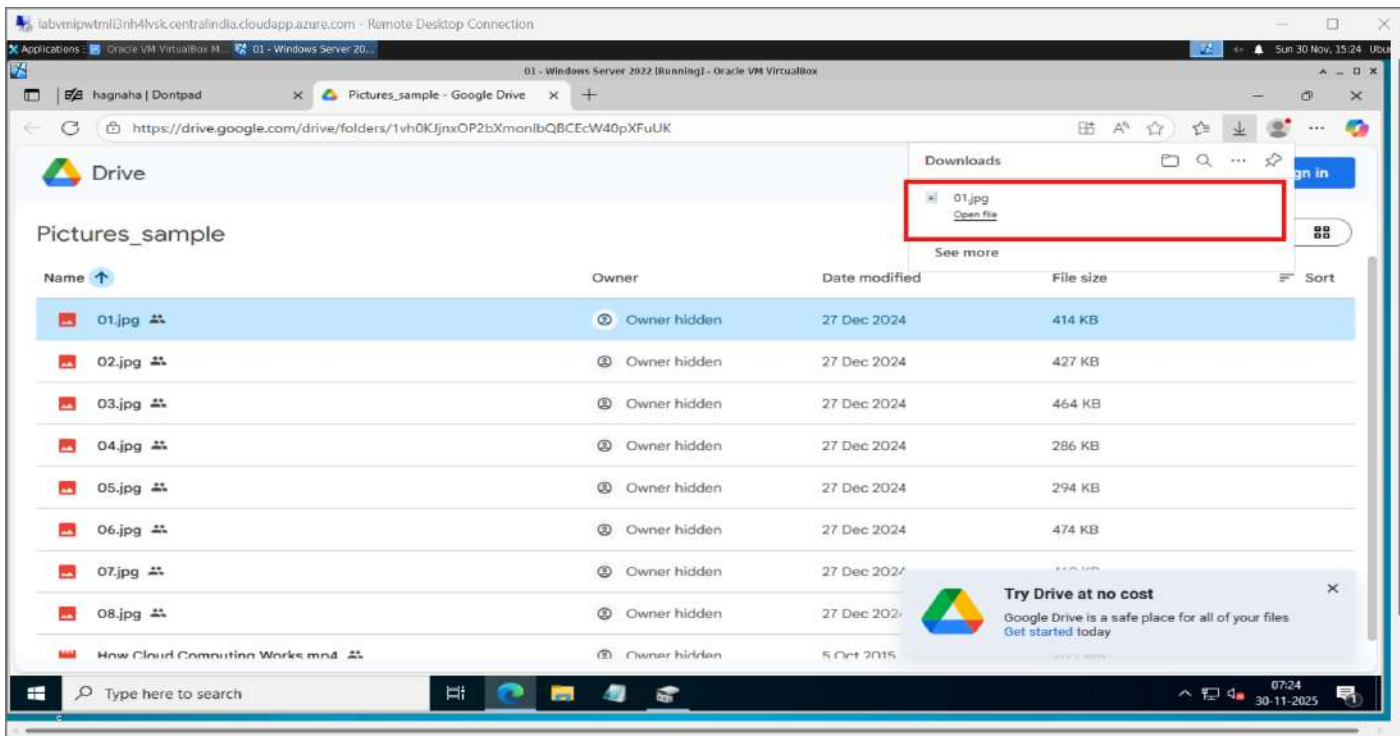
Step 2: Download and Prepare the Data to be tested

2.1 Download all files from the given link. Link:

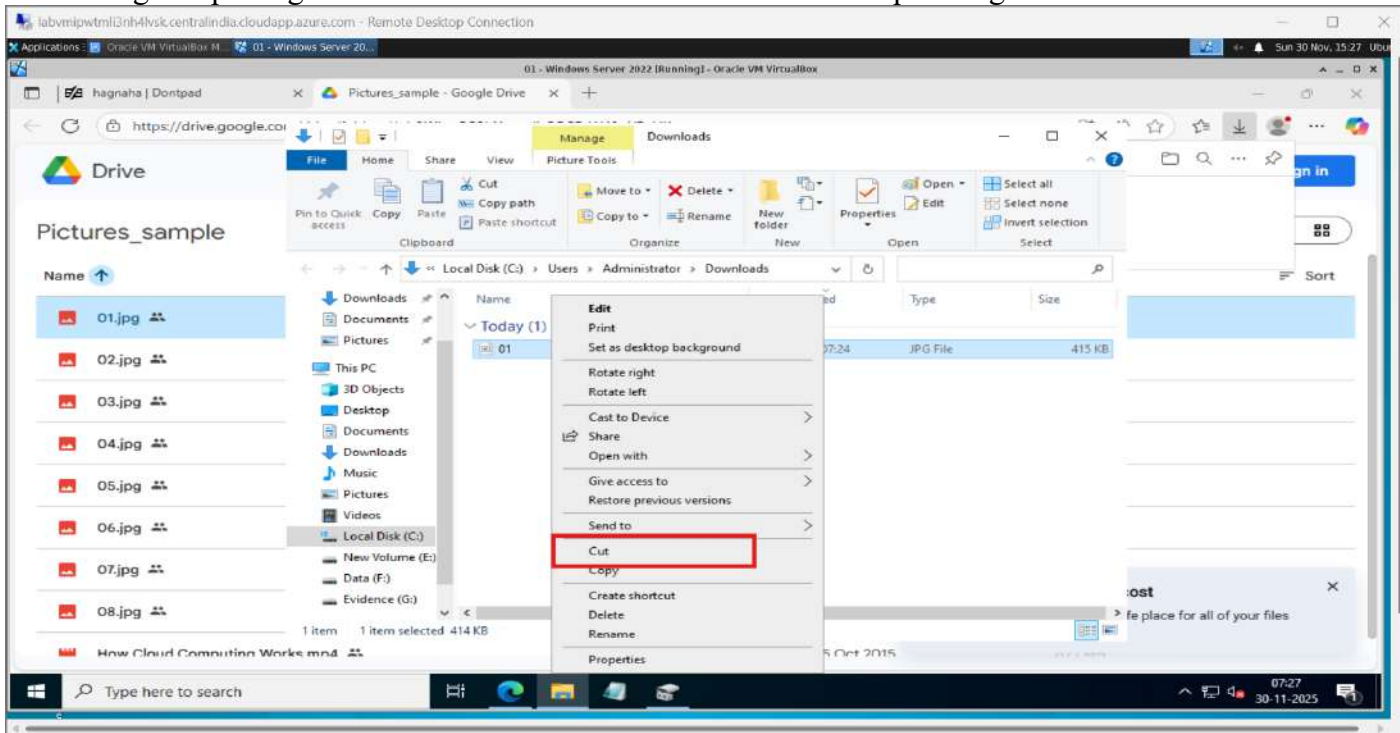
<https://drive.google.com/drive/folders/1vh0KJjnxOP2bXmonlbQBCEcW40pXFuUK?usp=sharing>

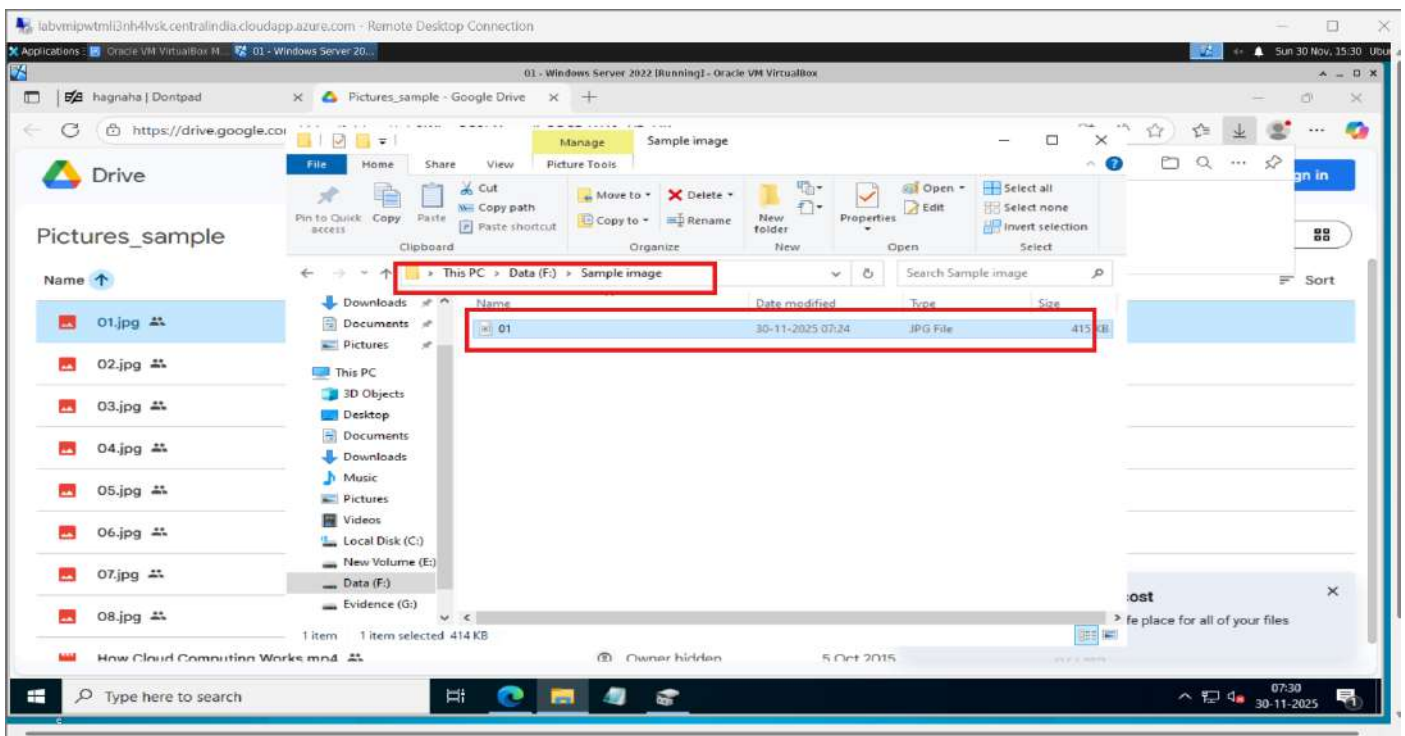
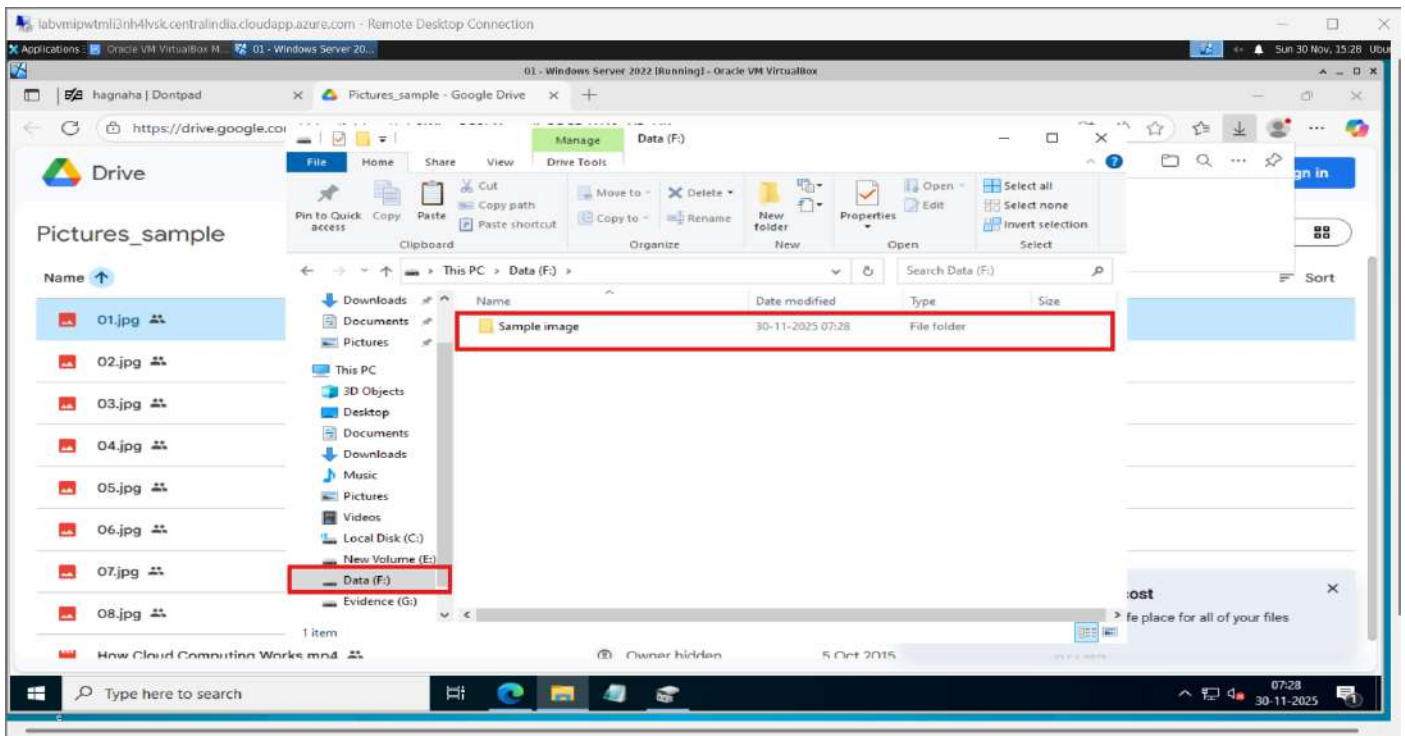


2.2 Downloaded file 01.jpg



2.3 Cutting and pasting this file into Data drive into a new folder sample image

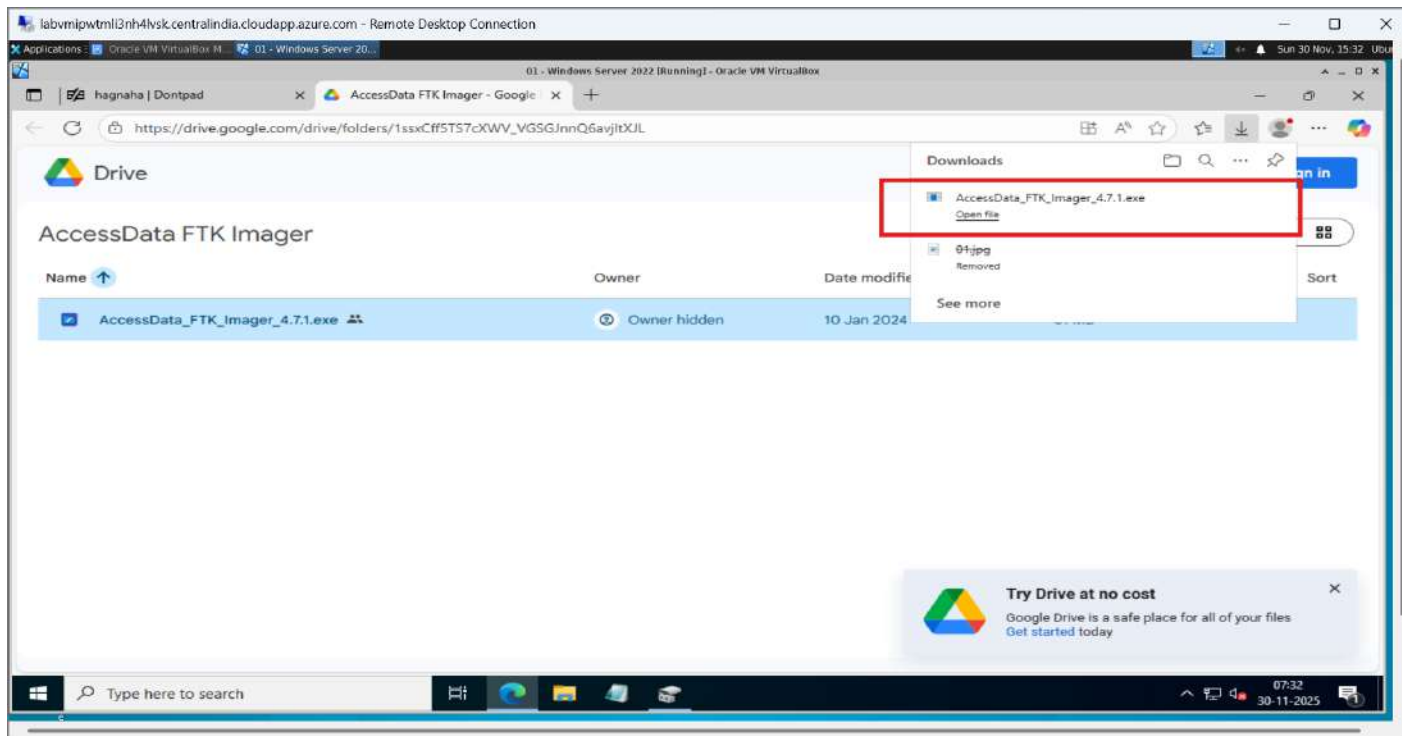




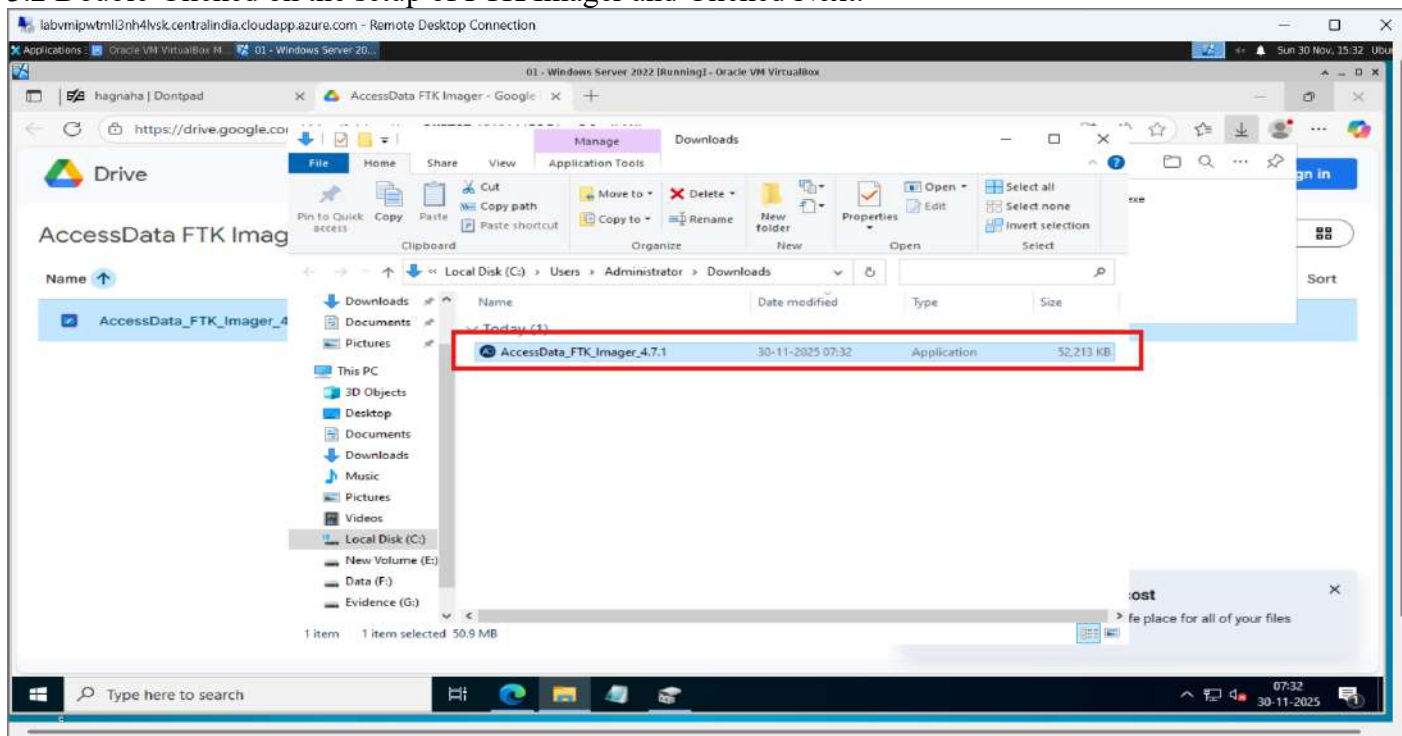
Step 3: Download Access data FTK Imager

3.1 Download Access data FTK Images from the following link. Link:

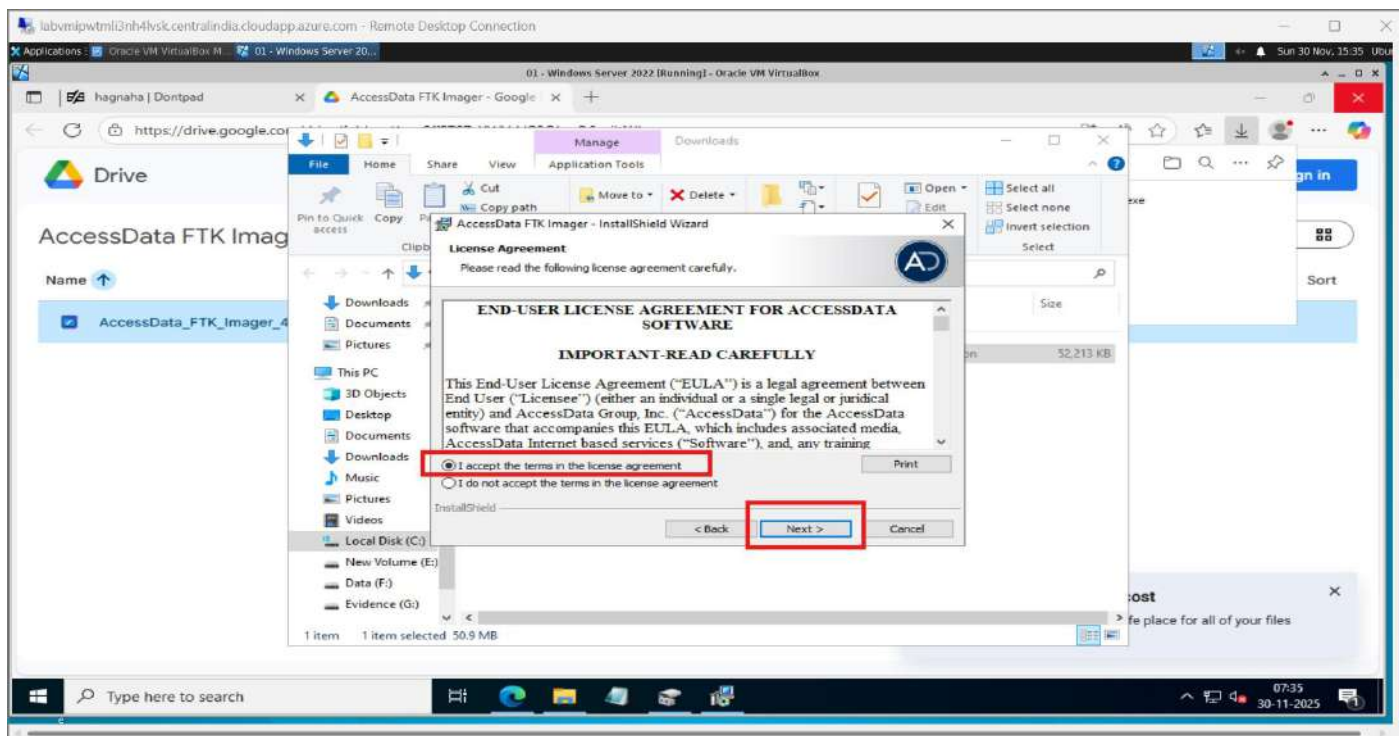
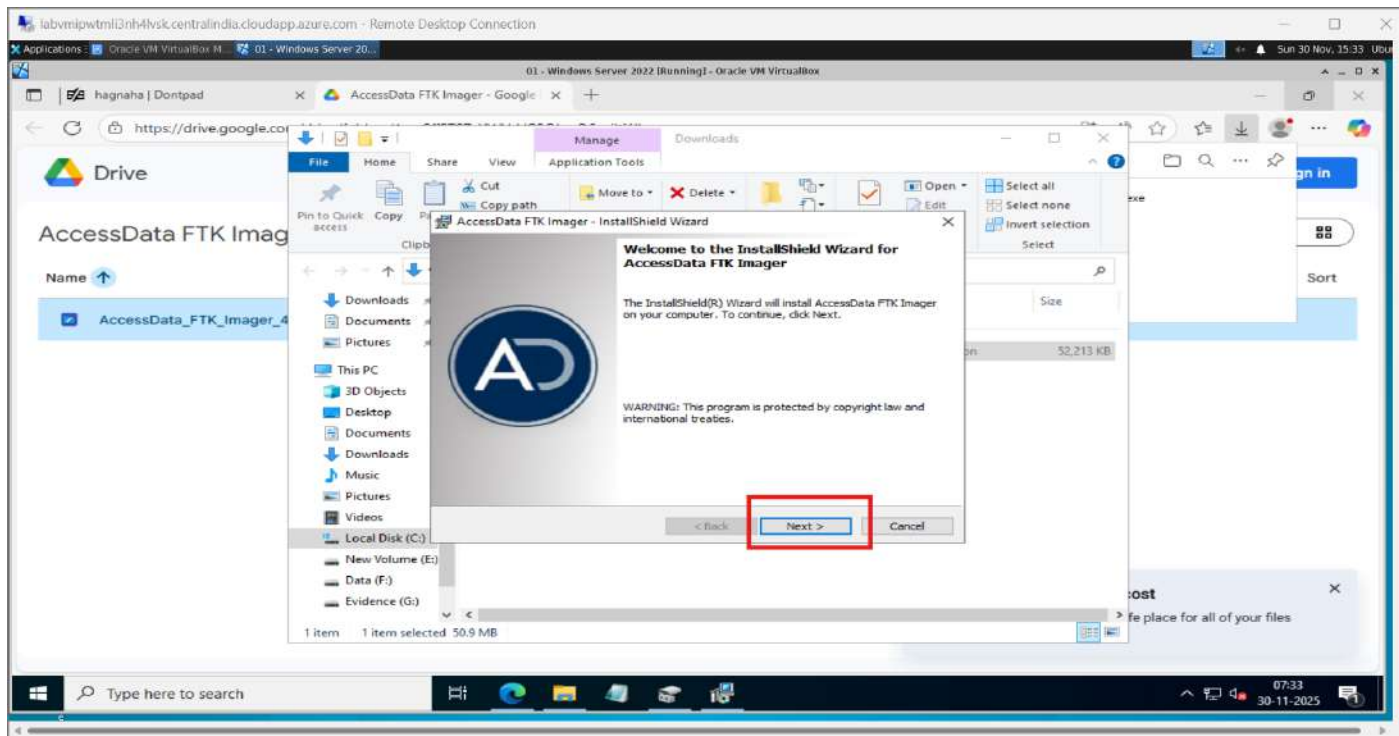
https://drive.google.com/drive/folders/1ssxCff5TS7cXWV_VGSGJnnQ6avjltXJL

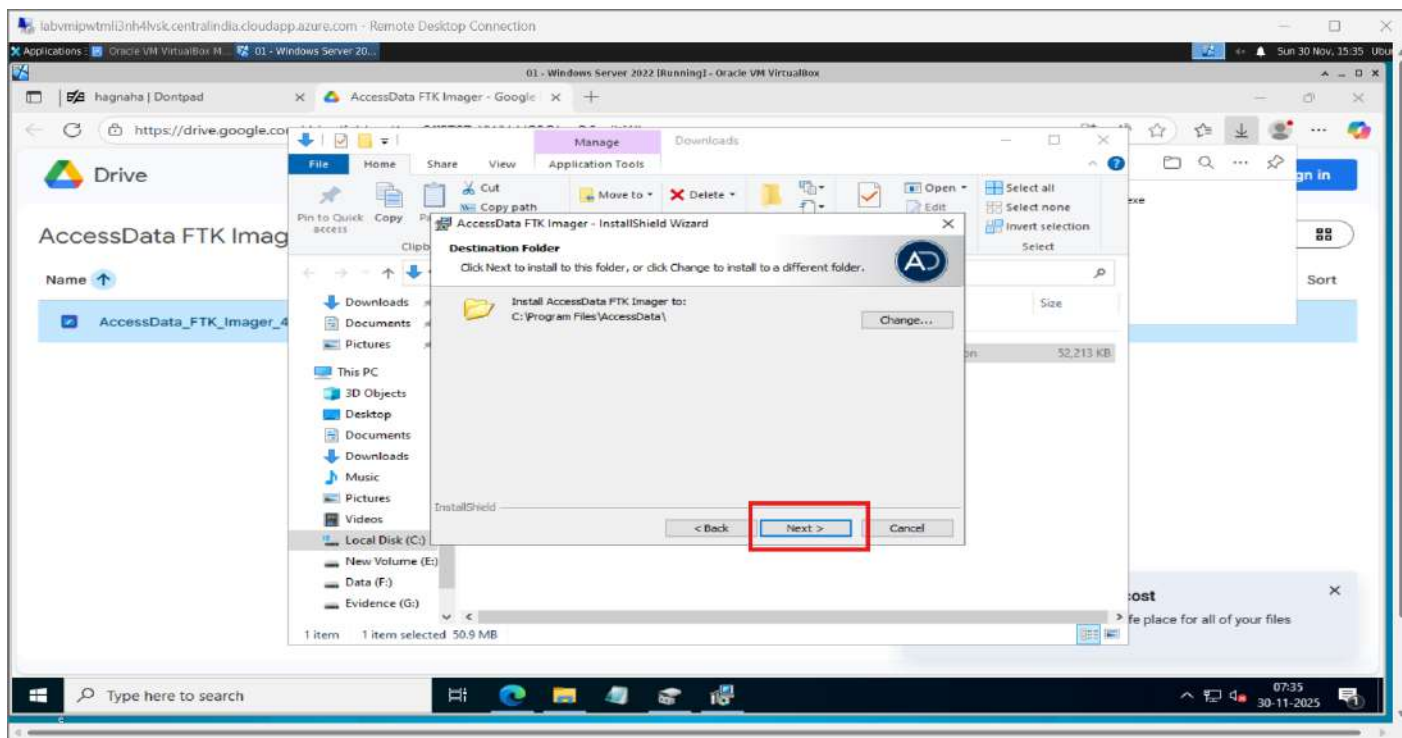


3.2 Double-Clicked on the setup of FTK Imager and Clicked Next.

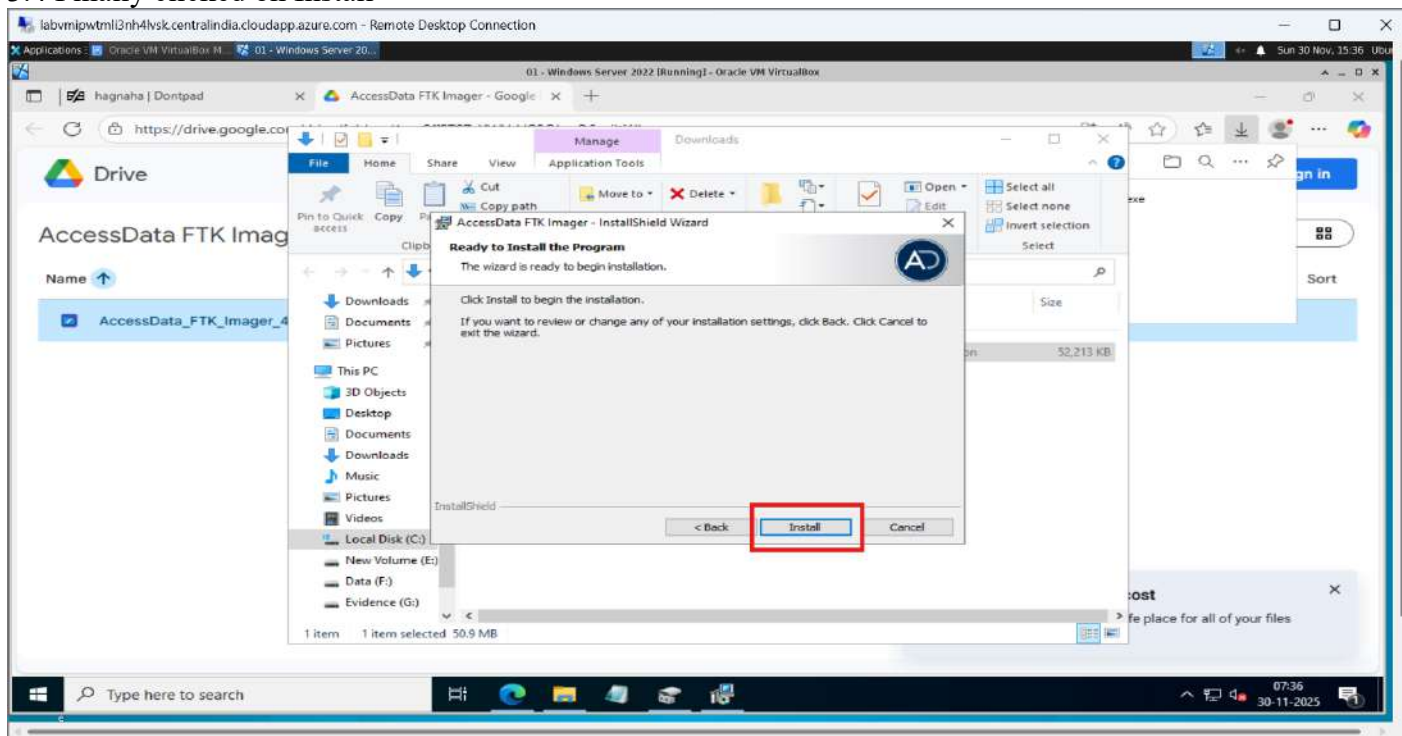


3.3 Clicked next and proceeding forward for installing the software.

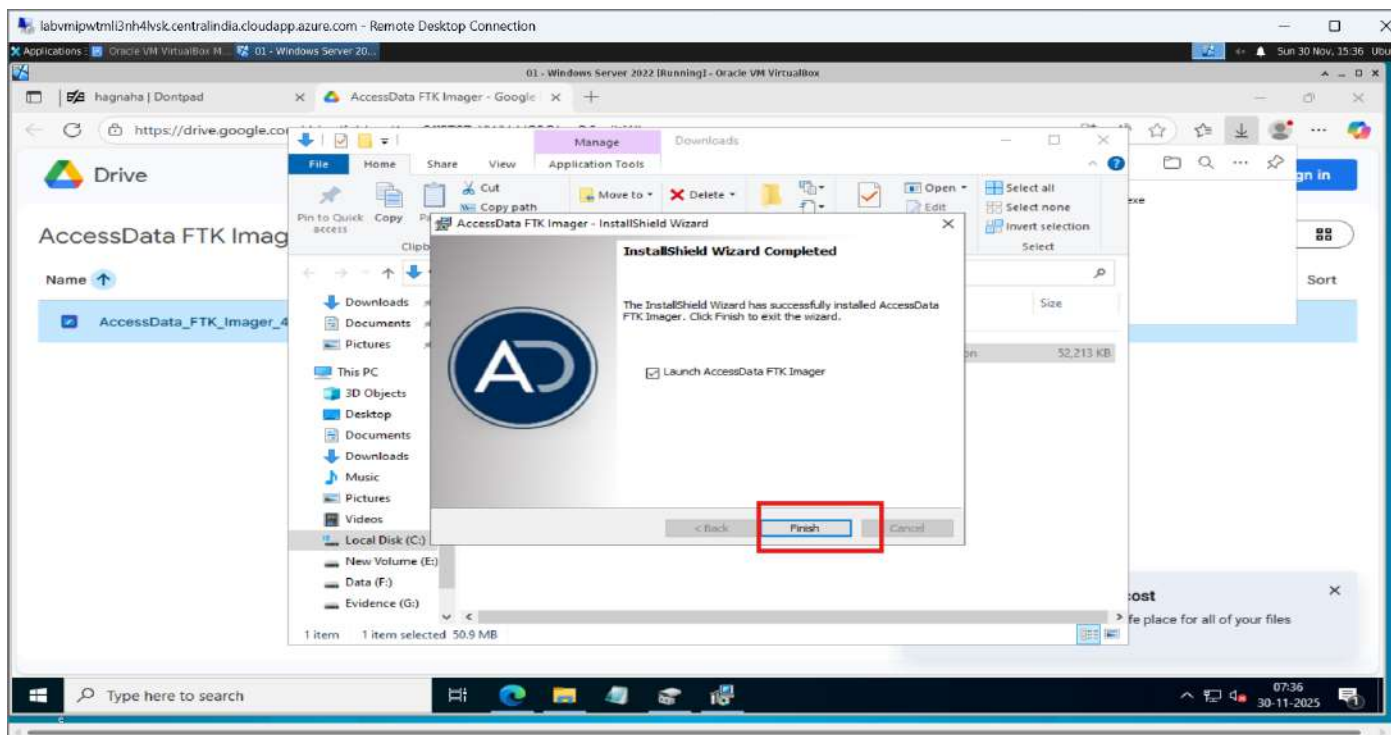




3.4 Finally clicked on Install

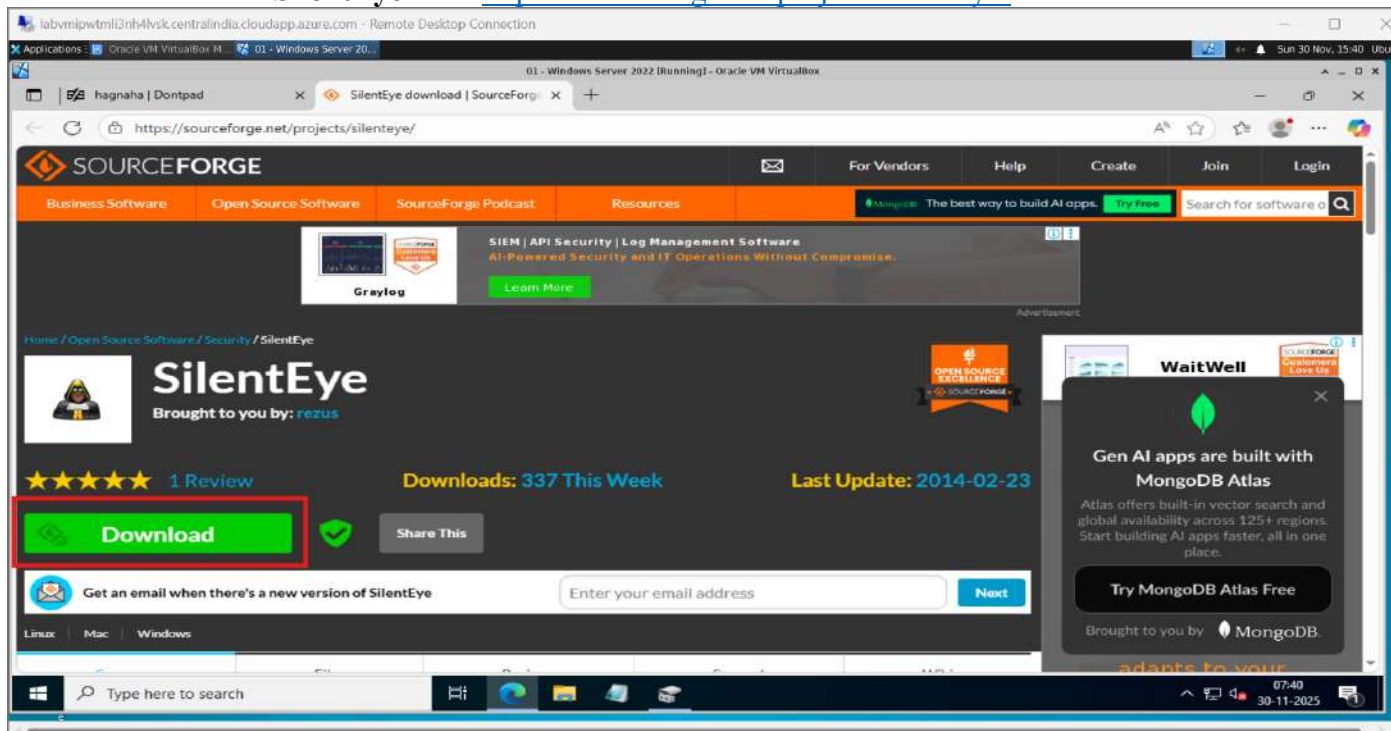


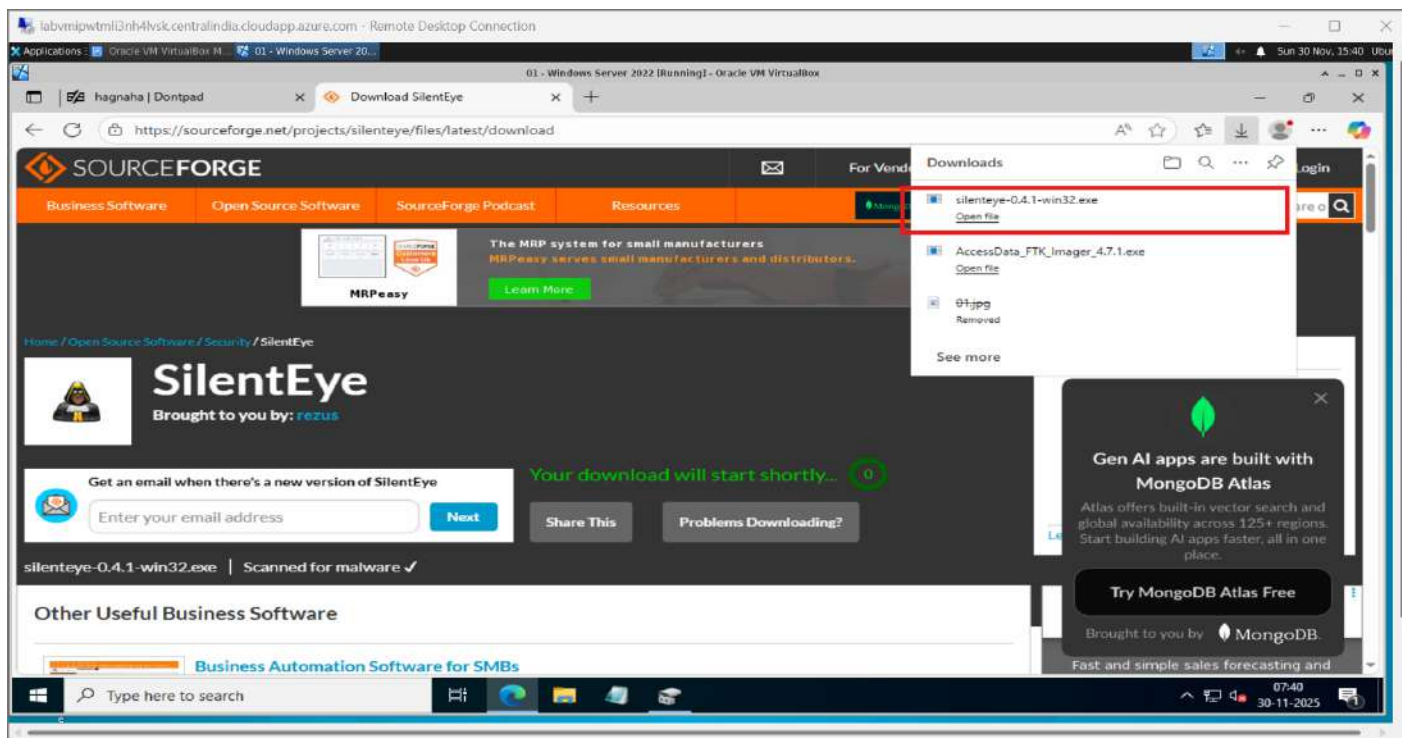
3.5 Clicked finish



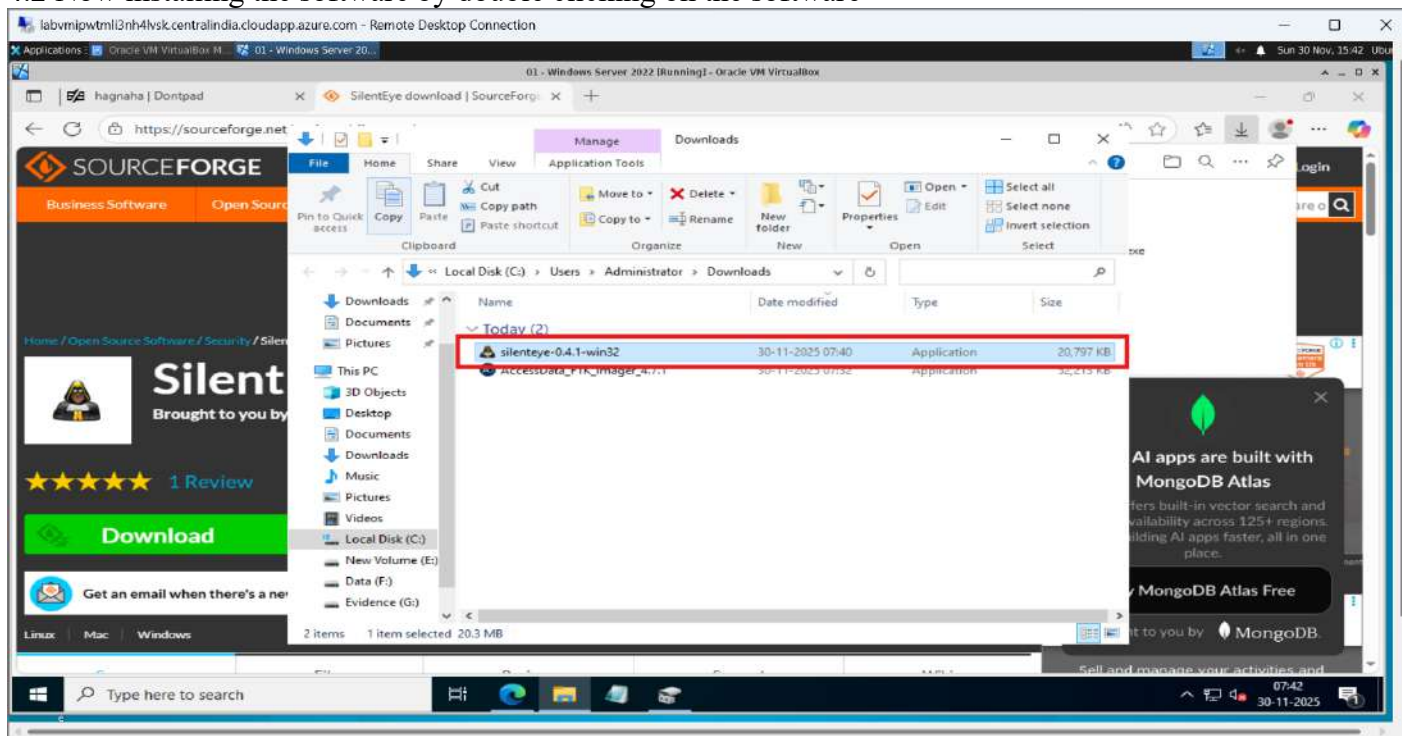
Step 4: Hide text in the image using Steganography

4.1 Downloaded the **SilentEye** from <https://sourceforge.net/projects/silenteye/>

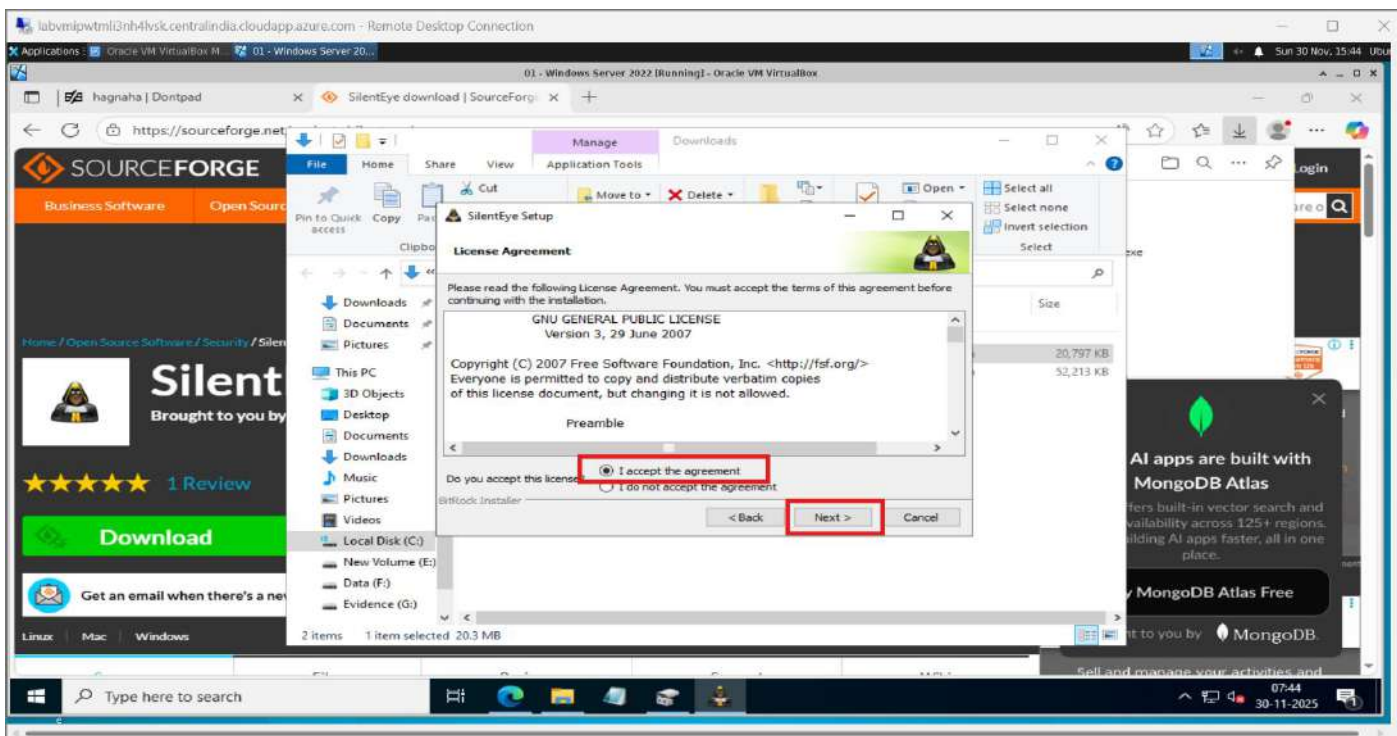
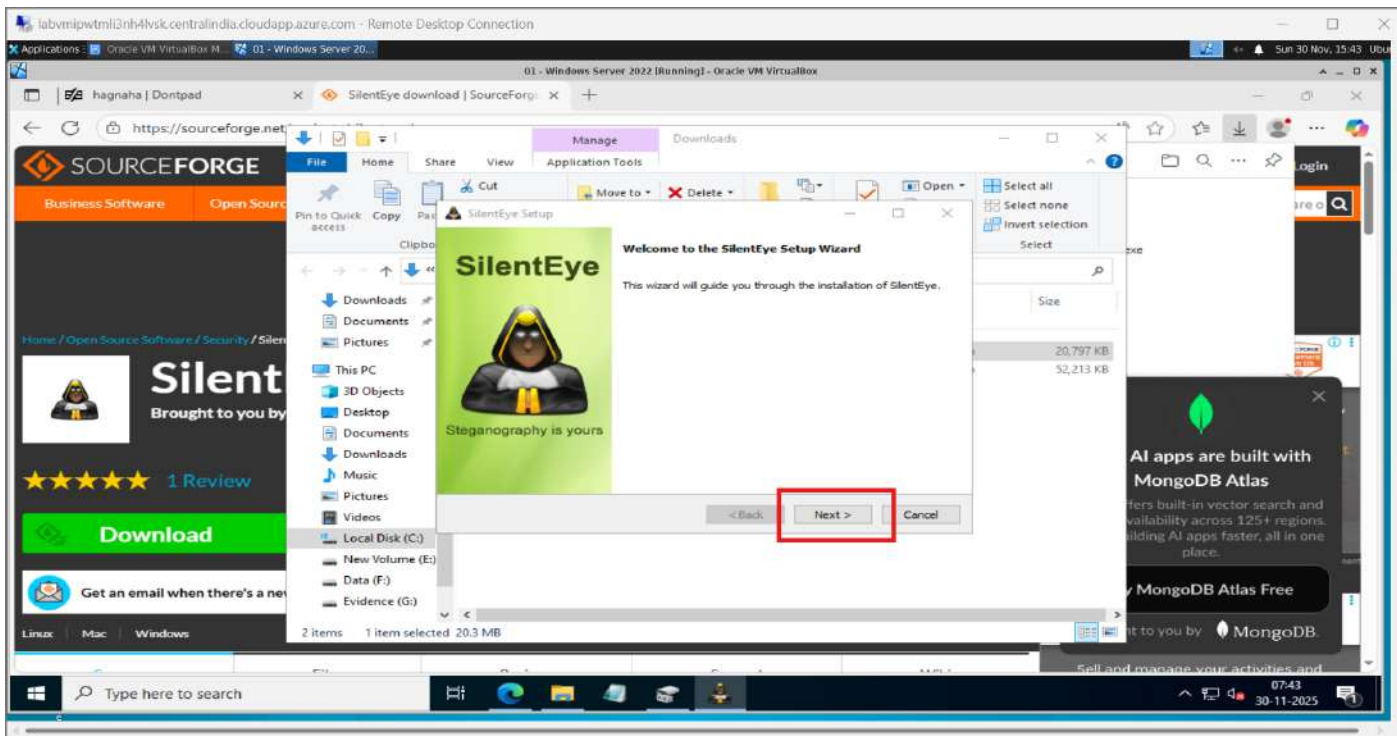


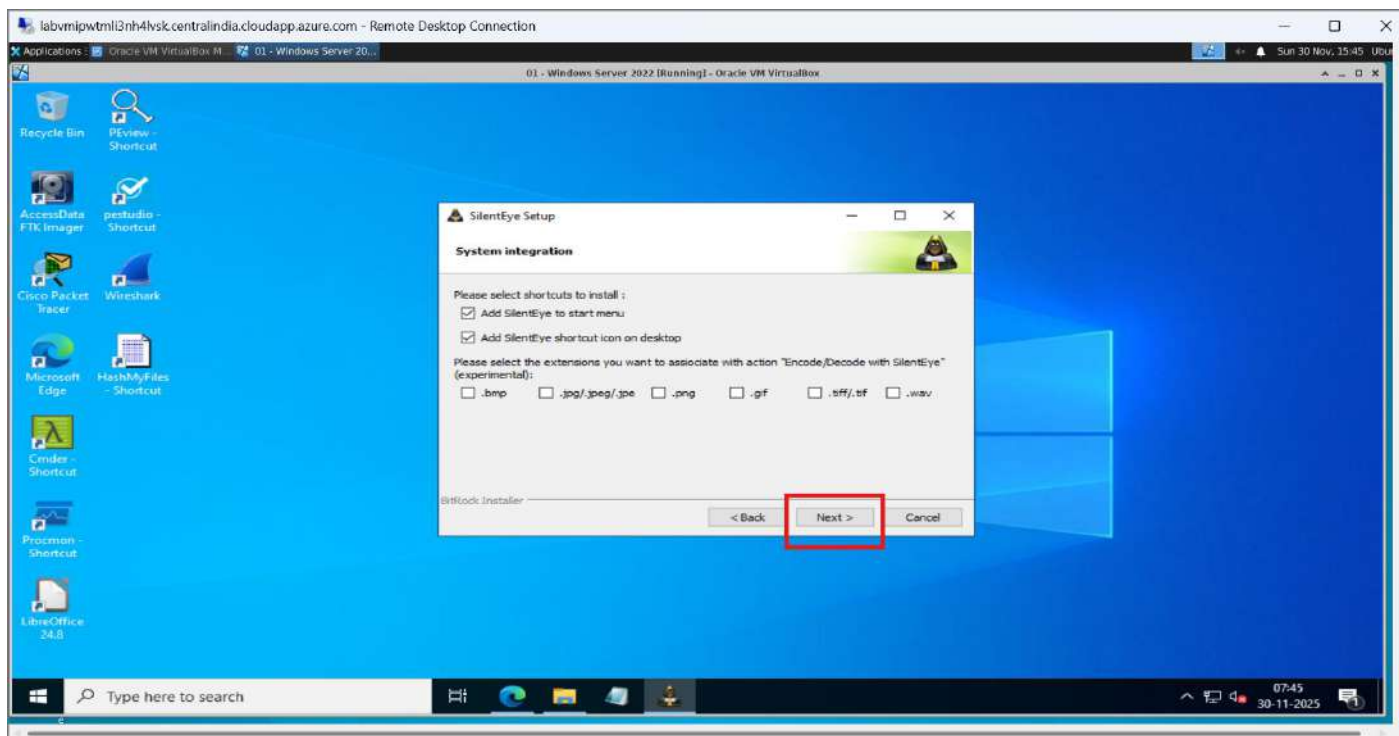
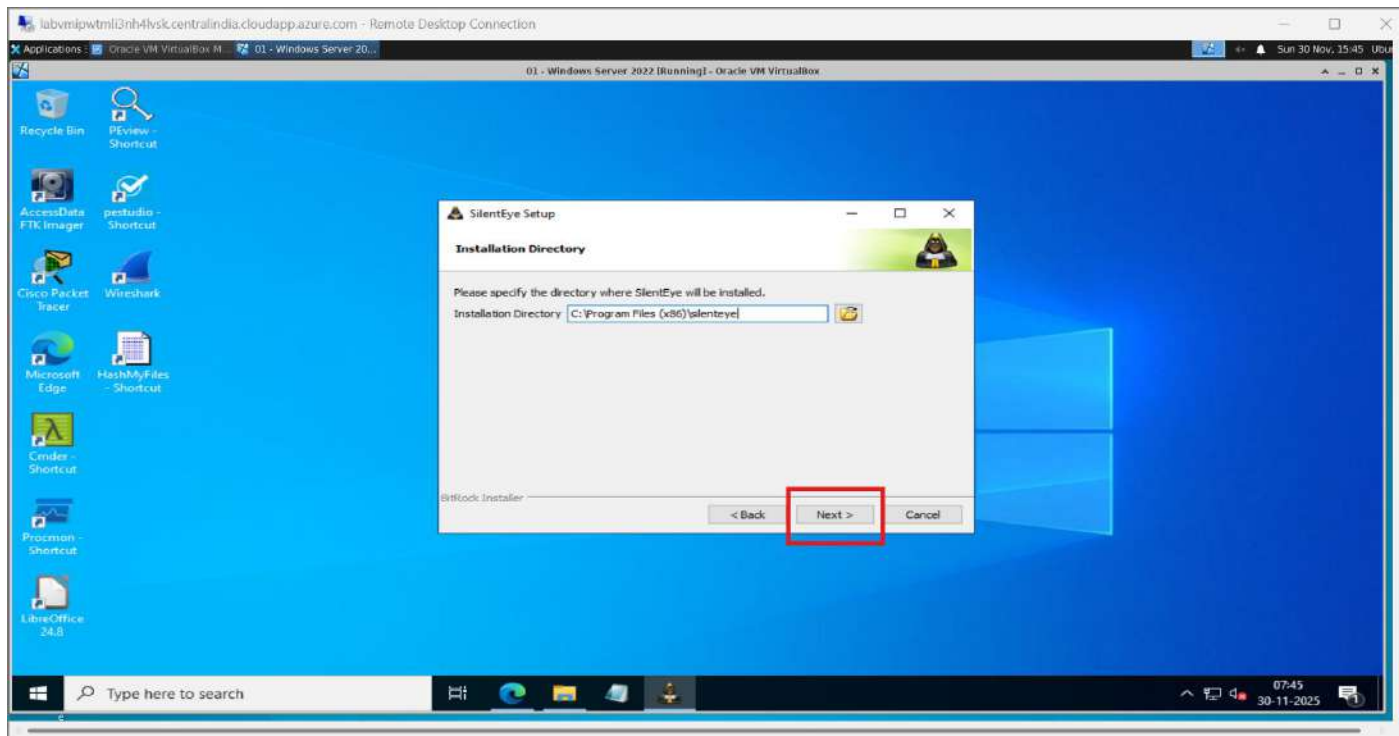


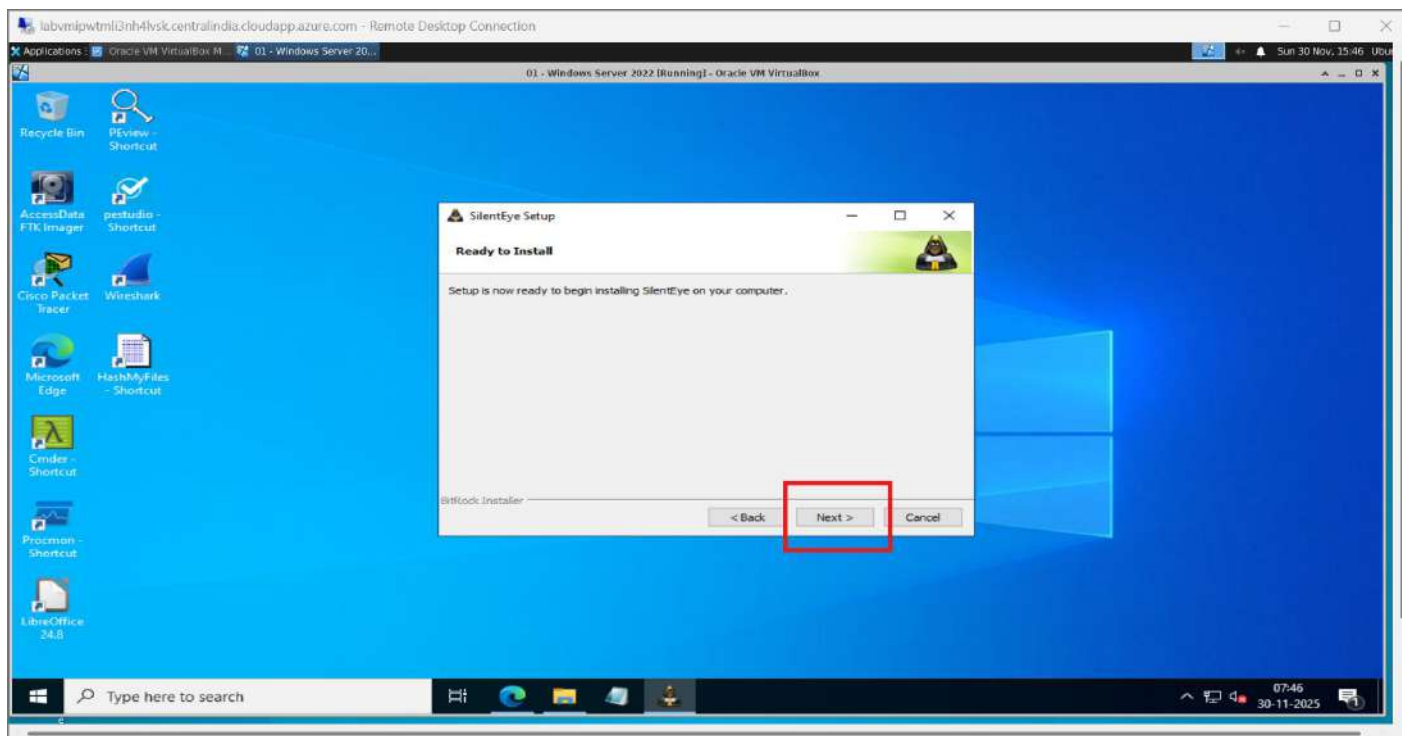
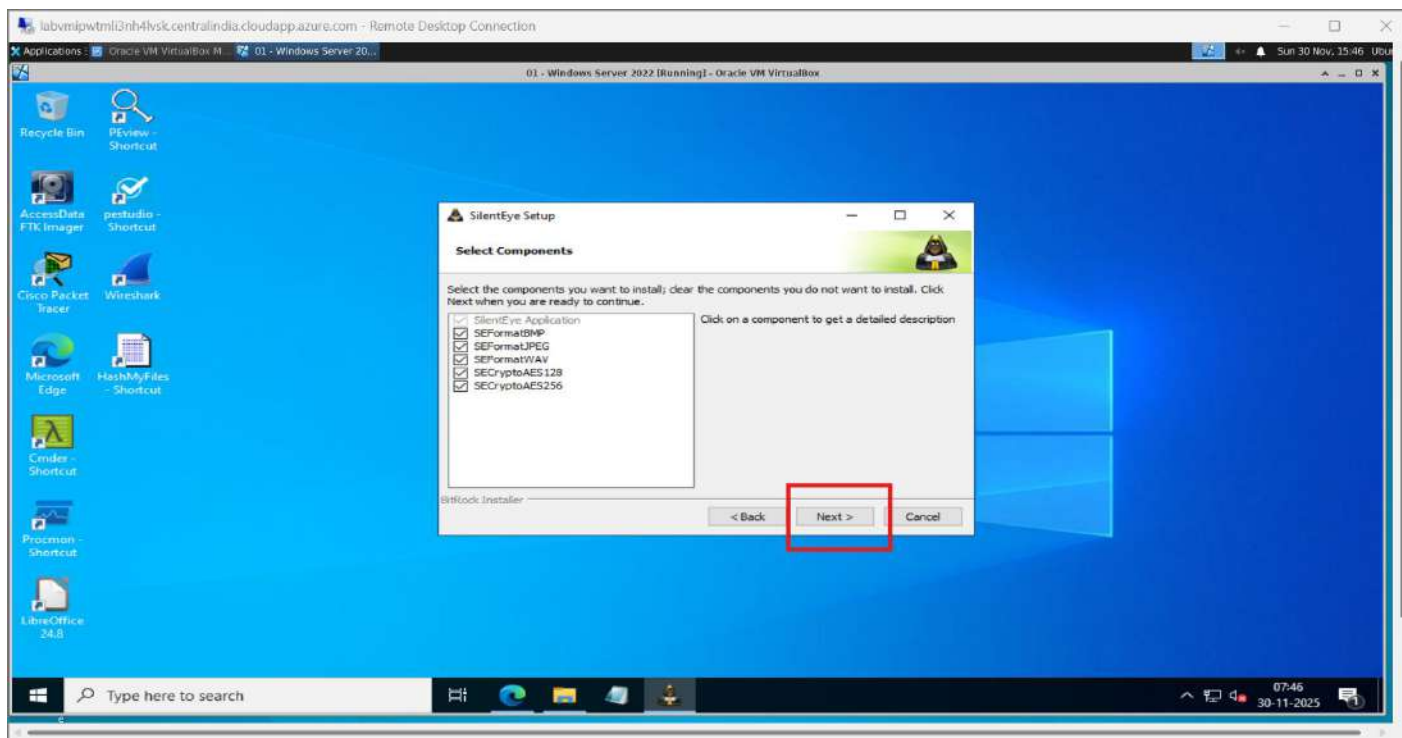
4.2 Now installing the software by double clicking on the software



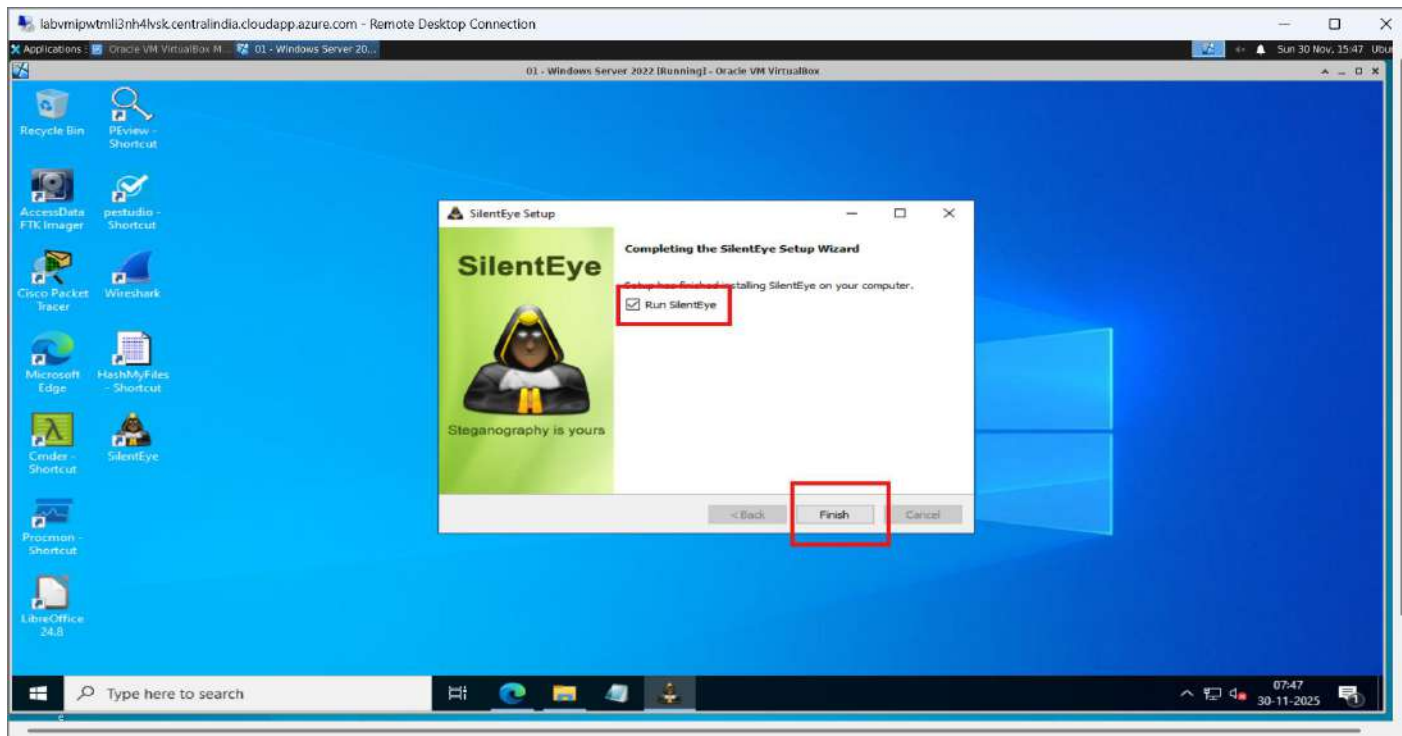
4.3 Clicking on **Next** and following the steps to install the software.



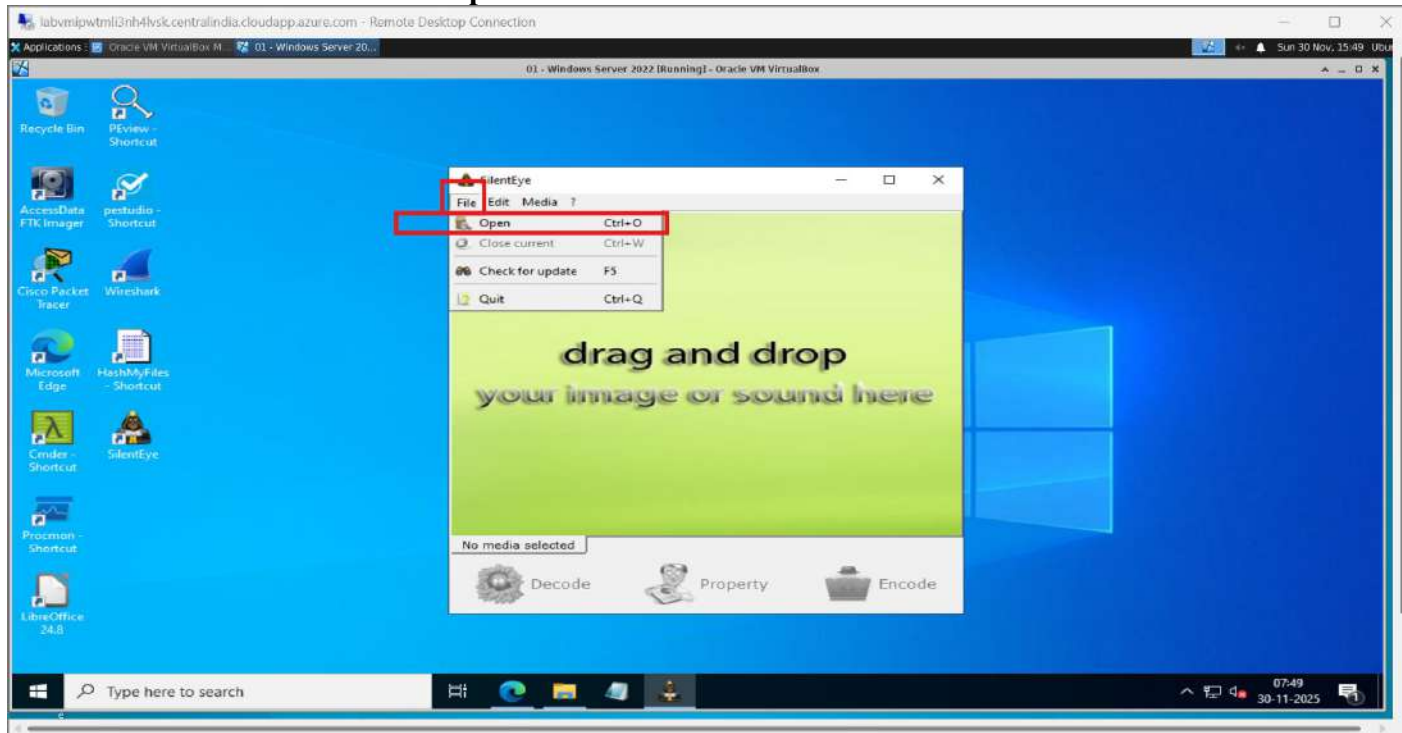




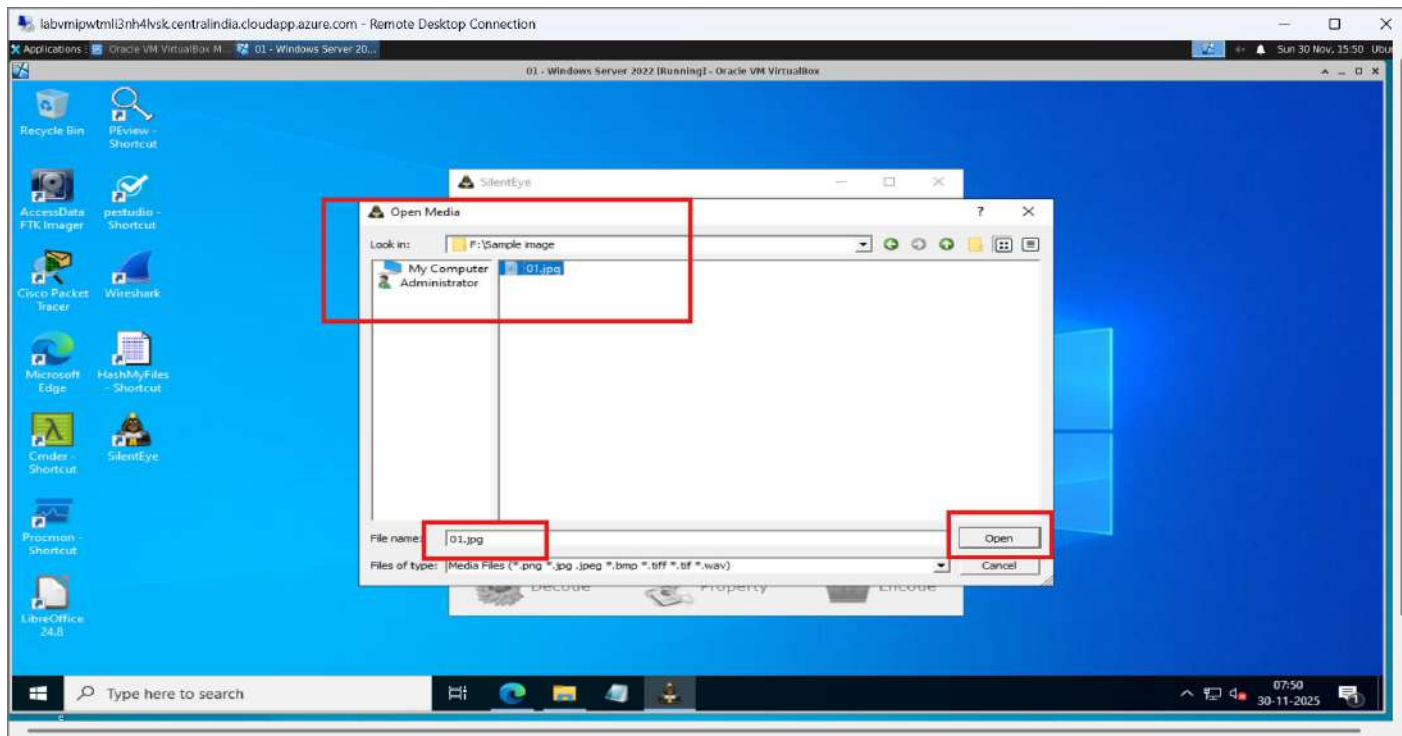
4.4 Finally last step of installation clicked on finish by checking the radio button of Run ScilentEye



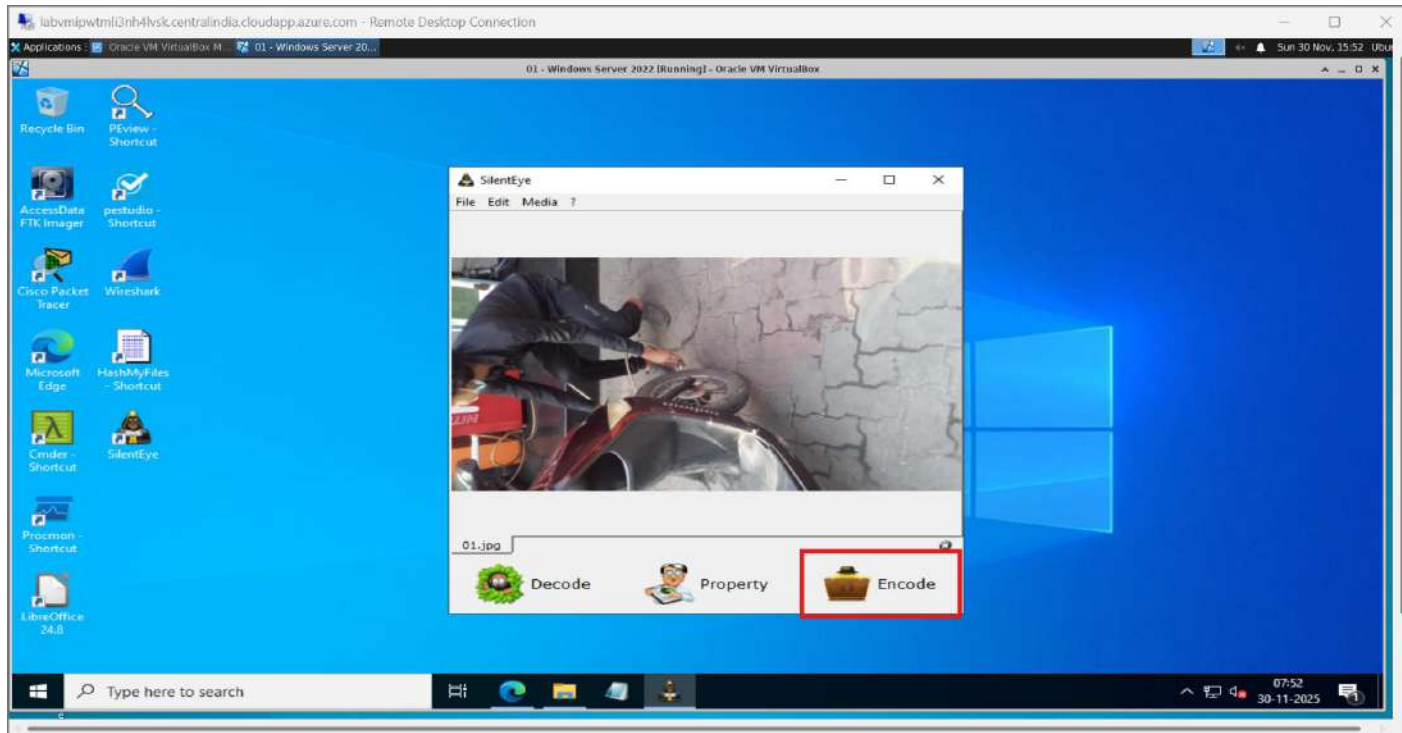
4.5 Clicked on File and selected Open.



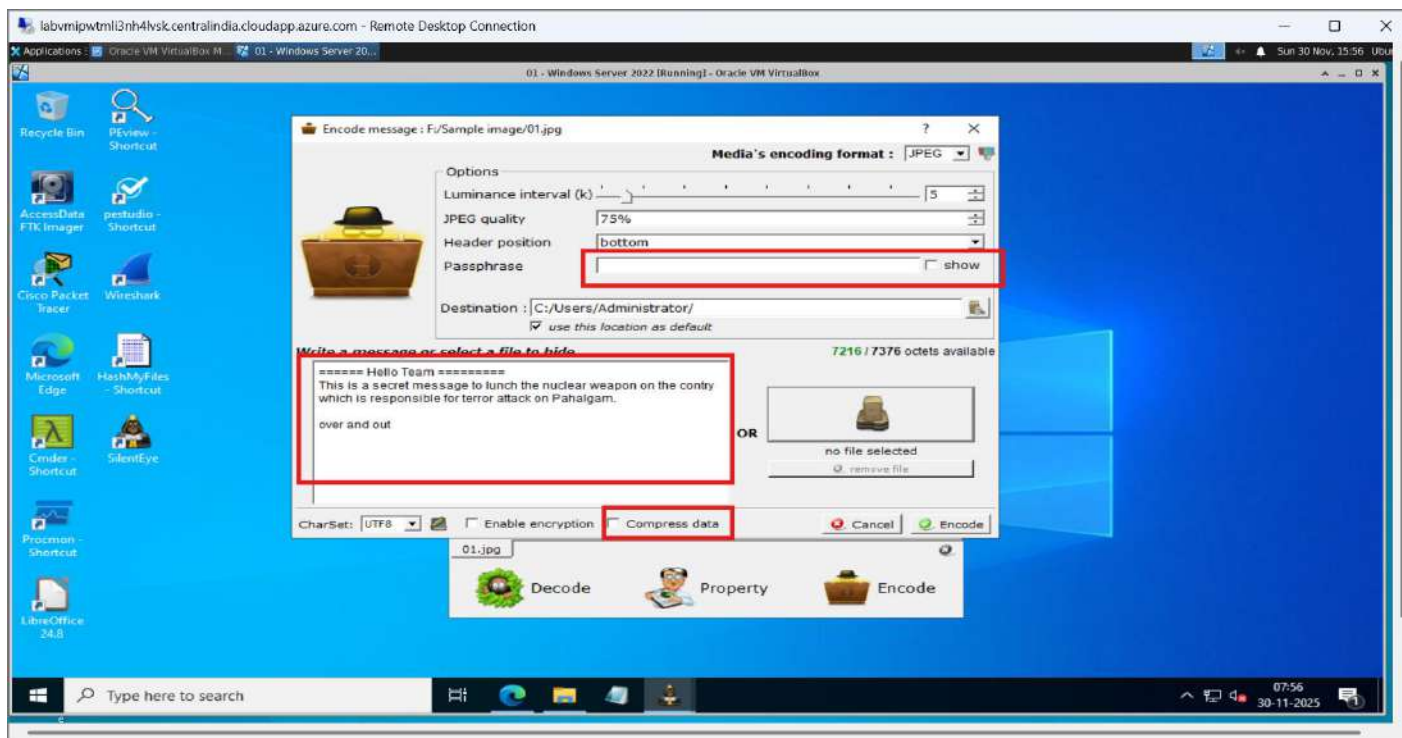
4.6 Navigated to Pictures_sample folder and selected the 01.jpg and Clicked on Open.



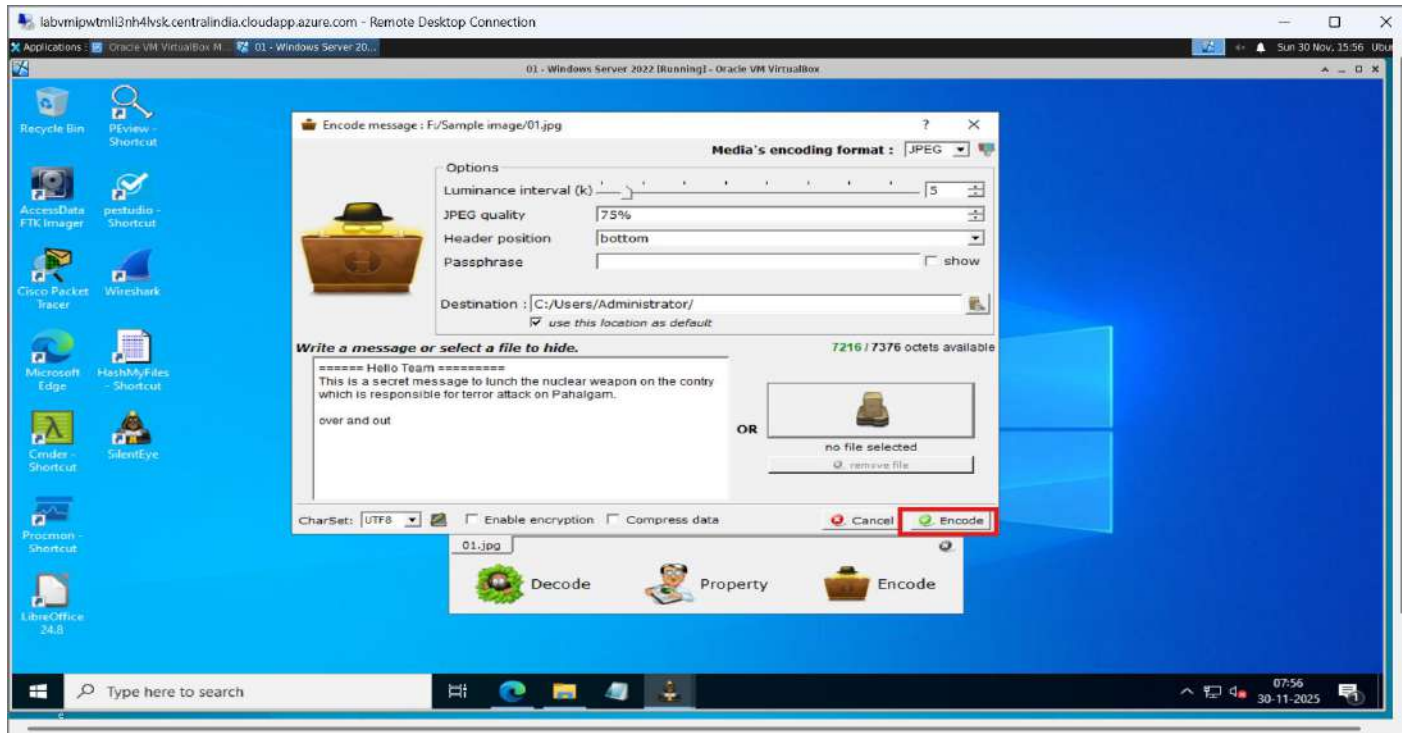
4.7 Clicked on **Encode**.



4.8 Remove the passphrase and uncheck the compress data button and write the message to be hidden.



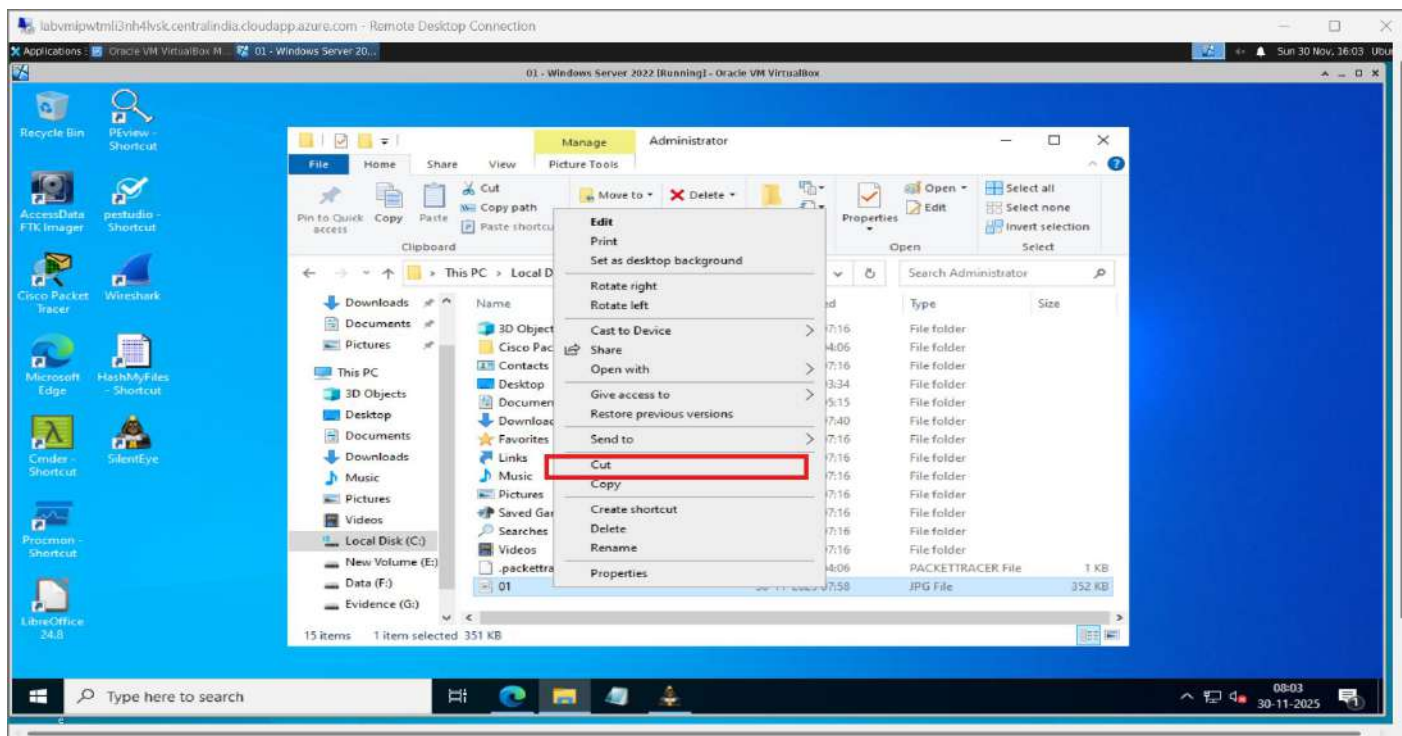
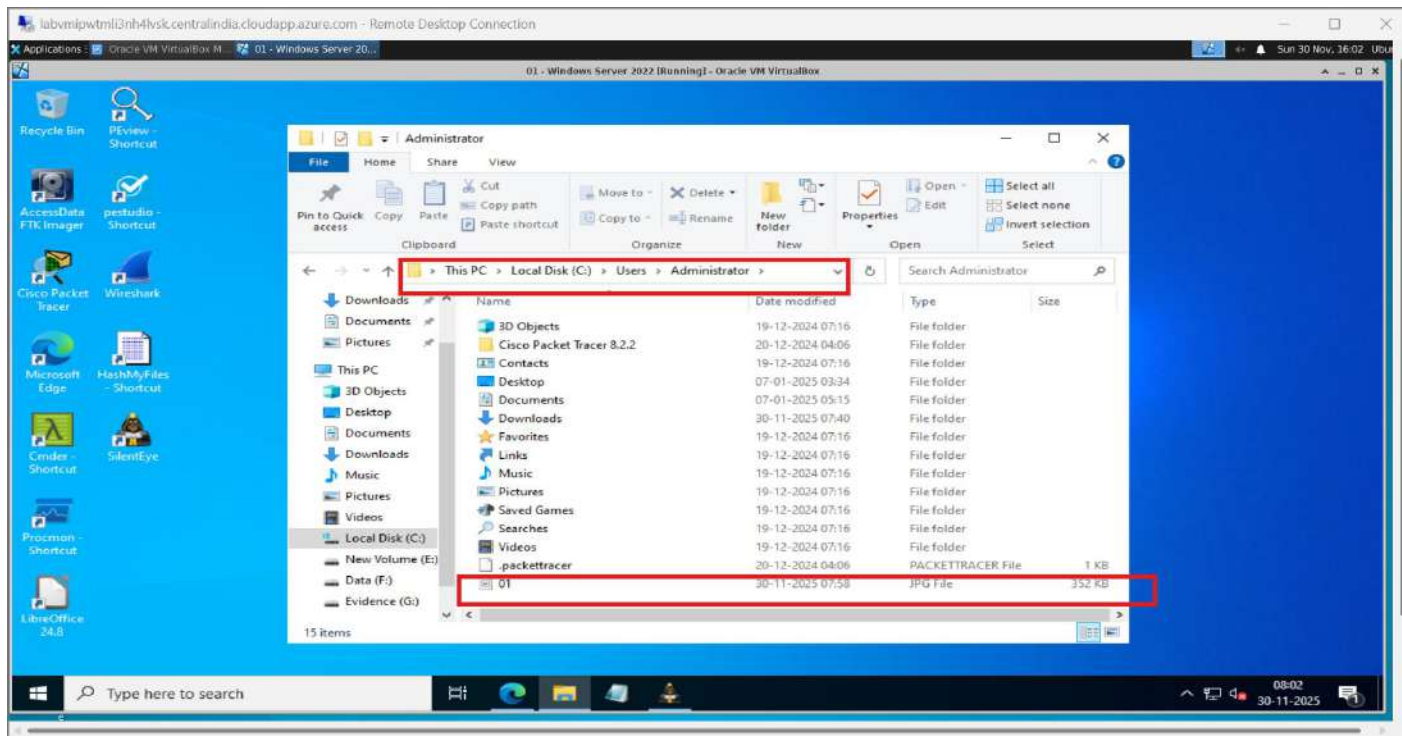
4.9 Clicked on Encode.



Task 2: Store the Image on a 2GB Disk in NTFS:

Step 1: Store the Image on Data drive.

1.1 Changing the Destination from user >> administrator to **Data drive(F)** and create a folder and give the name as **StegFiles**, then Clicked on **Encode**.

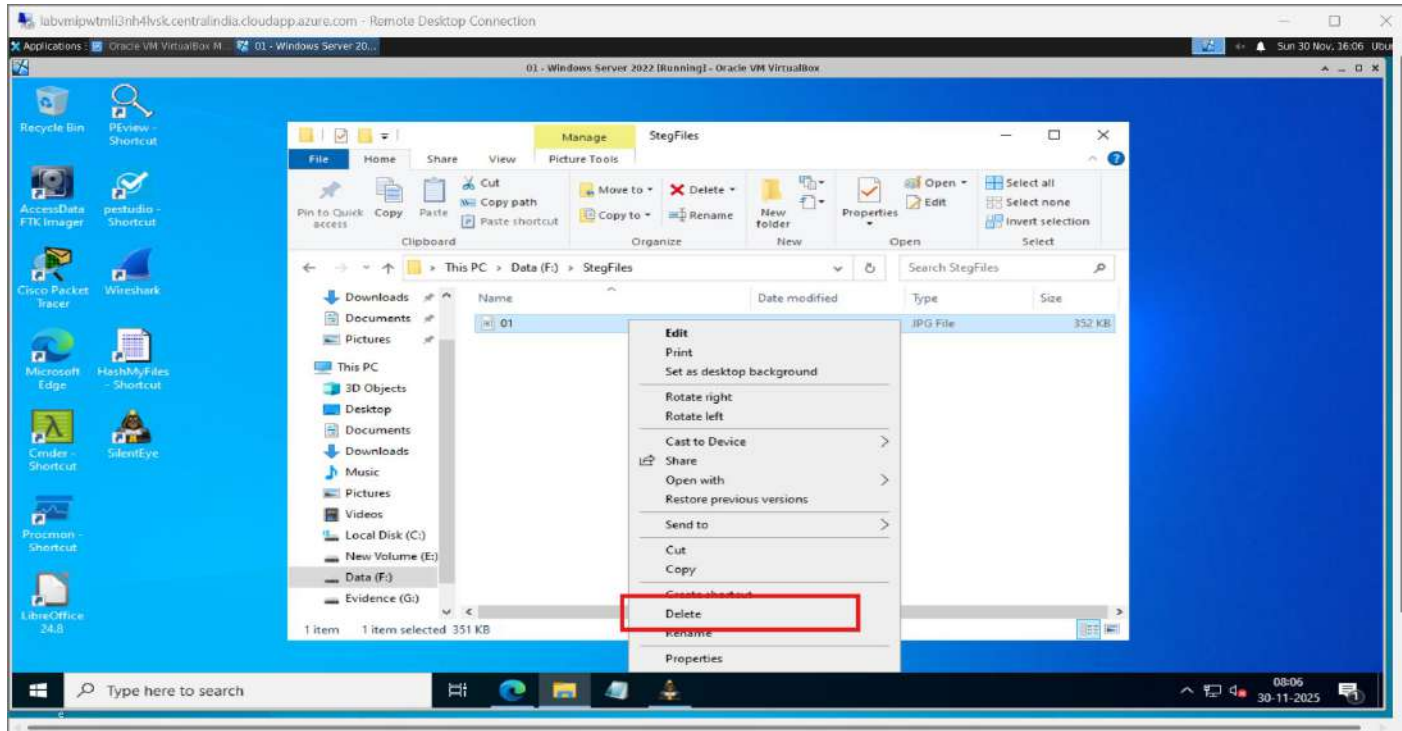


1.2 Navigate to **Data(F): > StegFiles**. Now we can see the file is created successfully.

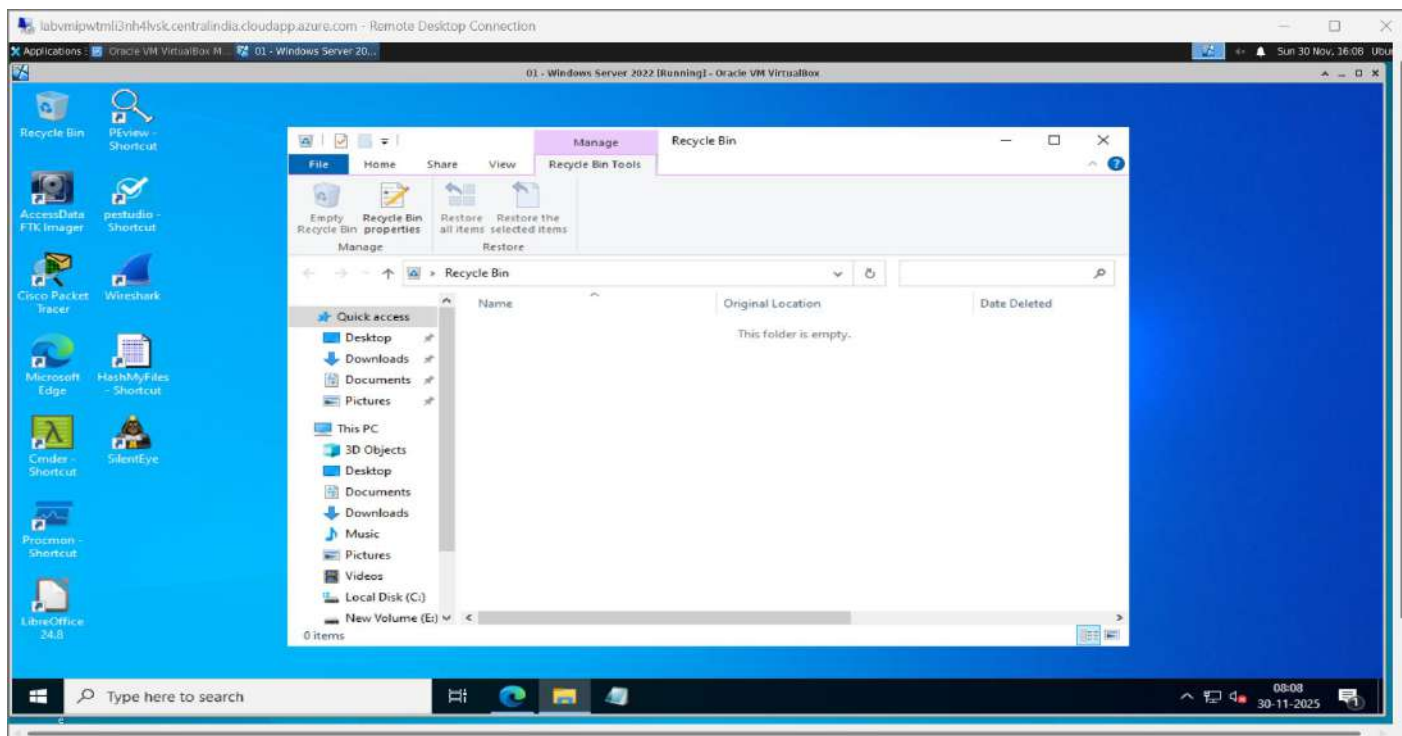


Task 3: Delete the File:

3.1 Delete the file

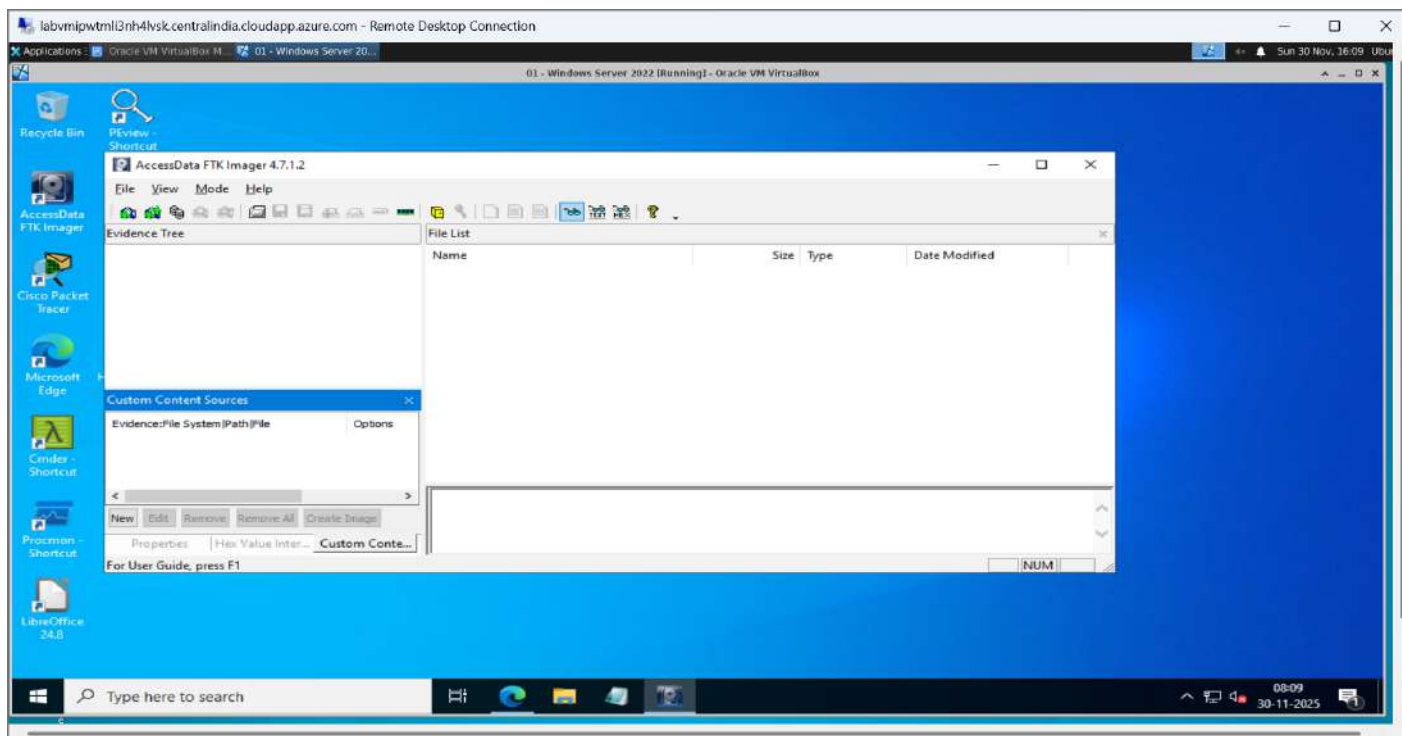
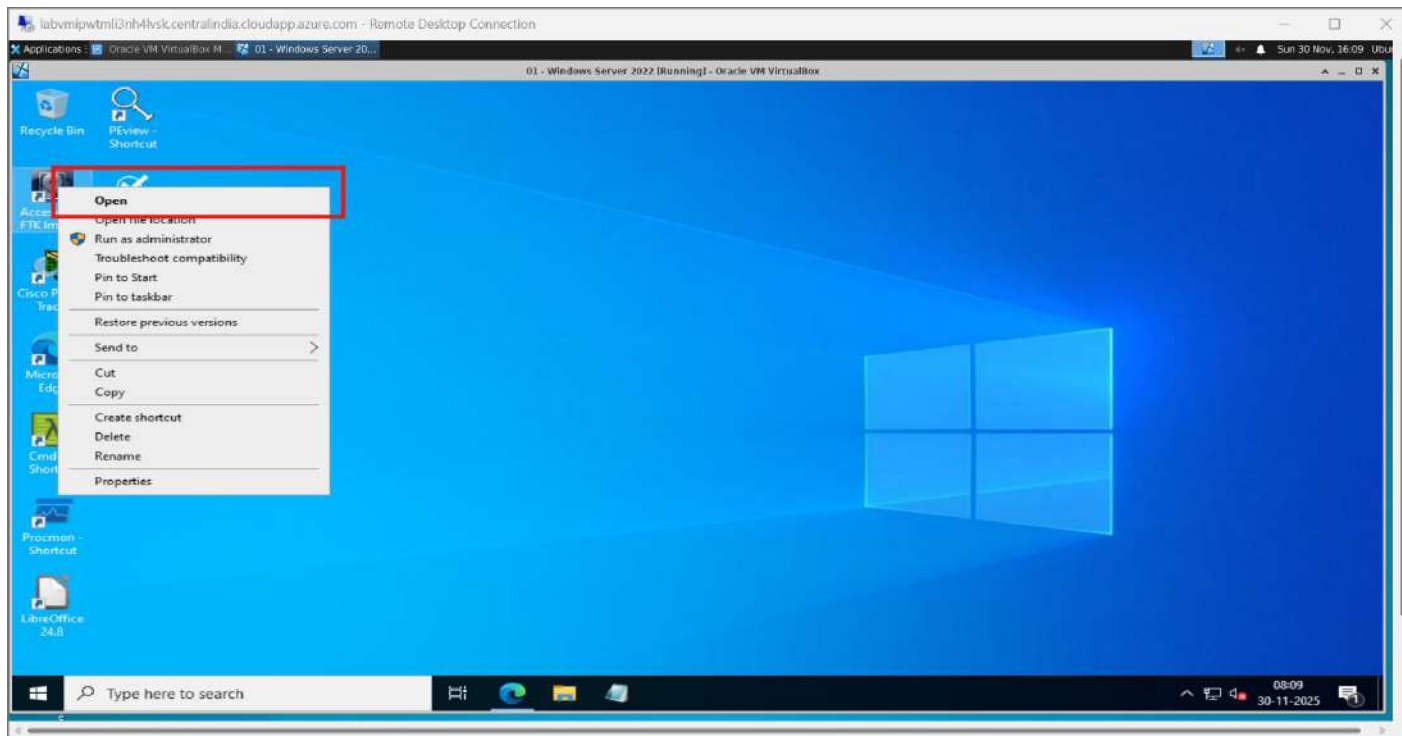


3.2 Deleting from Recycle bin

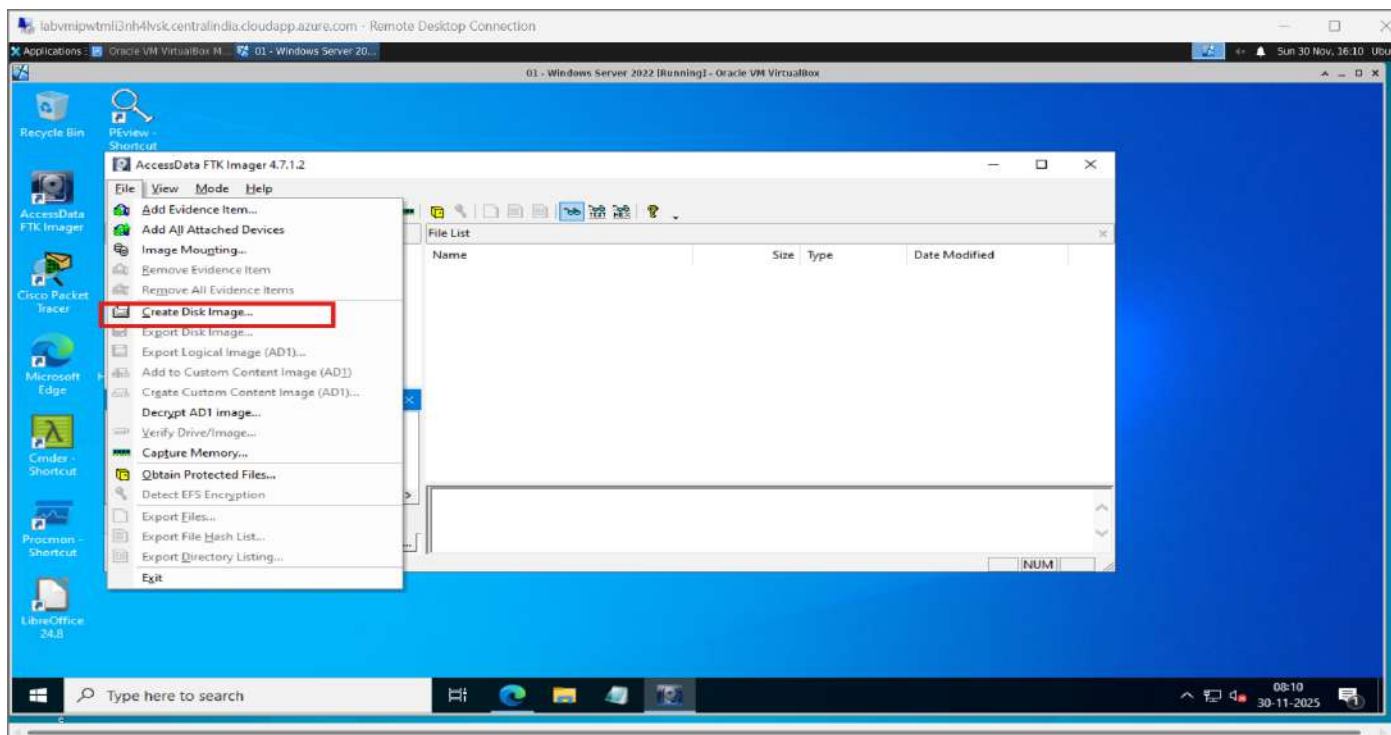


Task 4: Recover the Deleted File Using FTK

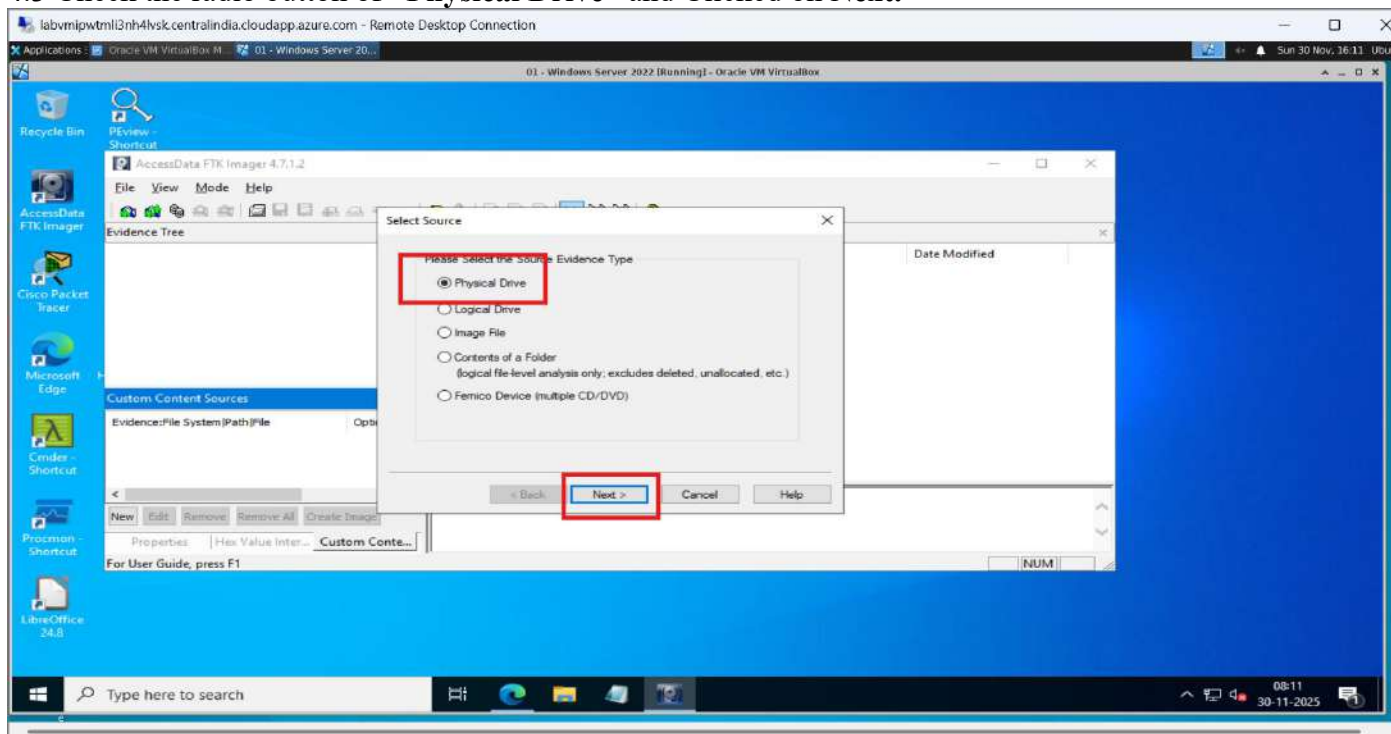
4.1 Open the Access Data FTK.



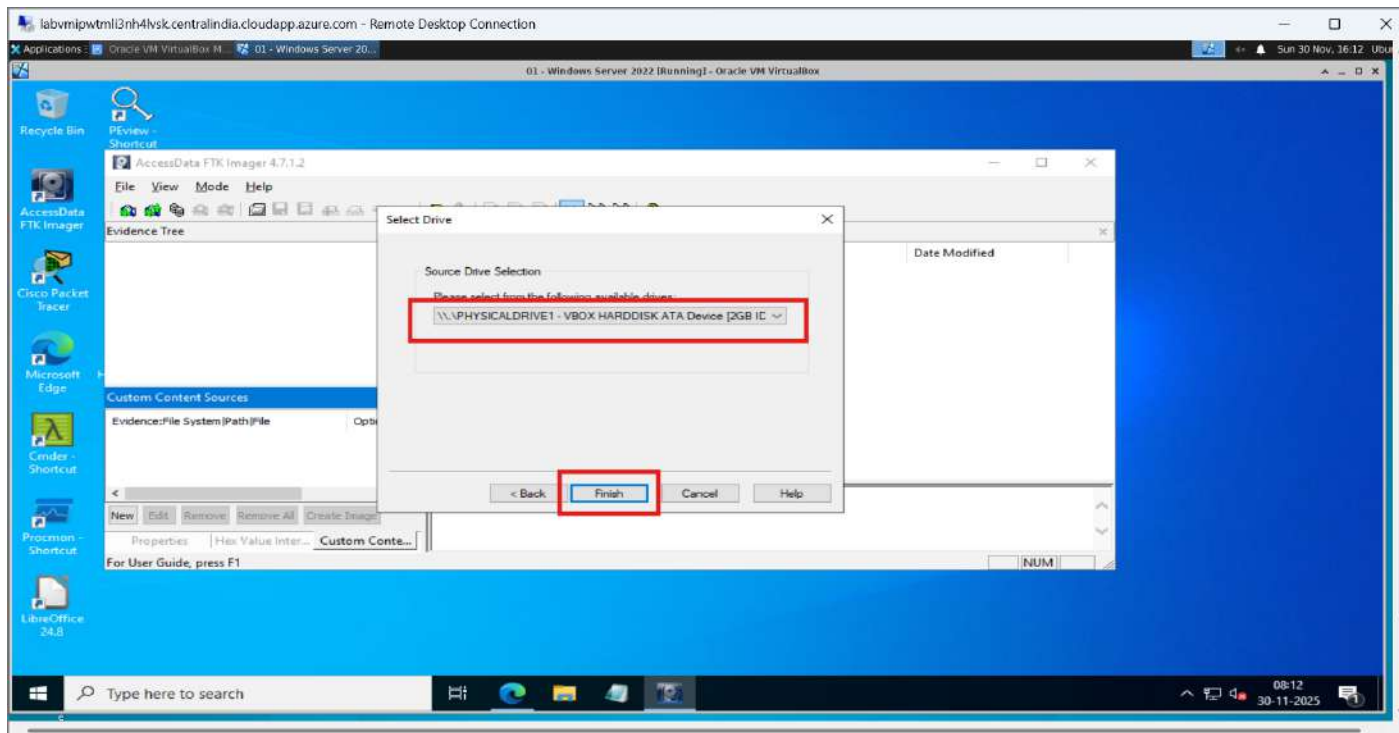
4.2 Clicked on **File** and selected the “**Create Disk image...**”



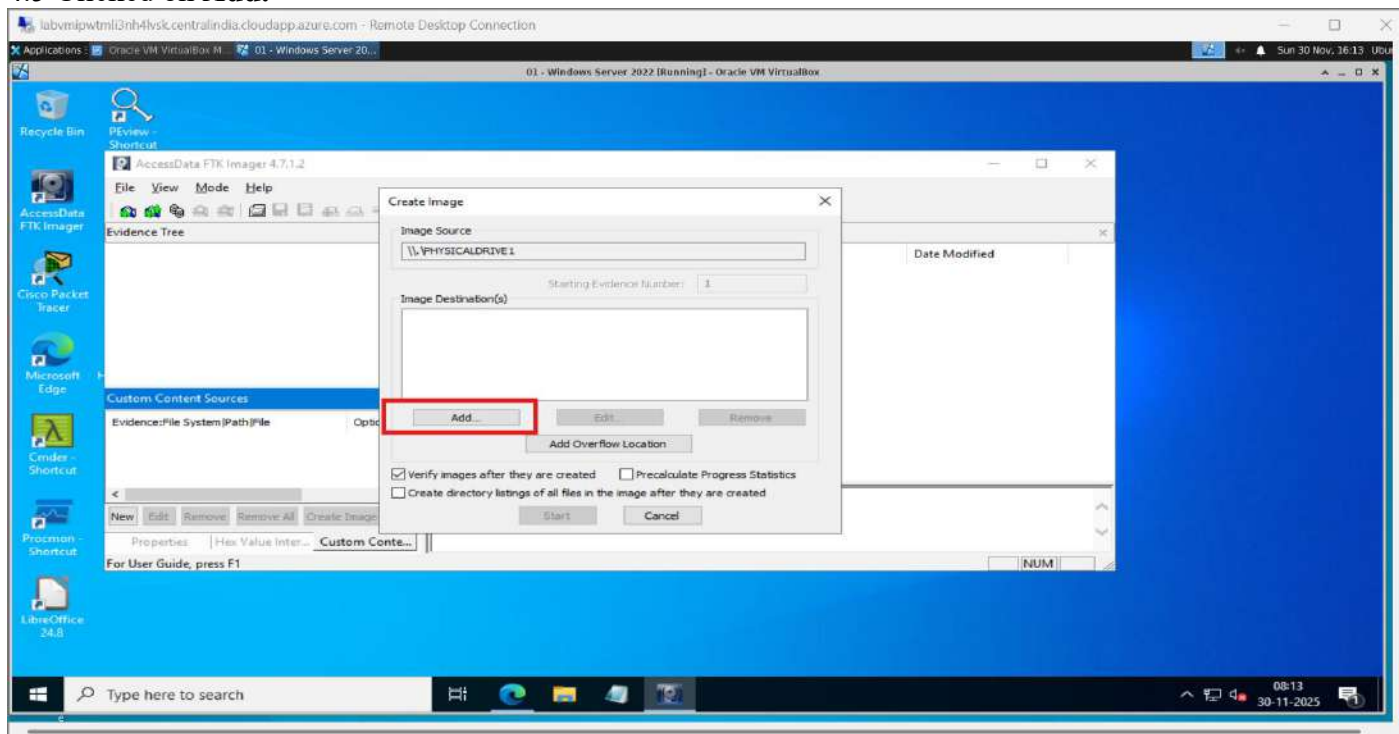
4.3 Check the radio button of “Physical Drive” and Clicked on Next.



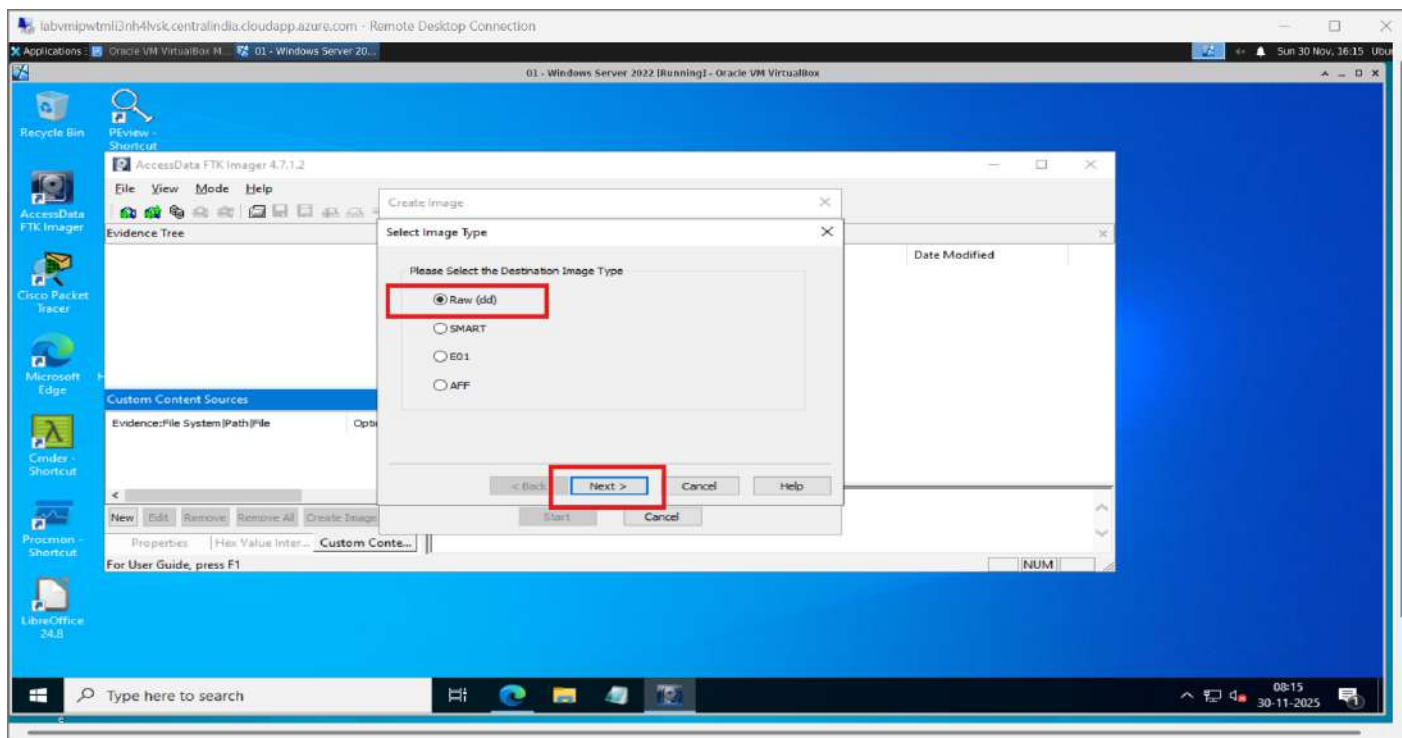
4.4 Choose the 2 GB of file from the drop-down button and Clicked on Finish.



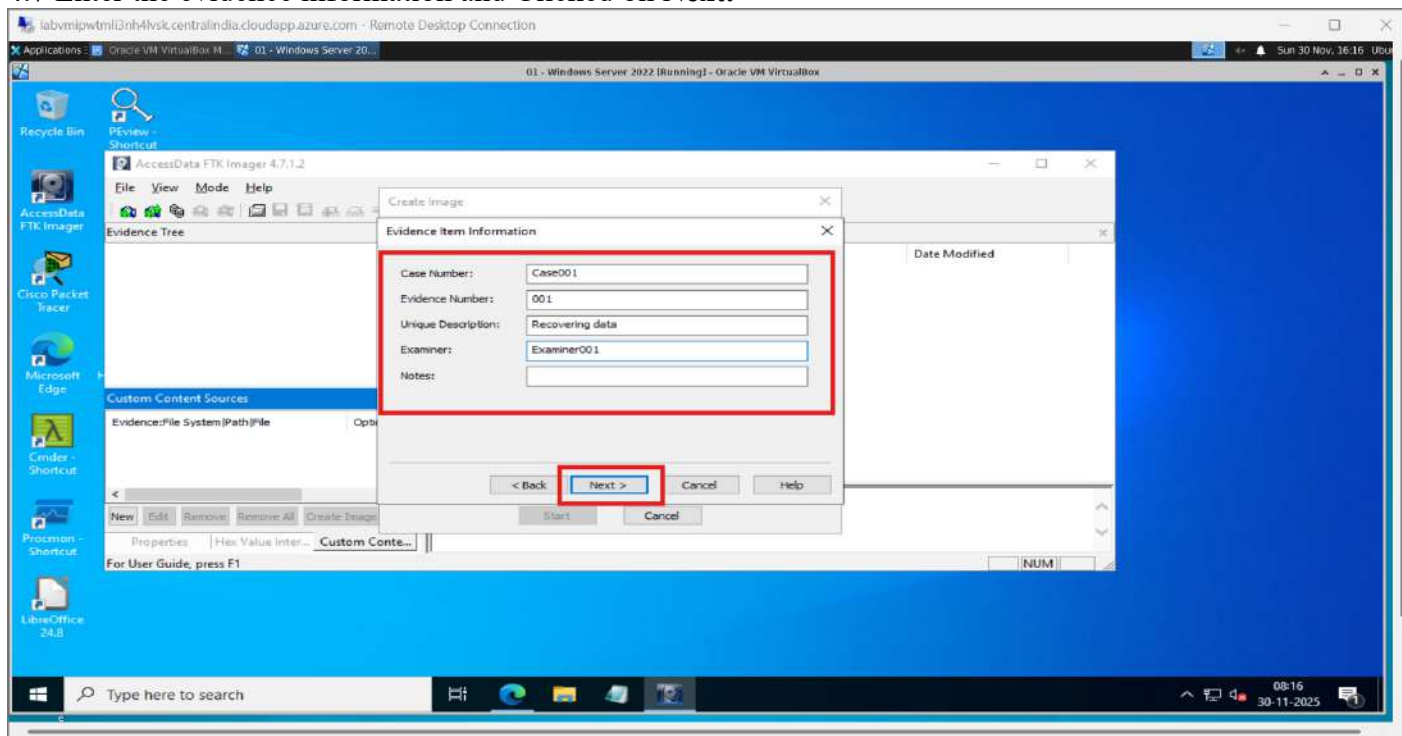
4.5 Clicked on **Add**.



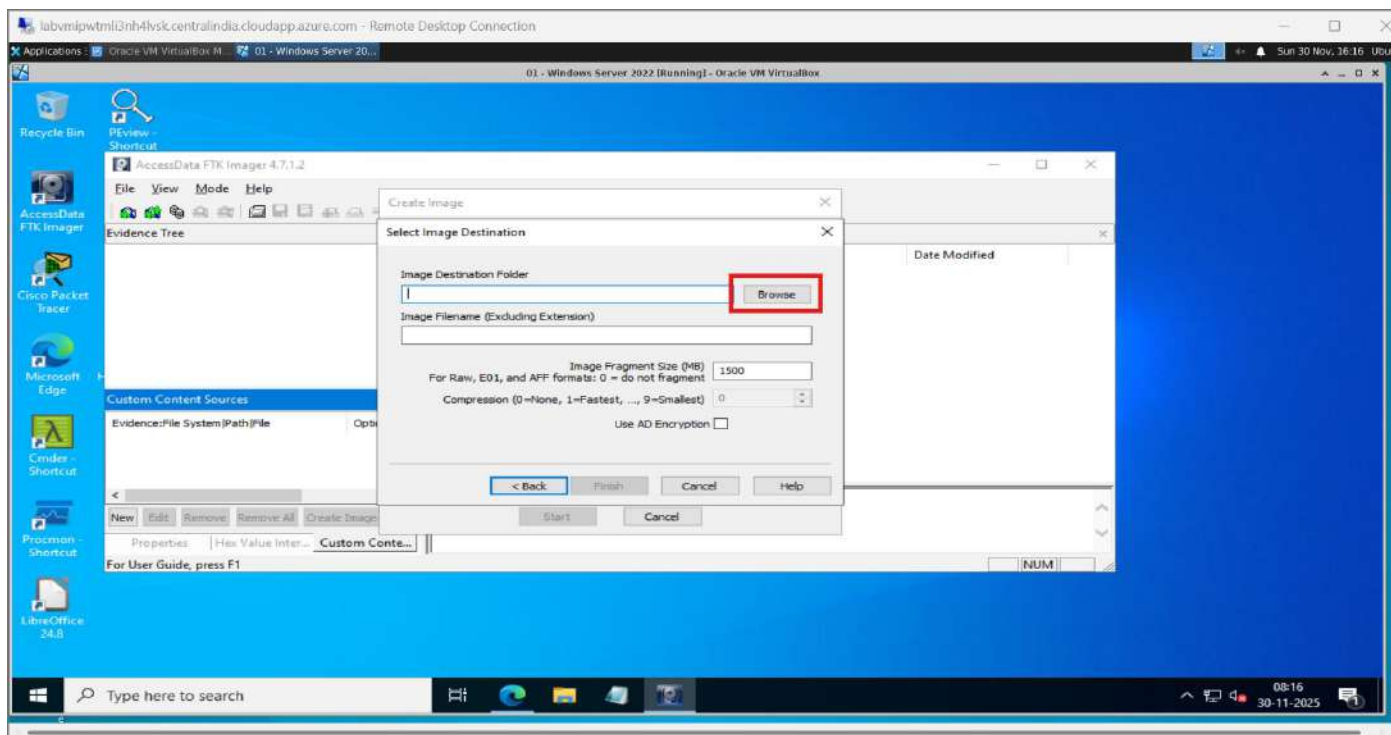
4.6 Check the radio button of **Raw** and Clicked on **Next**.



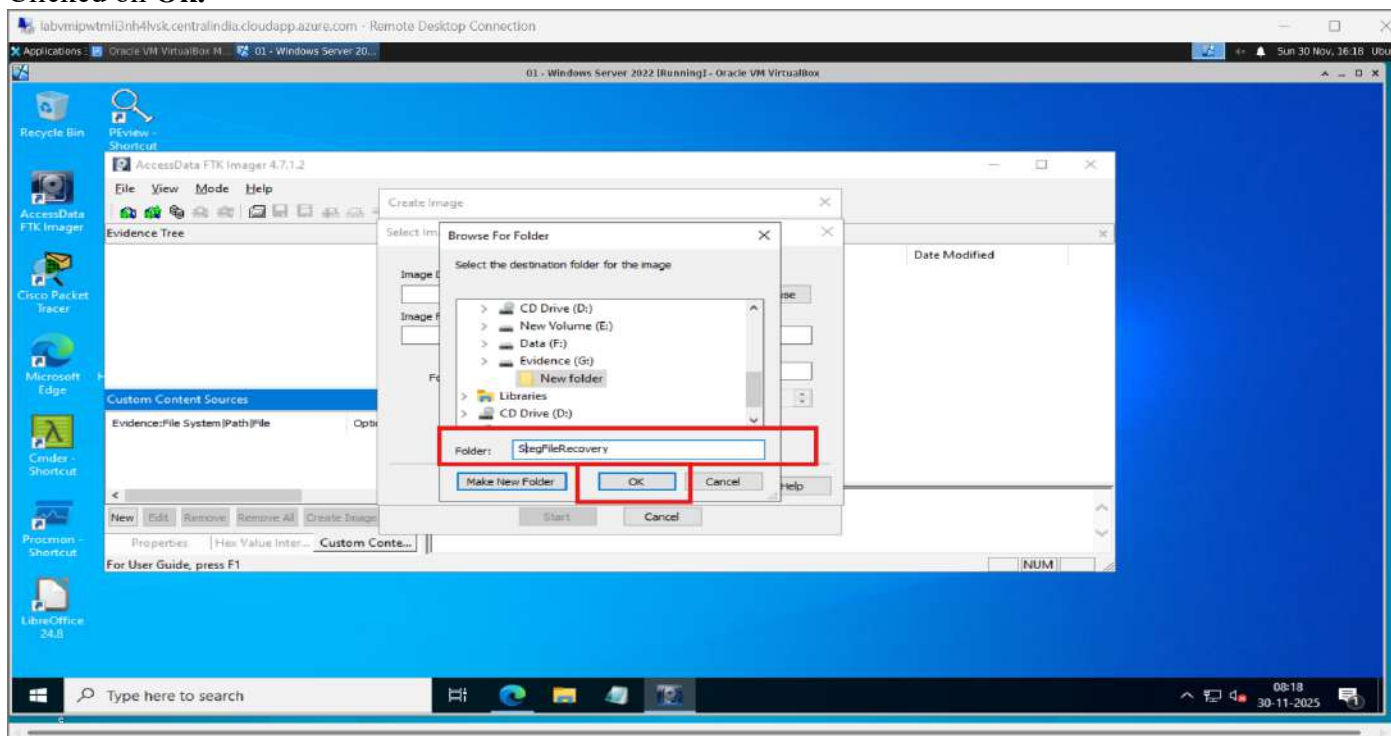
4.7 Enter the evidence information and Clicked on **Next**.



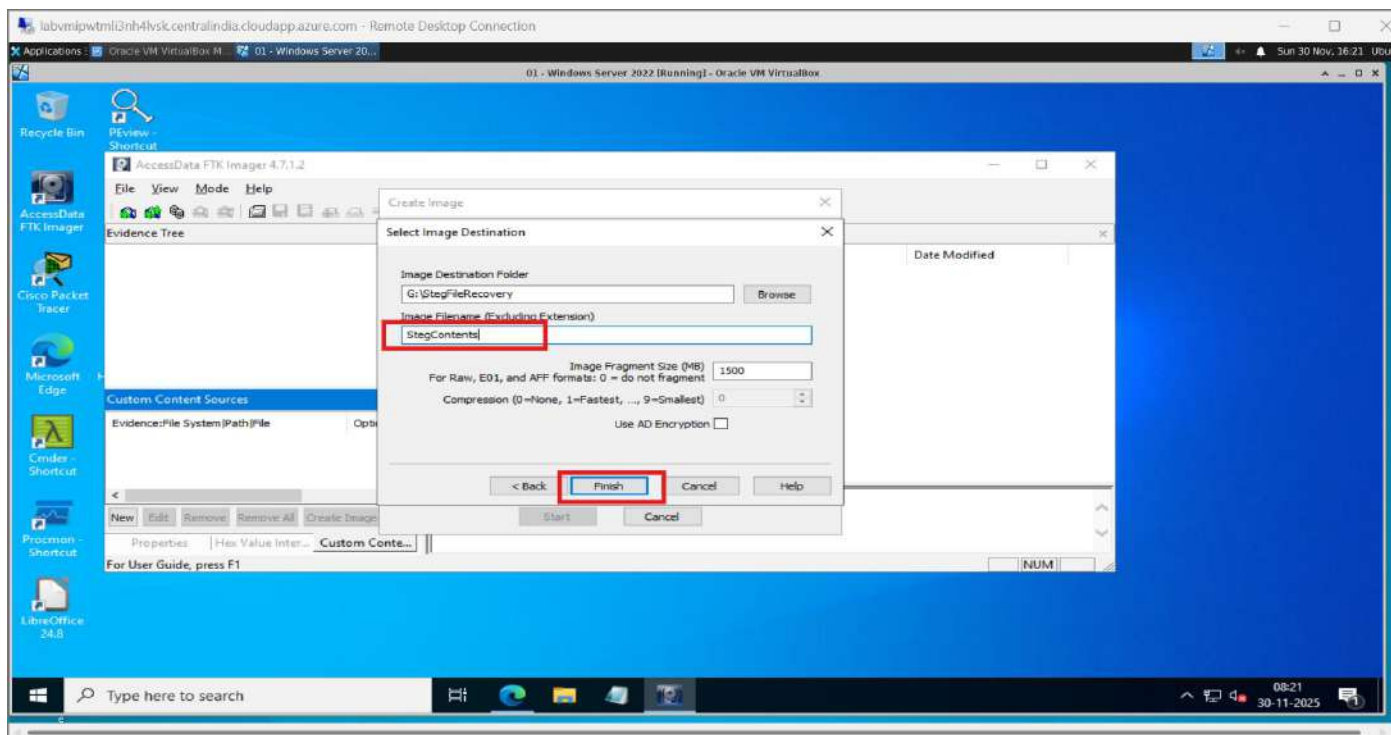
4.8 Clicked on **Browse**.



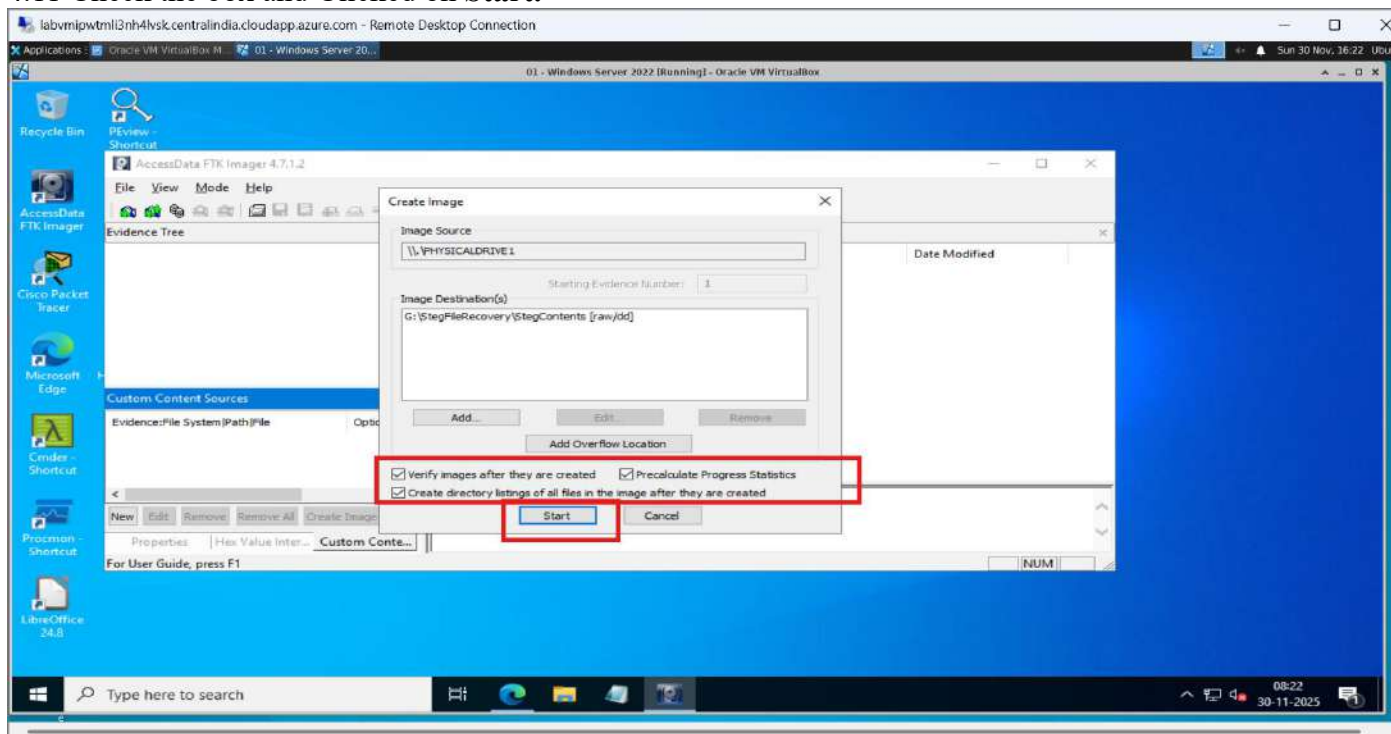
4.9 Created a new folder (**StegFilesRecovery**) inside the **Evidence** drive and selected the folder then Clicked on **Ok**.

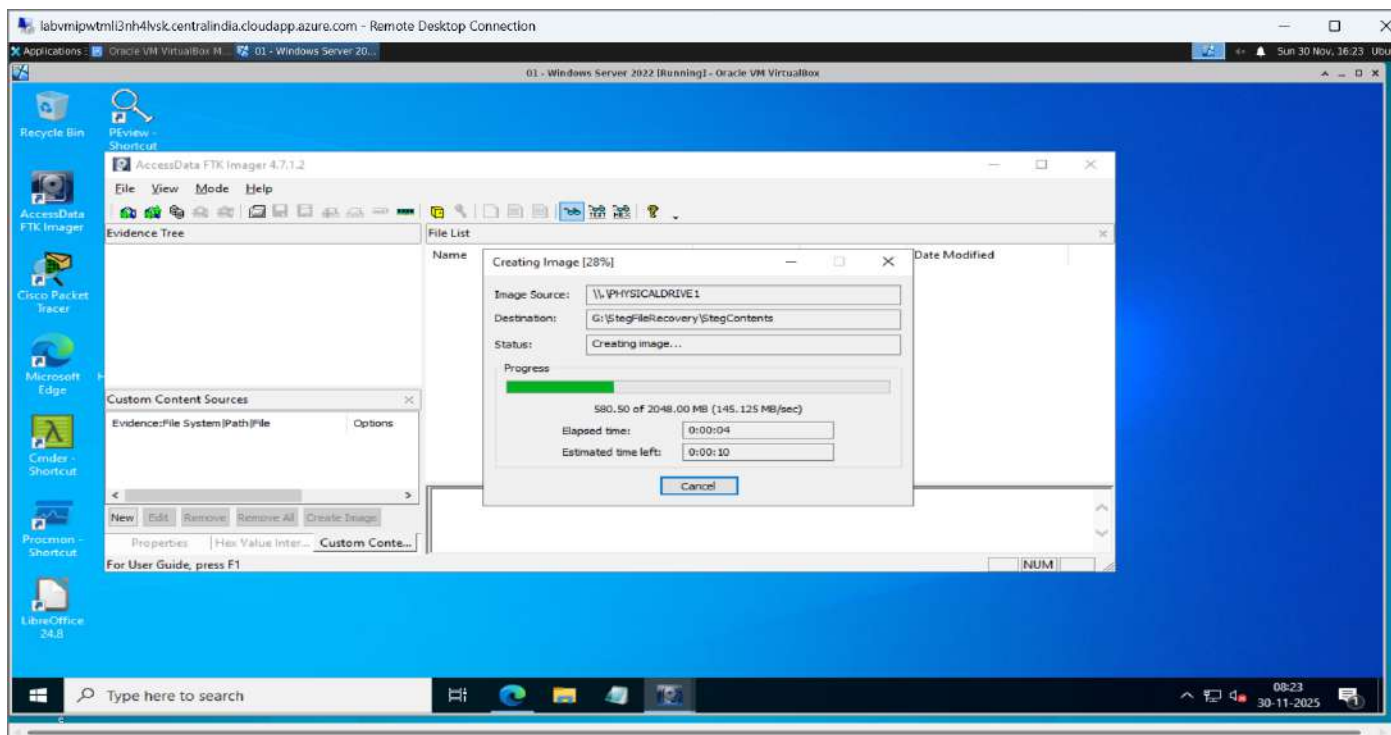


4.10 Enter the **Image Filename** and Clicked on **Finish**.

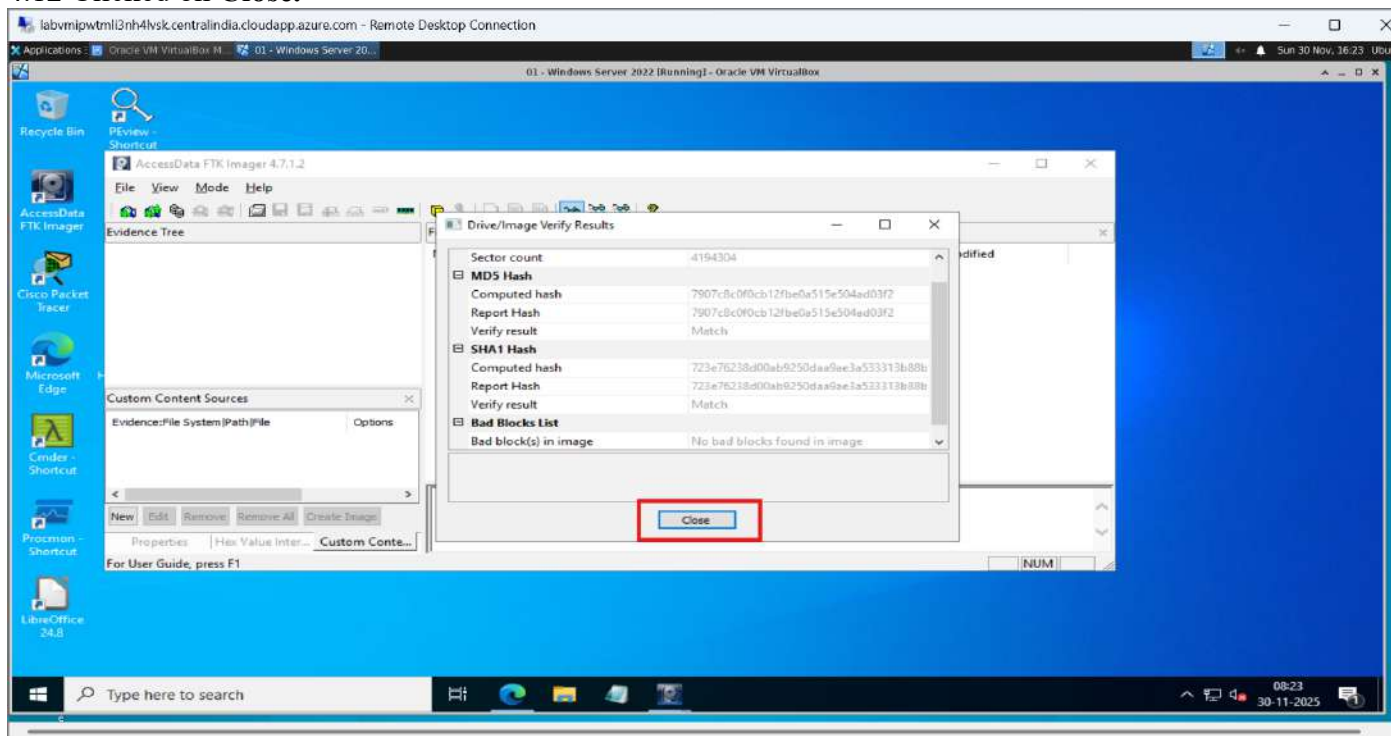


4.11 Check the box and Clicked on Start.

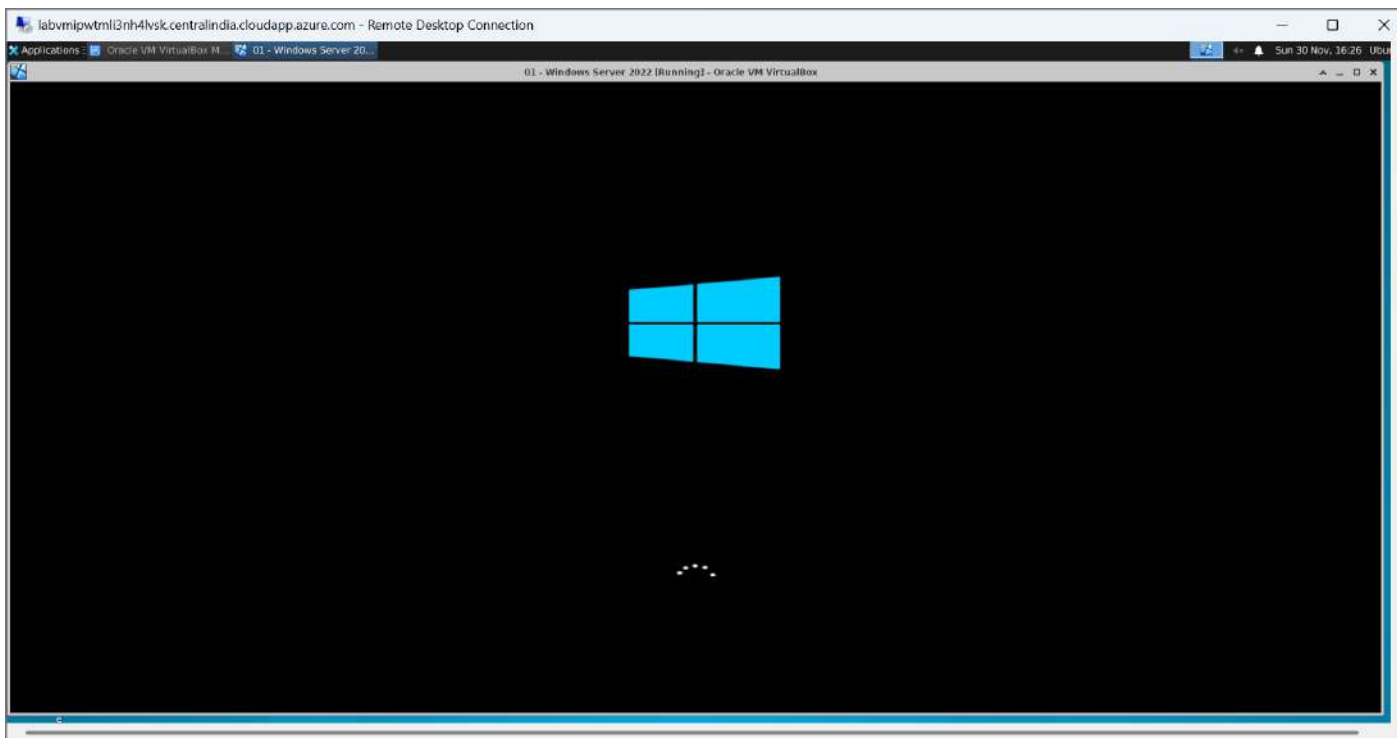
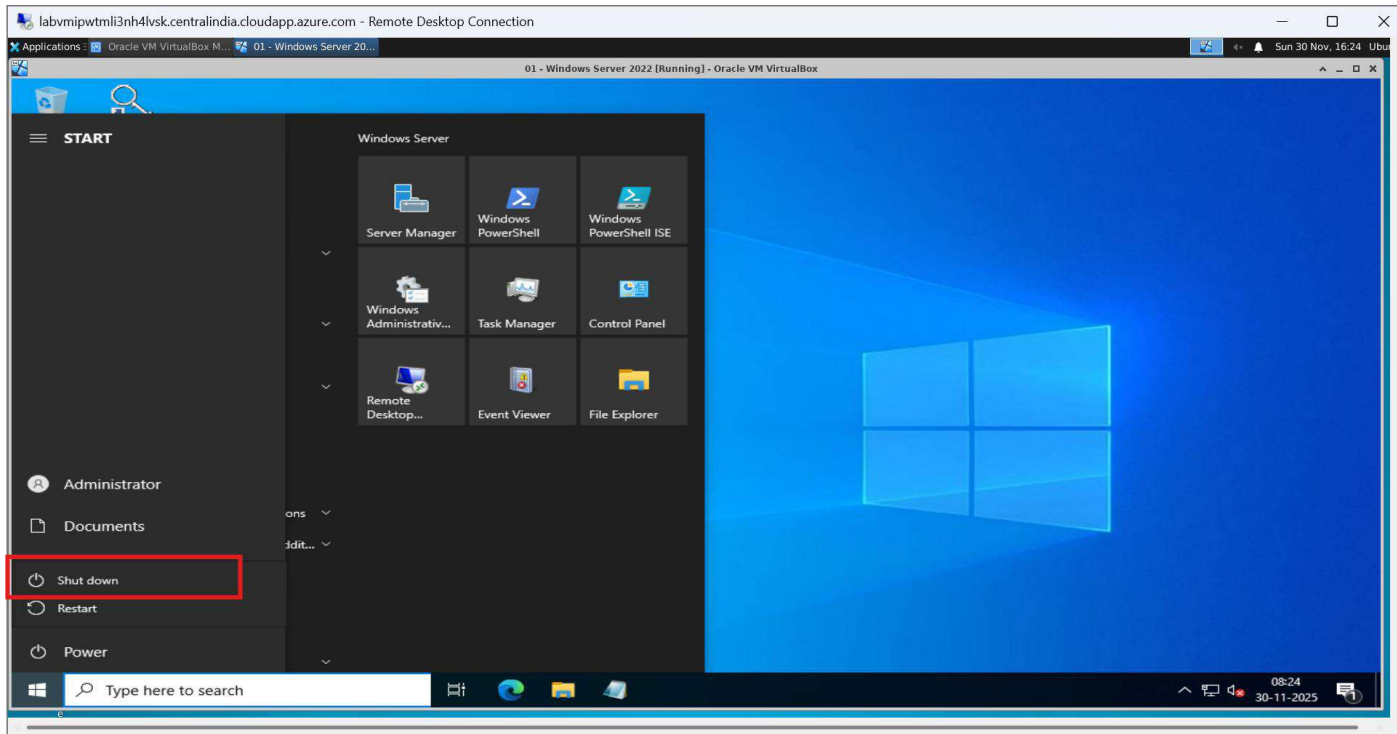


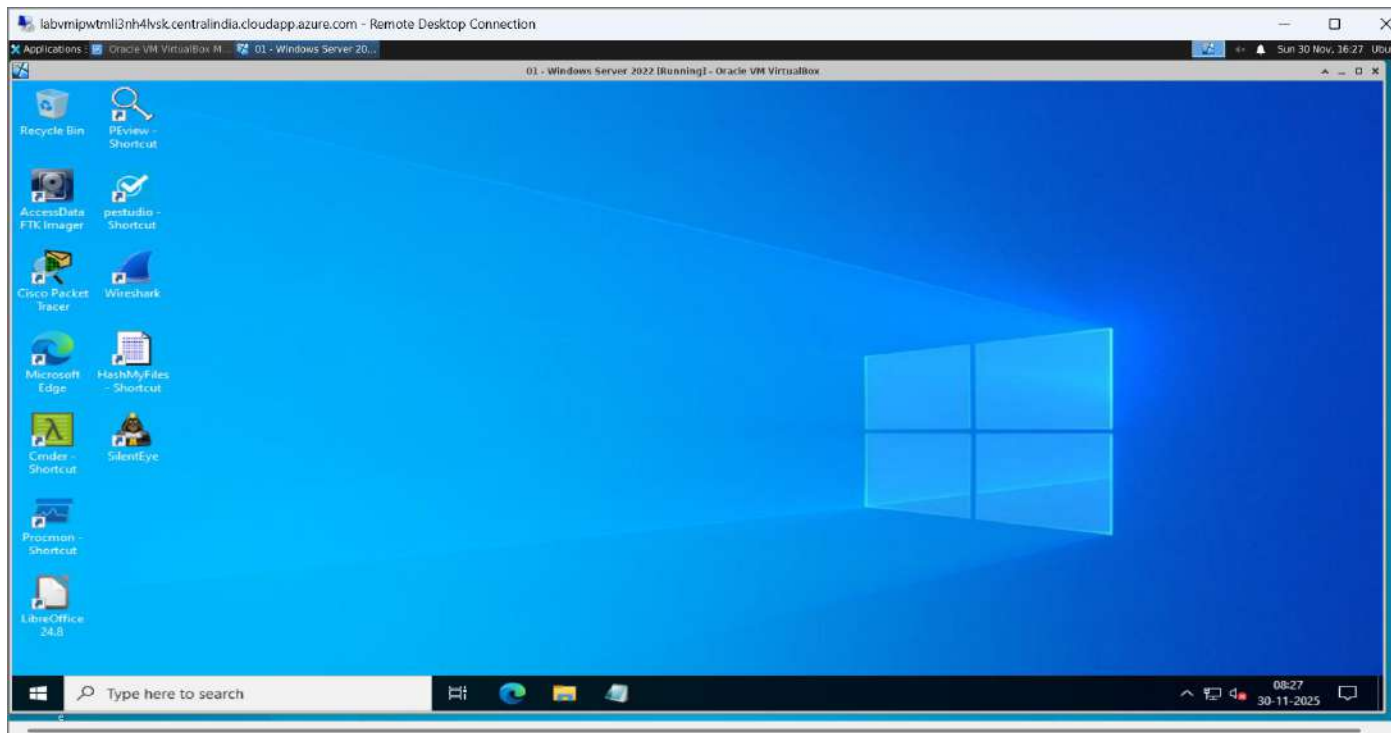


4.12 Clicked on Close.



4.13 Shut down and starting again the system



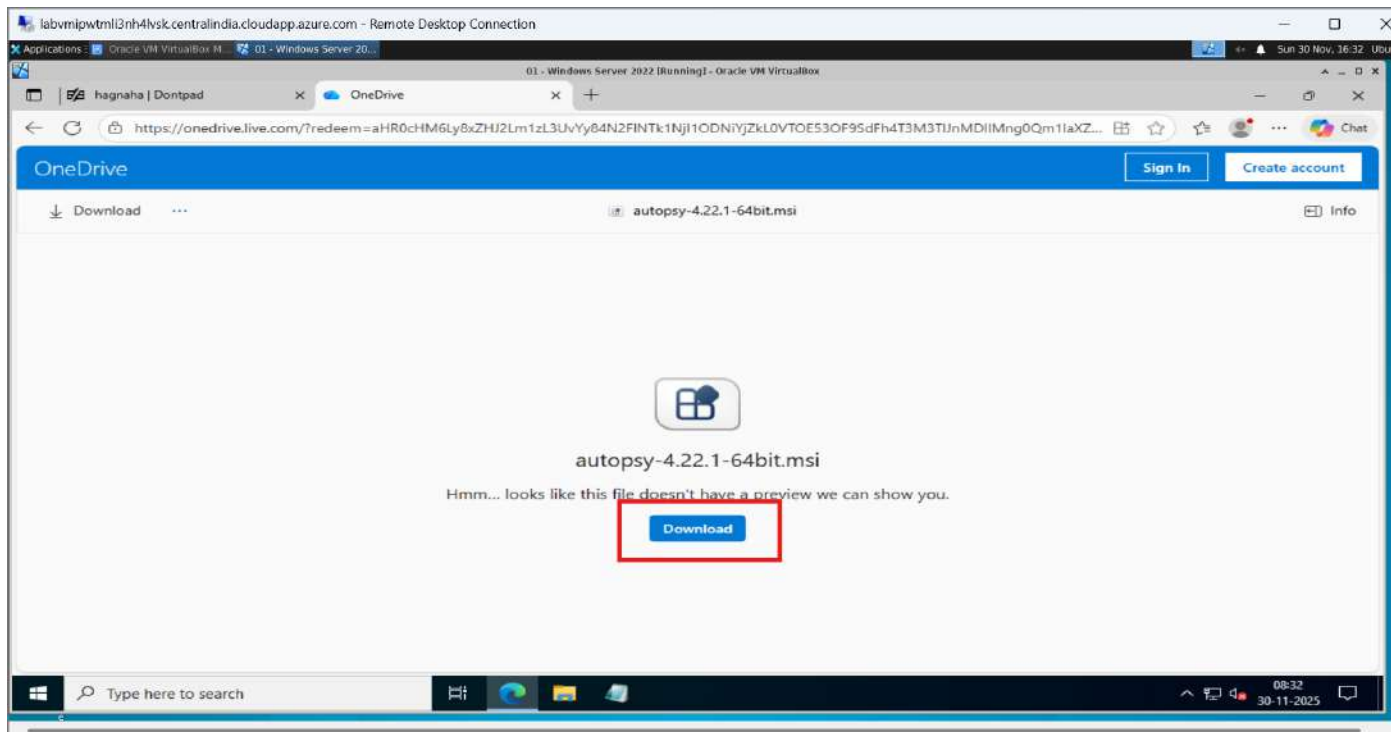


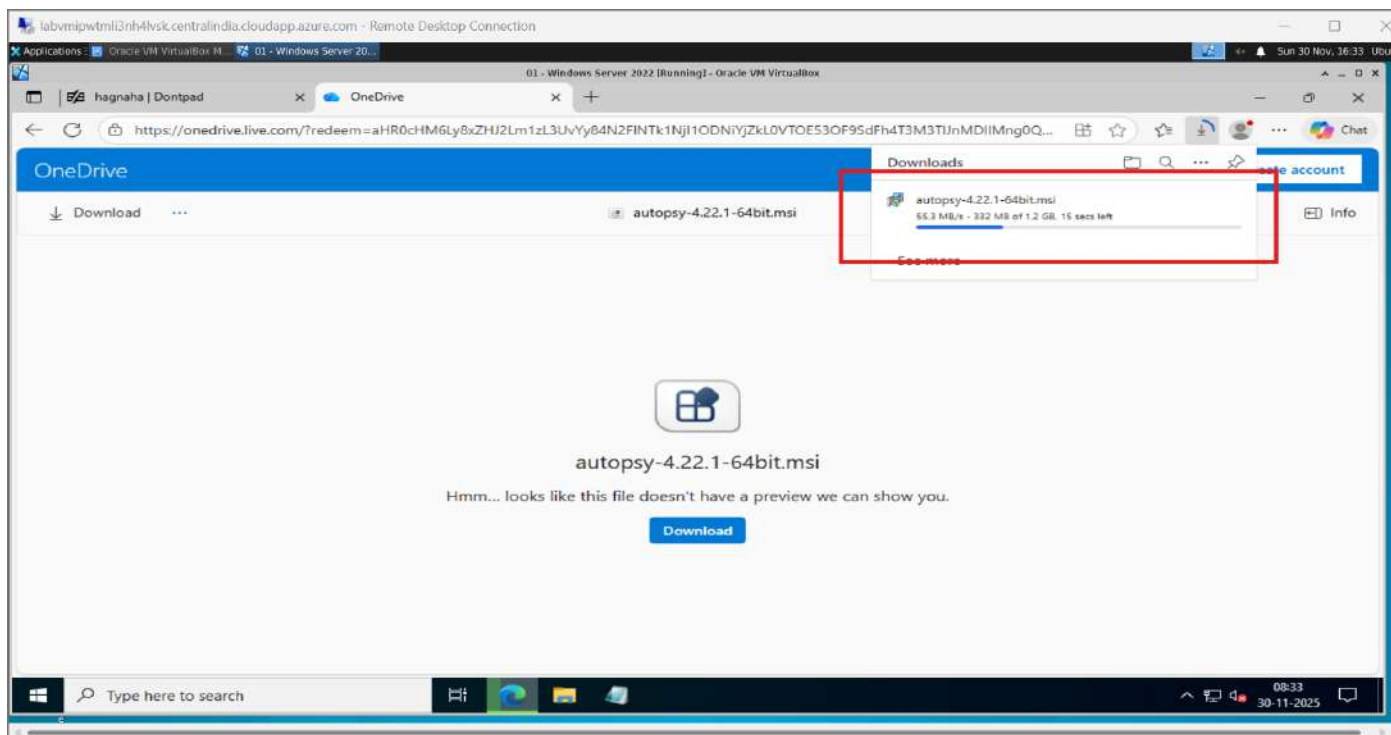
Task 5: Using Autopsy to recover data

5.1 Open the **Edge Browser** and follow this link to download **autopsy**.

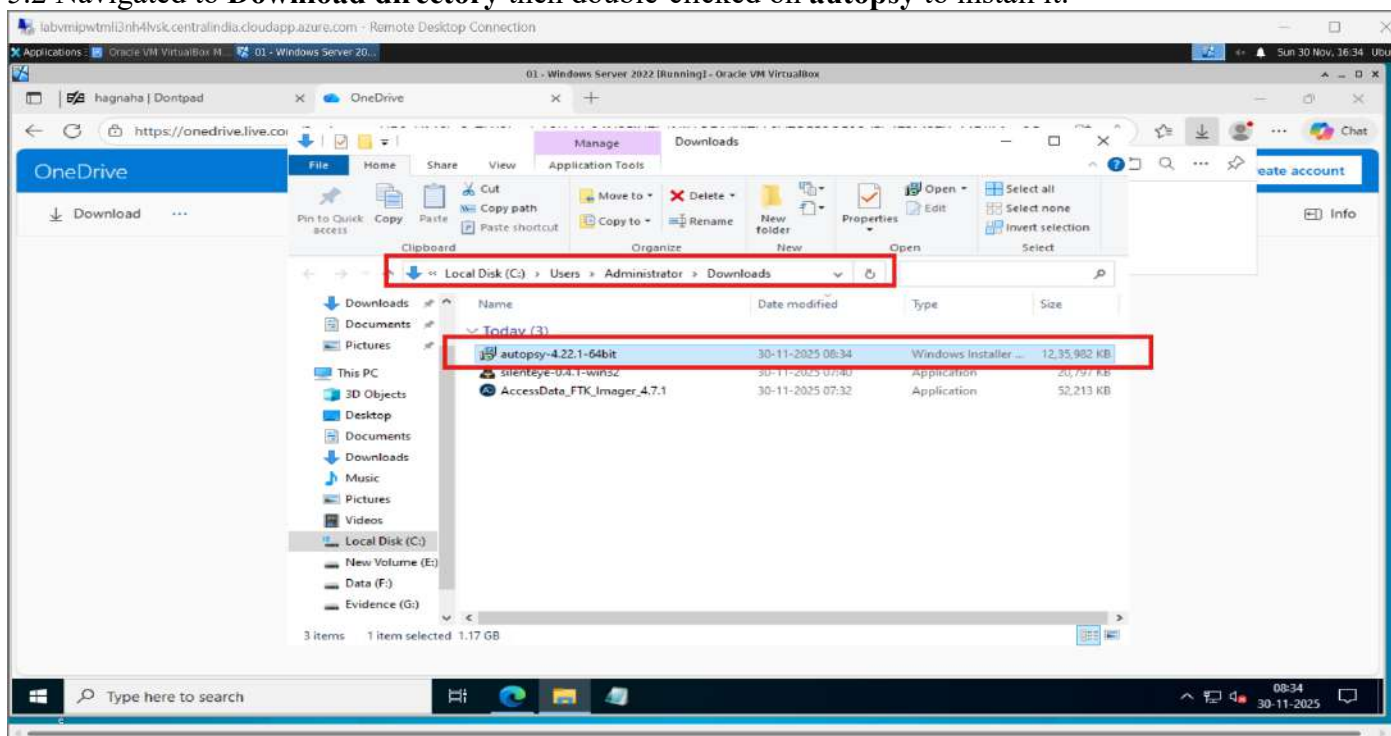
Link:

https://1drv.ms/u/c/87ae59562583bb6d/ES8Nw8_RtXxOs7NRg09H2x4BmeivQA7kLv7yuXKmt5LQZw?e=fkEDhc

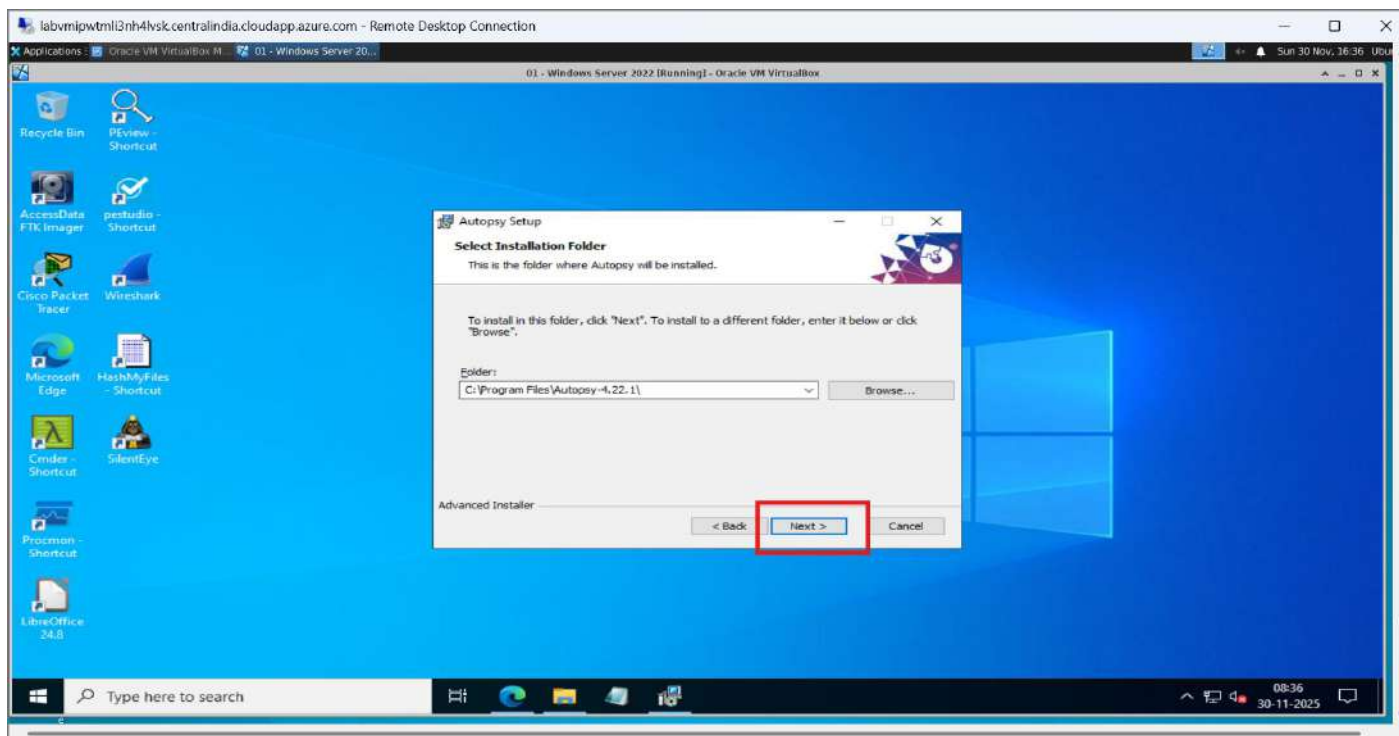
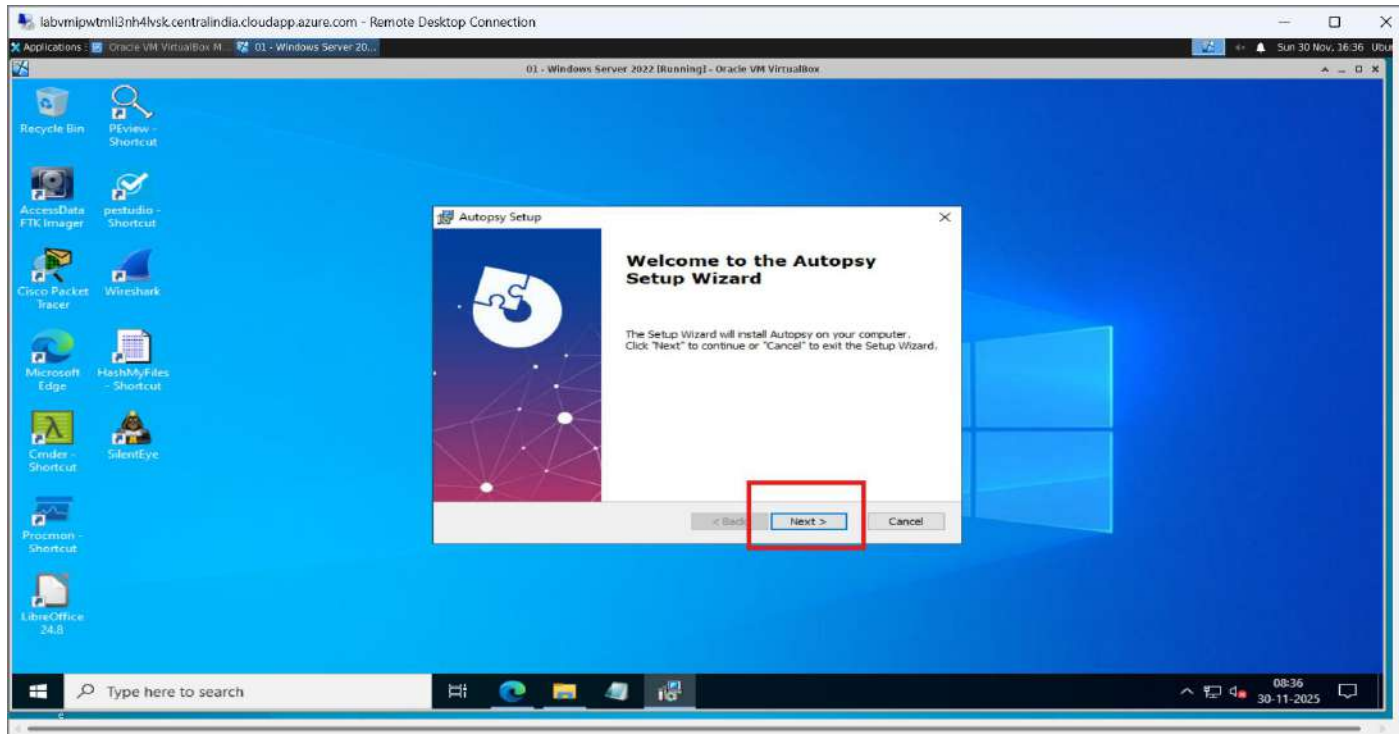


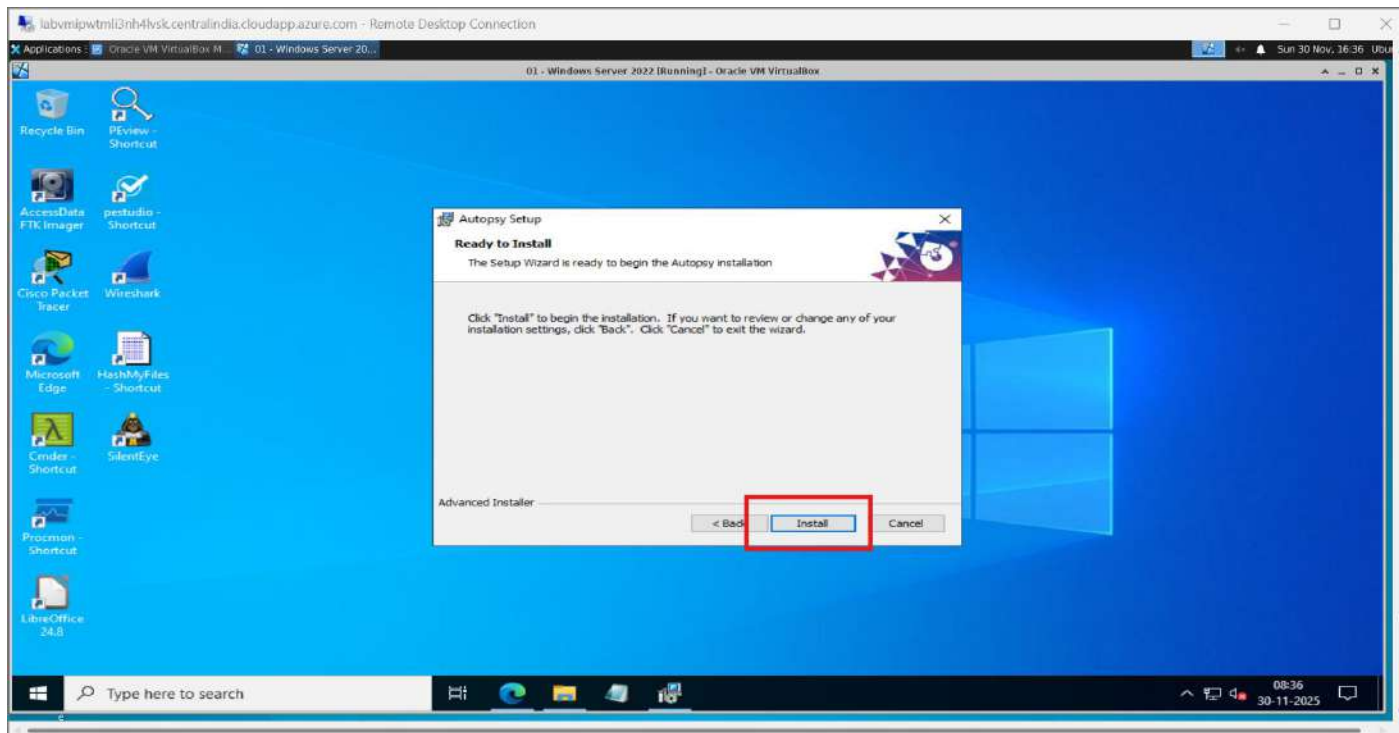


5.2 Navigated to **Download** directory then double-clicked on **autopsy** to install it.

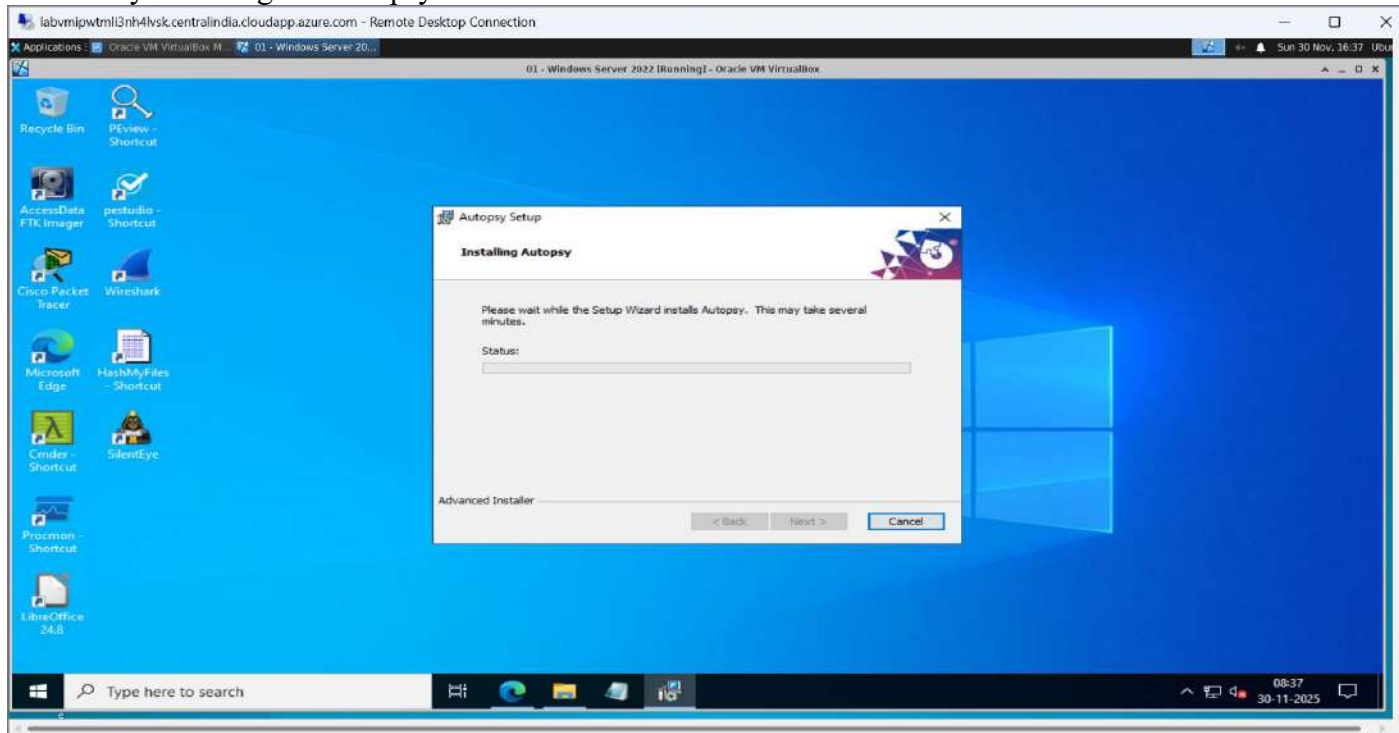


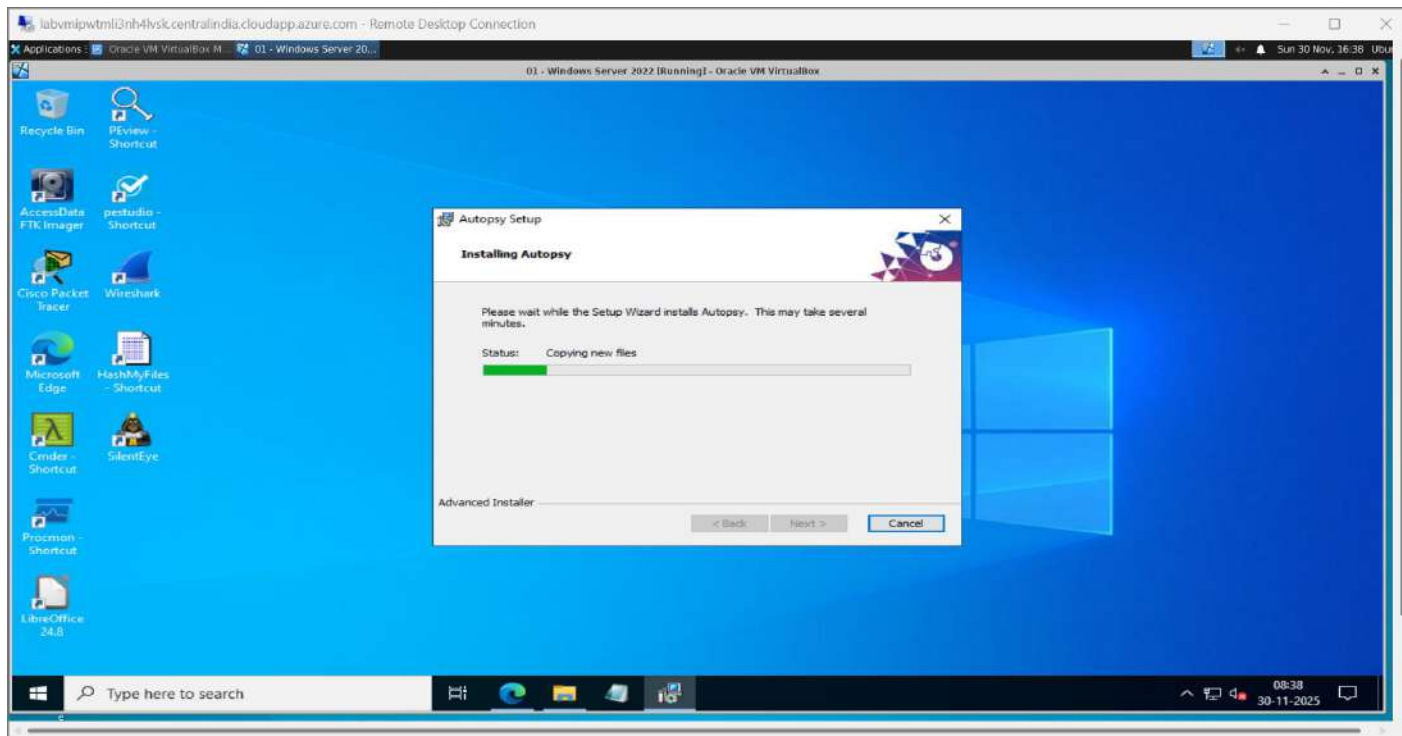
5.3 Following the procedure and installing the software



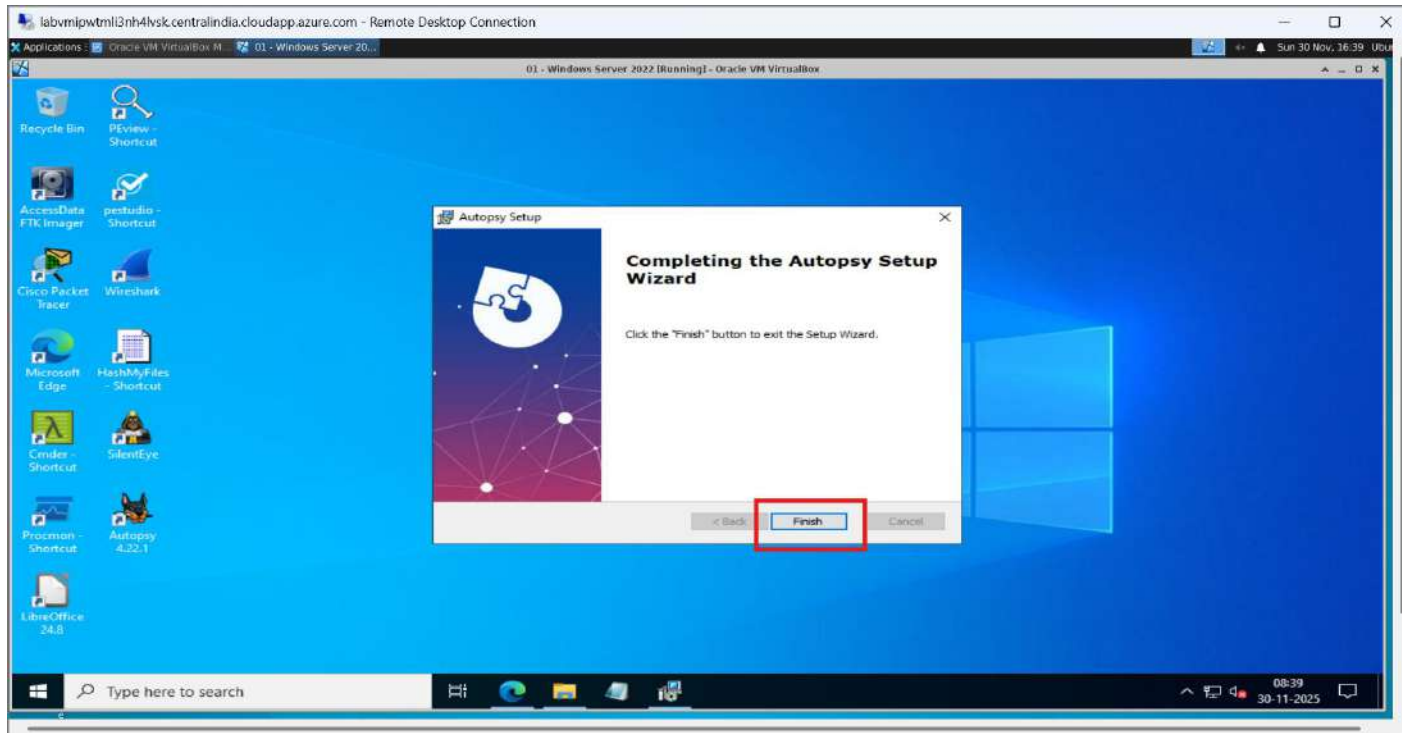


5.4 Finally installing the Autopsy

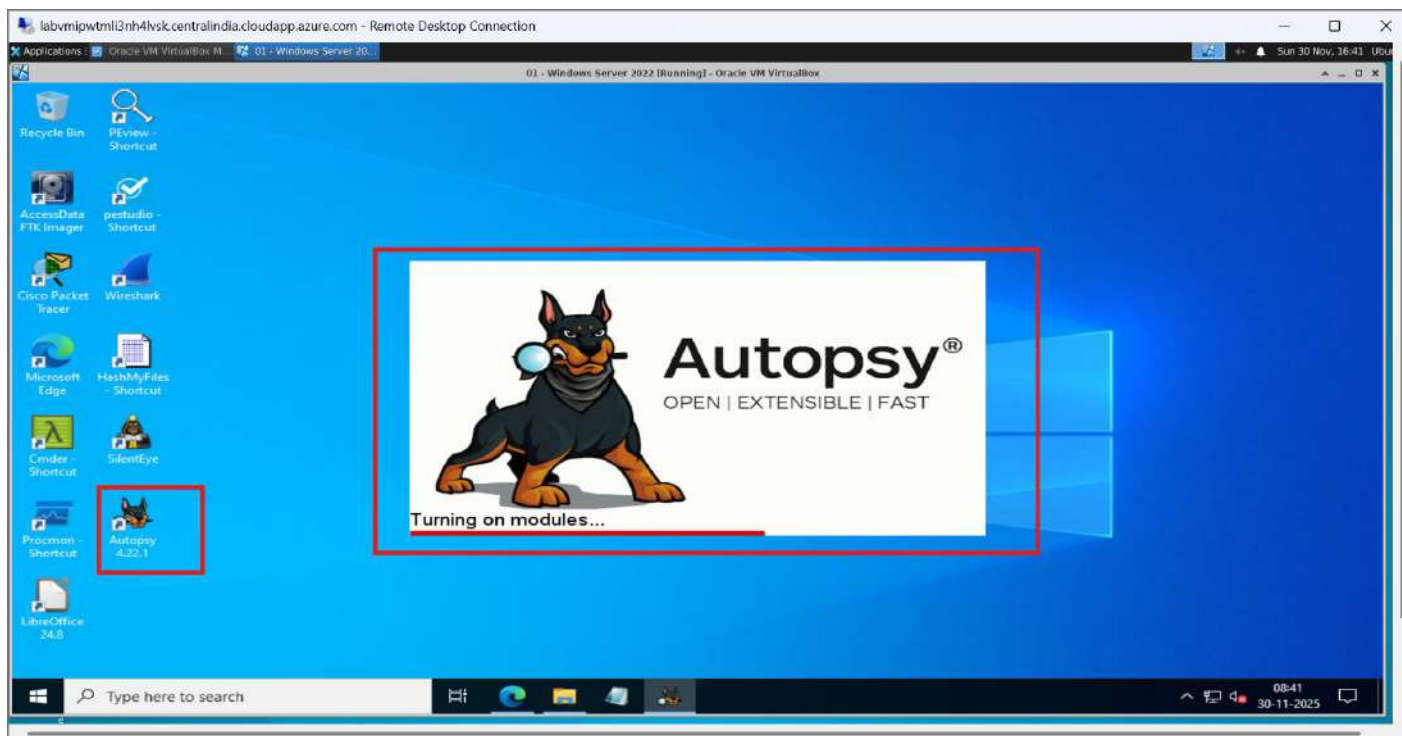
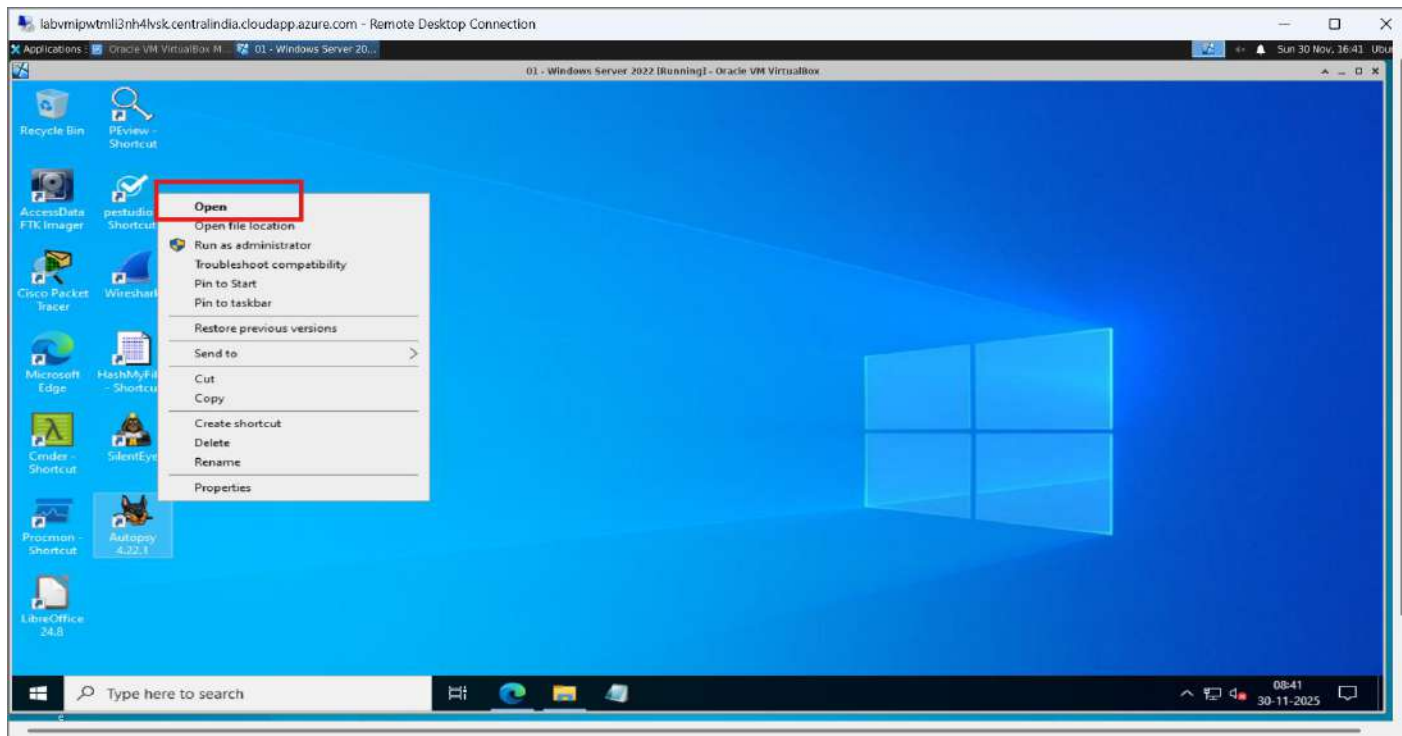




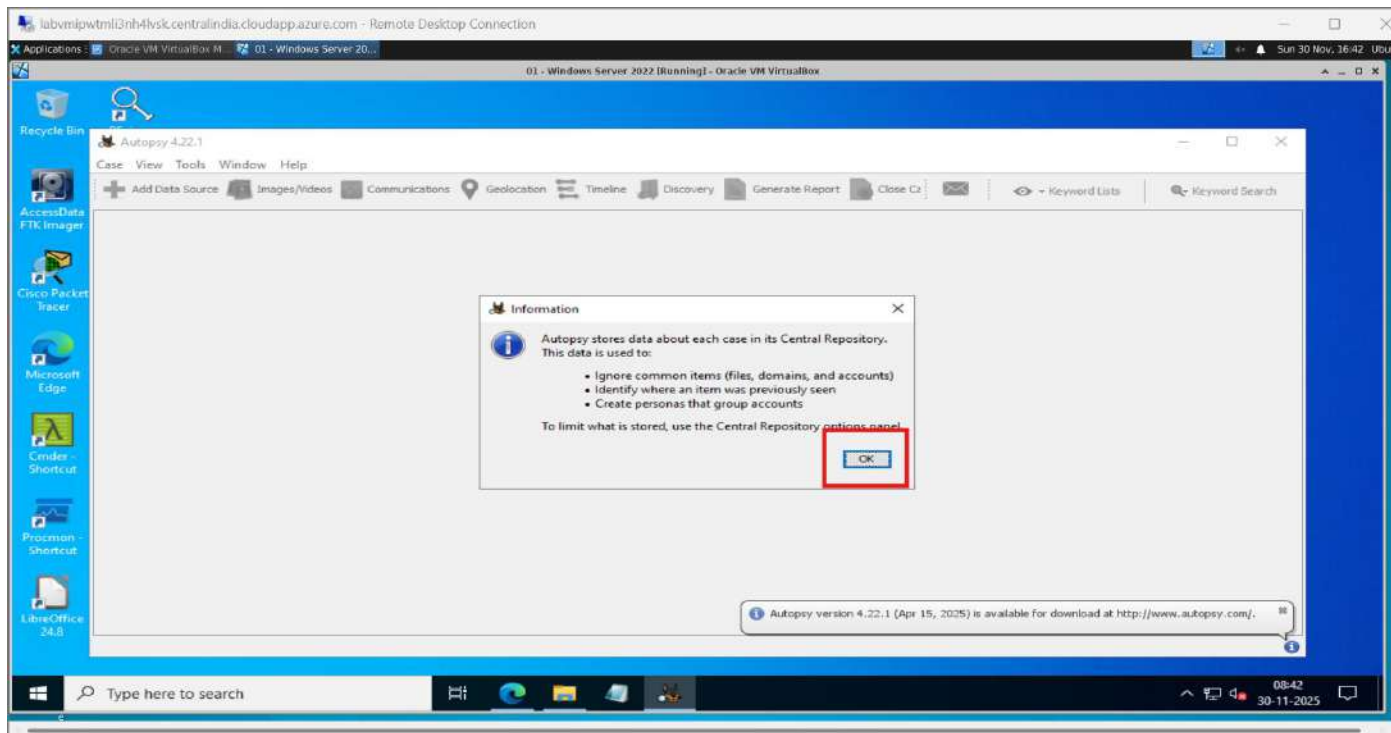
5.5 Clicked on Finish



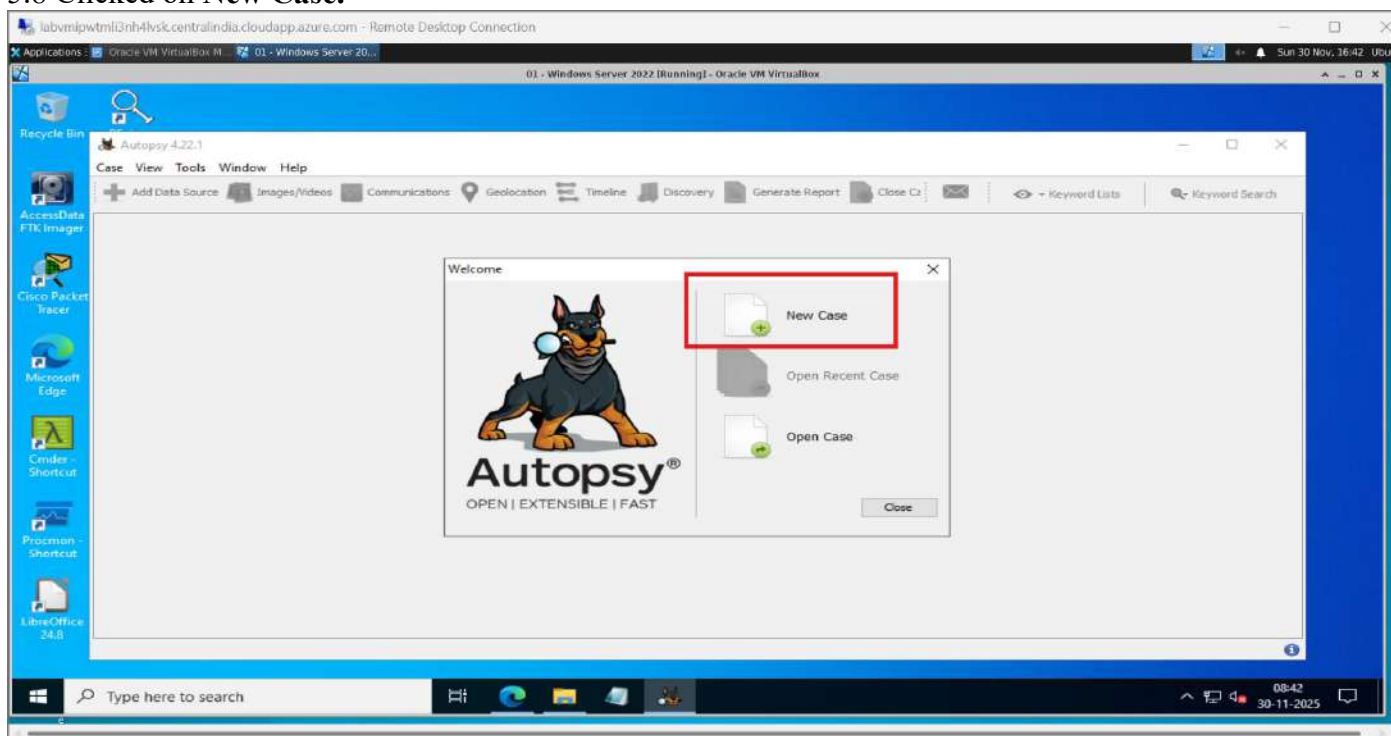
5.6 Open the Autopsy.



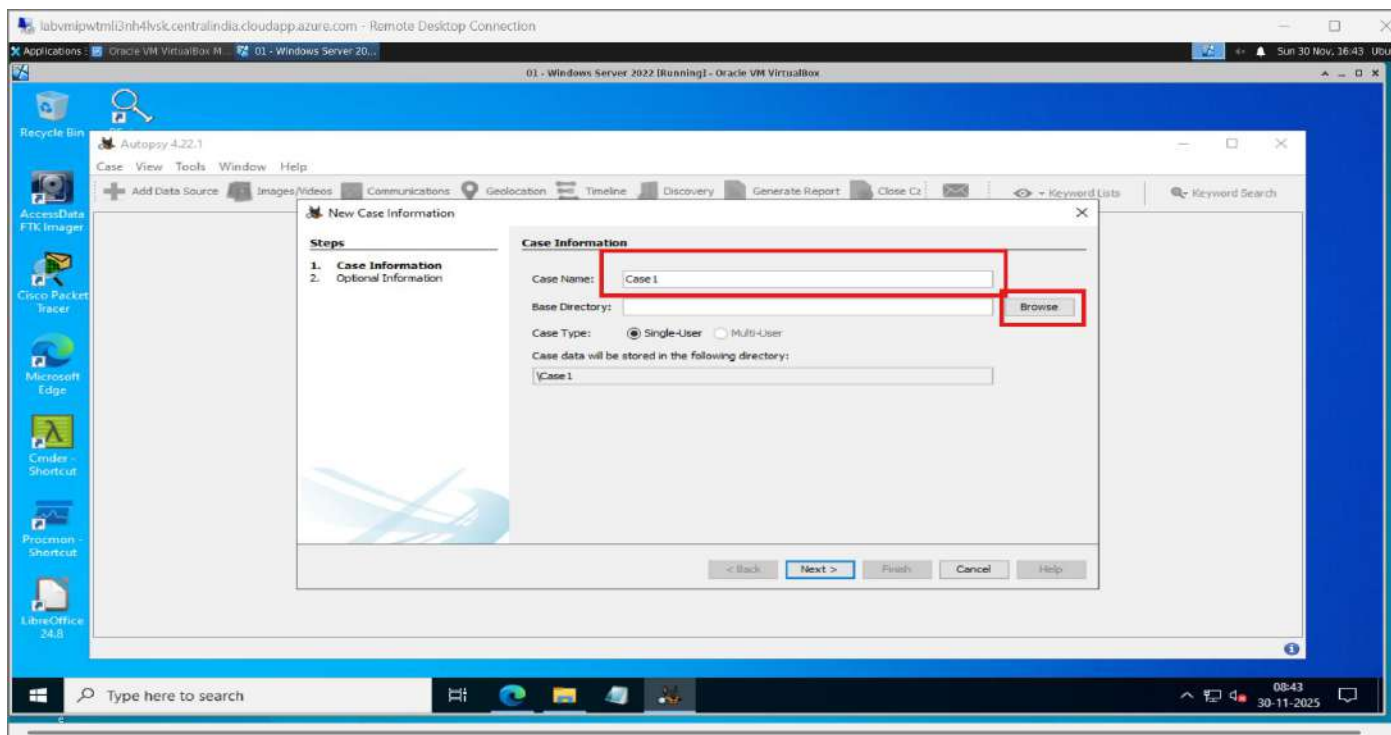
5.7 Clicked on **Ok**.



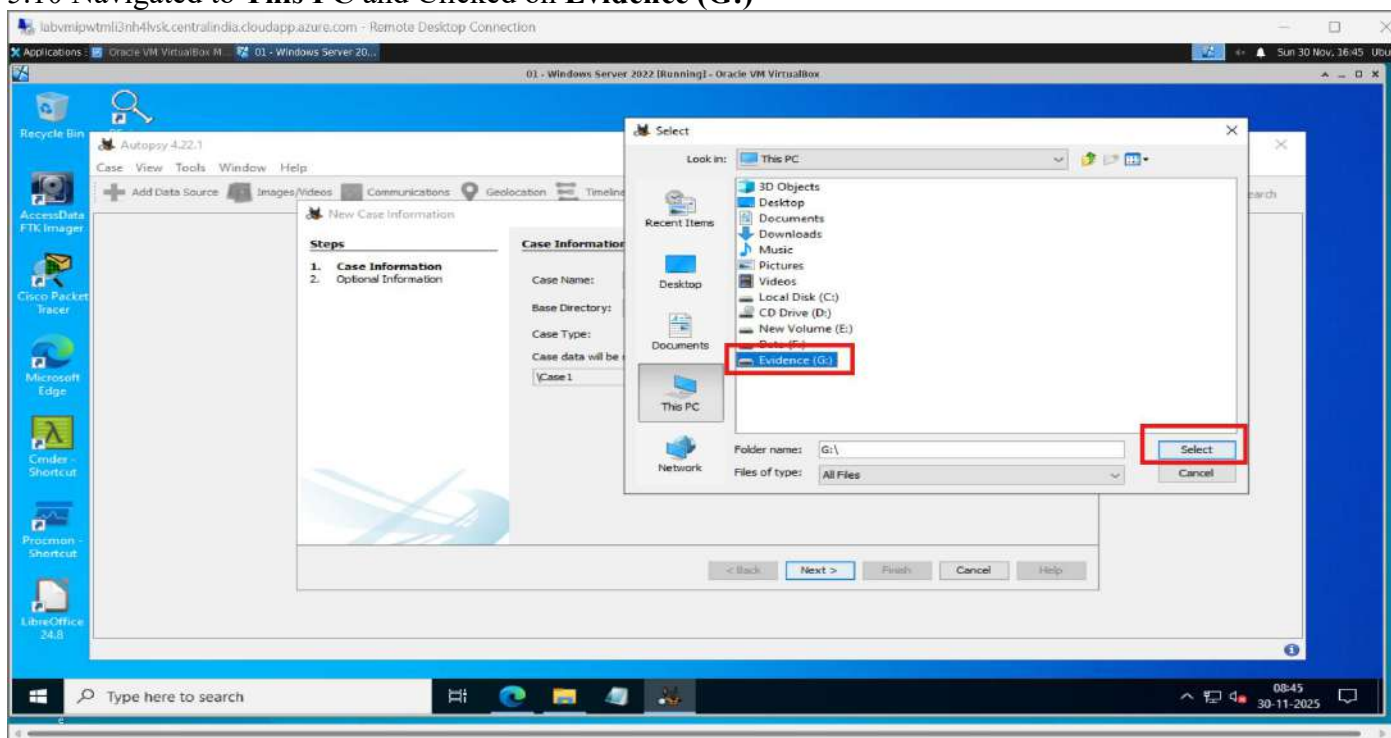
5.8 Clicked on New Case.



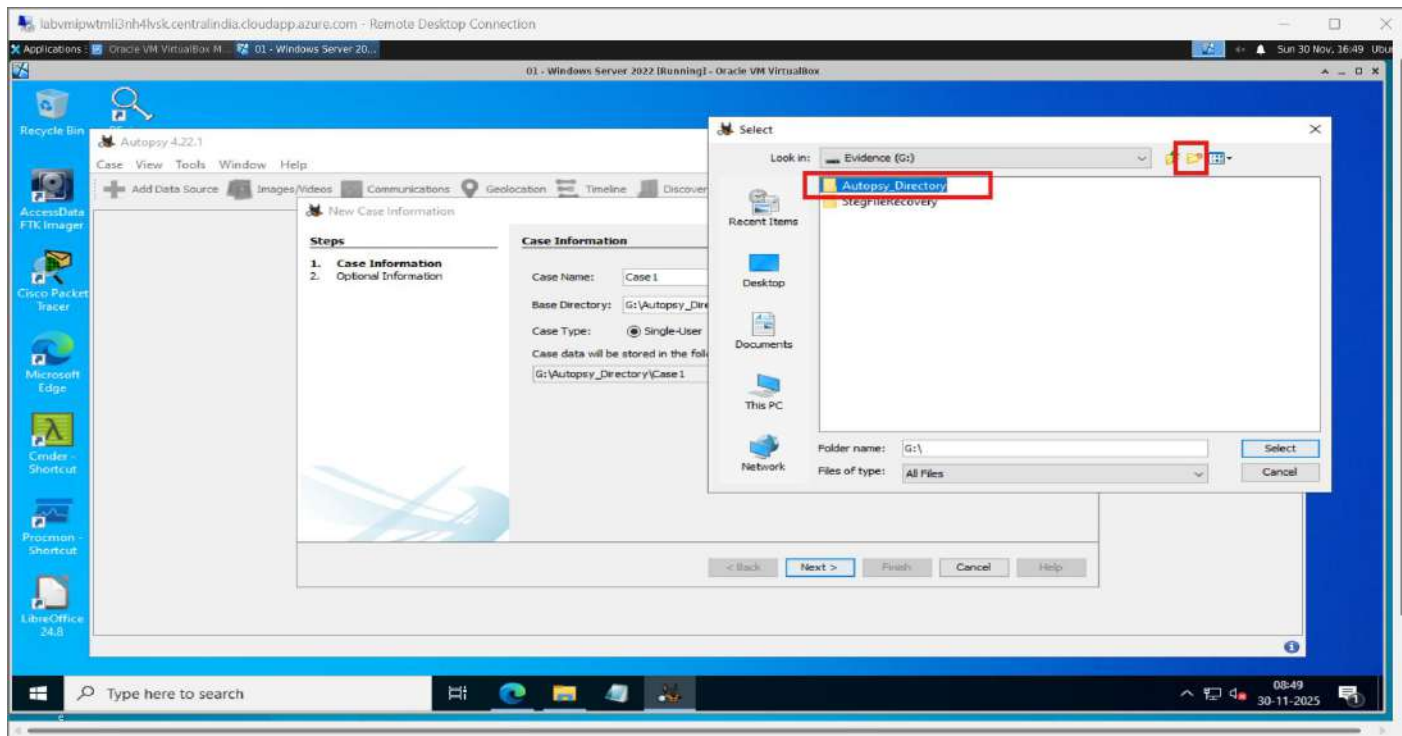
5.9 Typed the Case Name as “Case 1”, then Clicked on Browse.



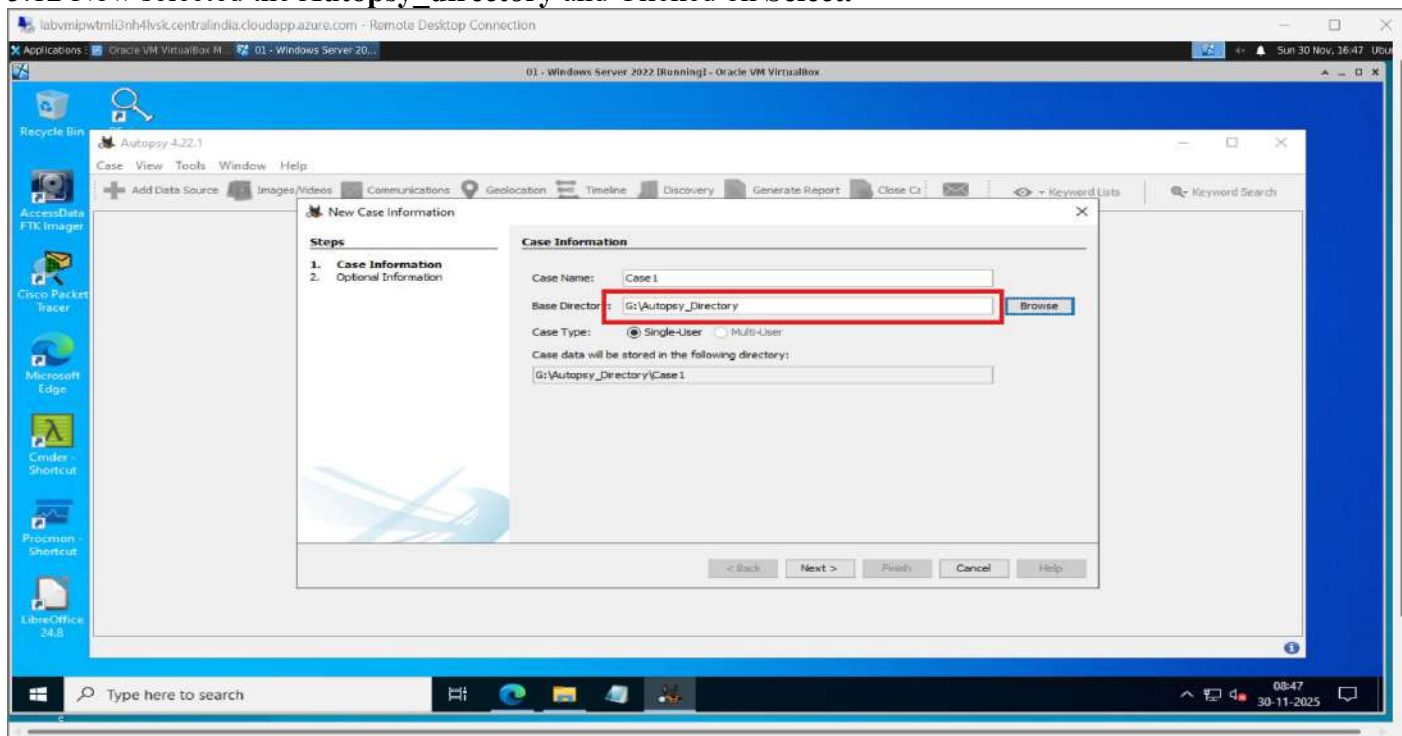
5.10 Navigated to **This PC** and Clicked on **Evidence (G:)**



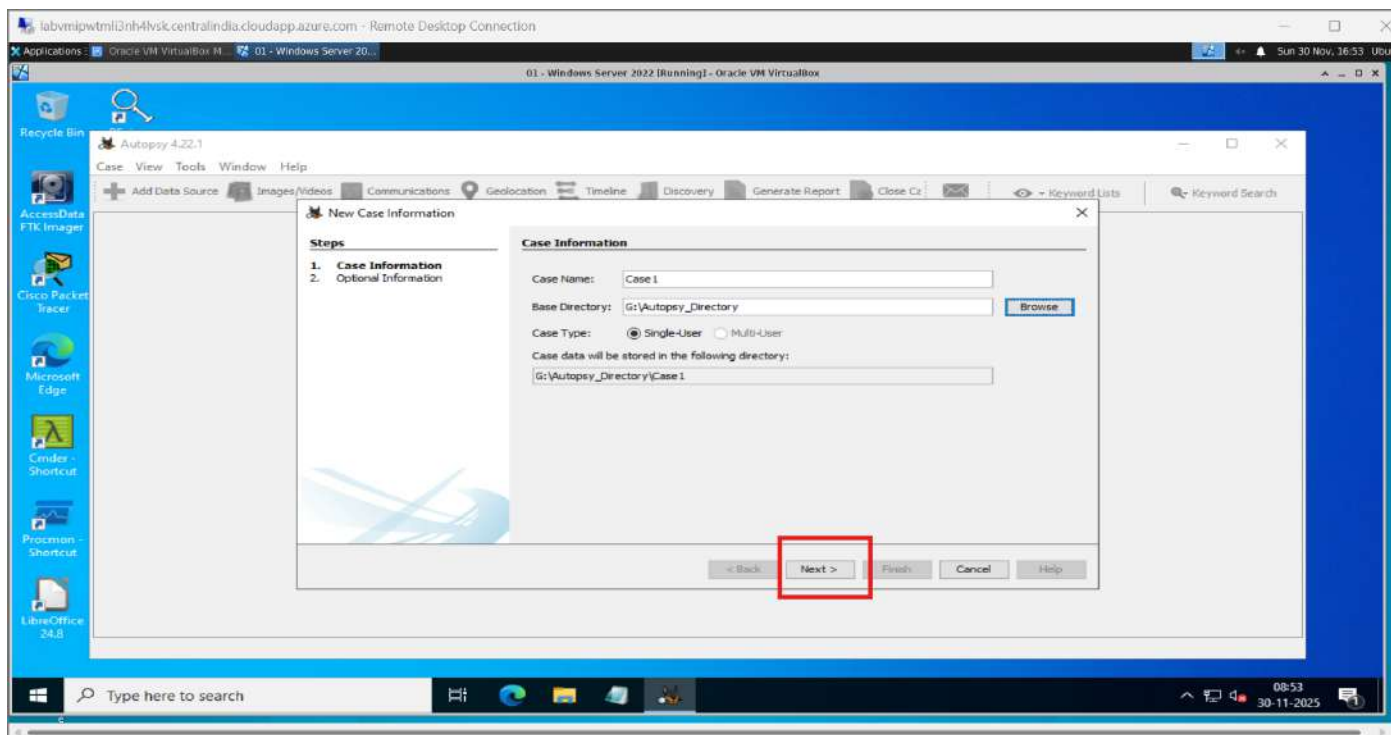
5.11 Clicked on “**Create a New Folder.**”



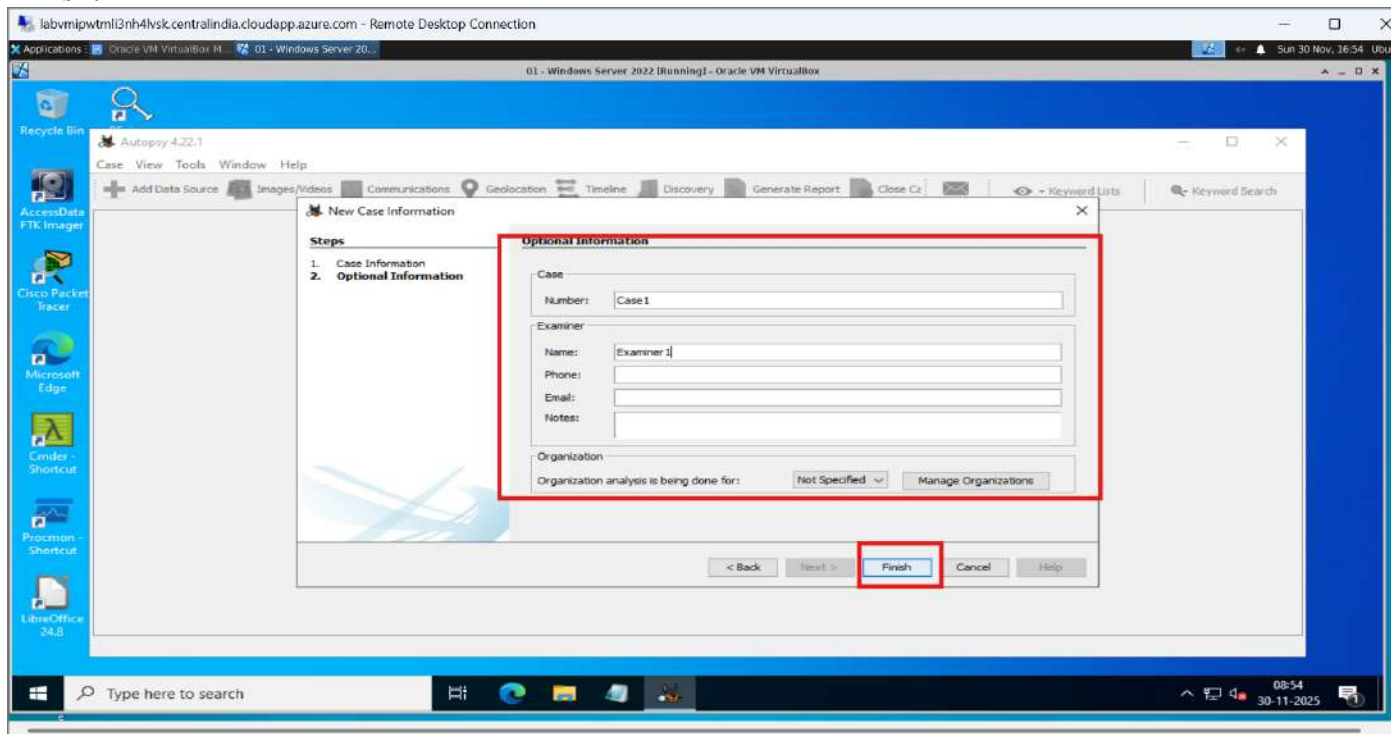
5.12 Now selected the **Autopsy_directory** and Clicked on **Select**.

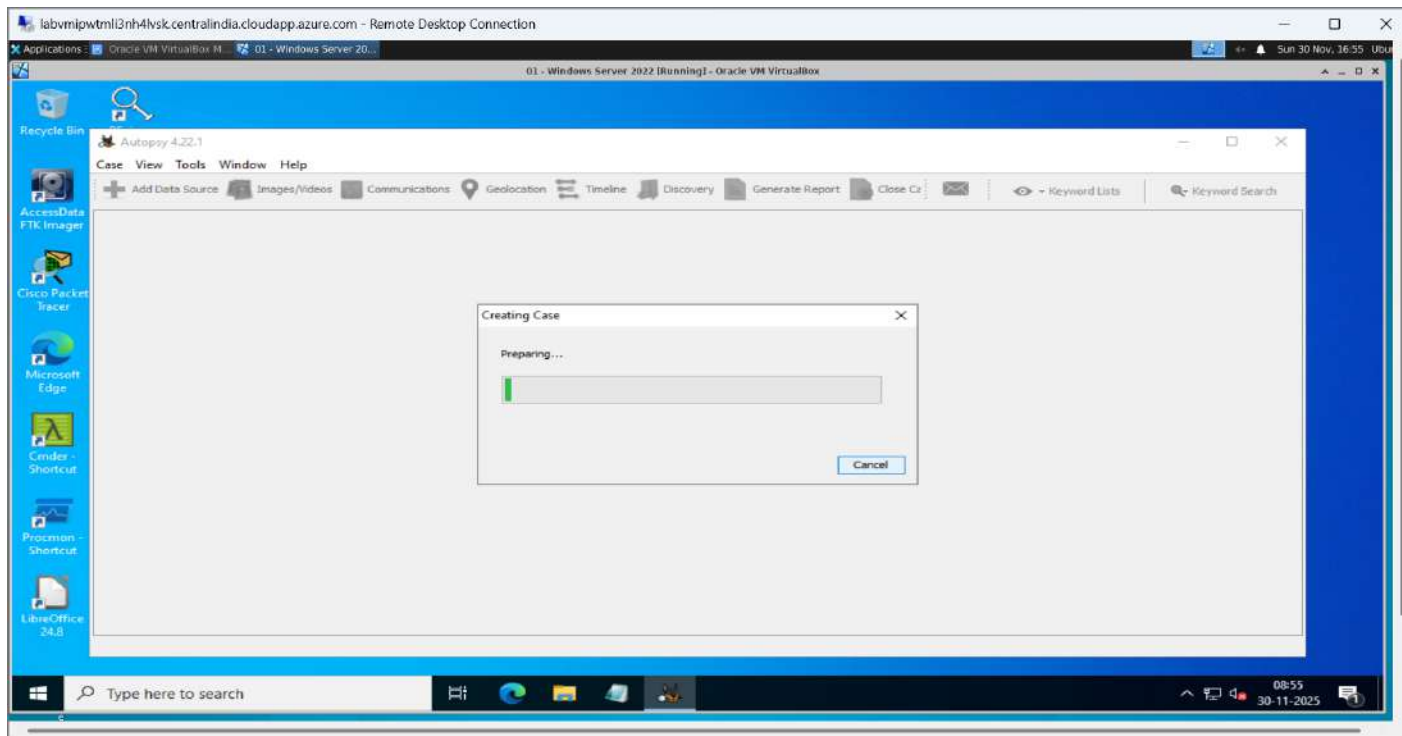


5.13 Clicked on **Next**.

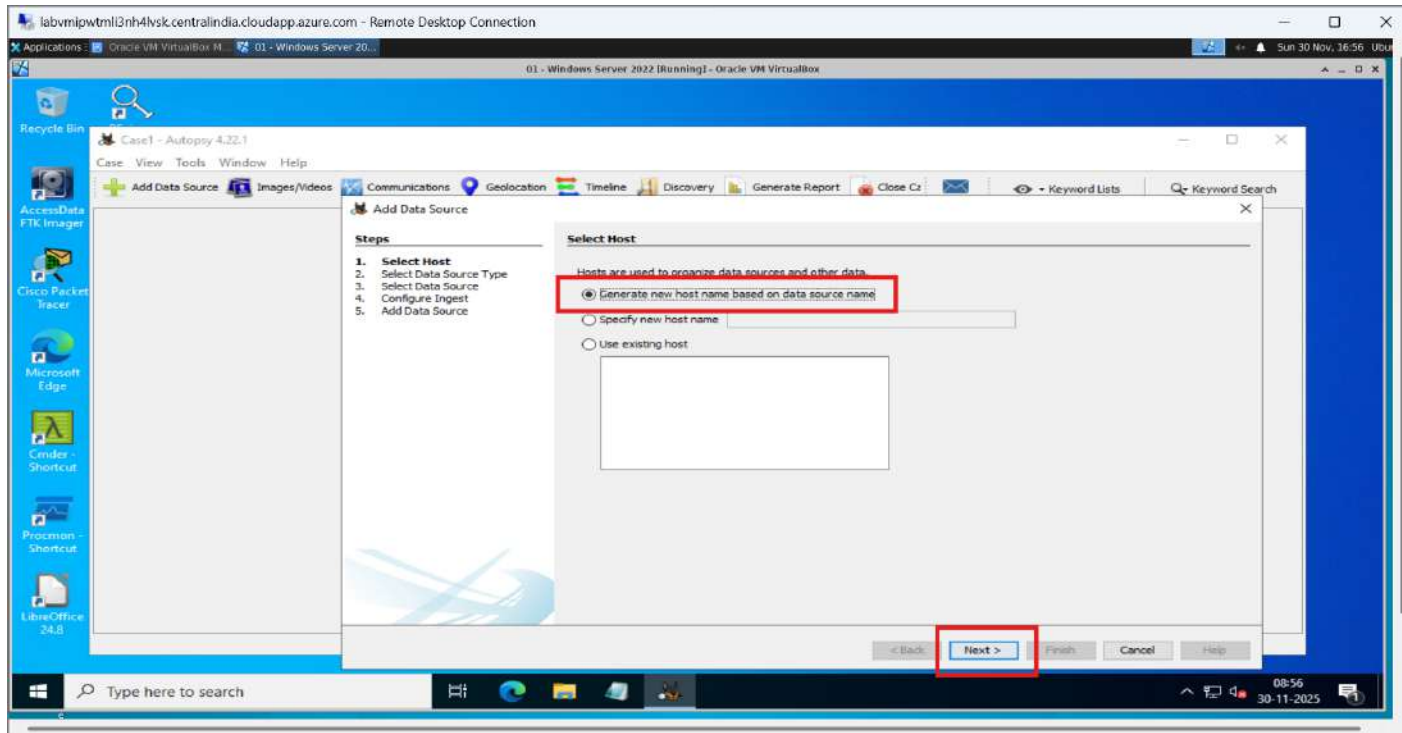


5.14 In the Case Number typed “Case 1” and in the Examiner Name typed “Examiner 01”, then Clicked on **Finish**.

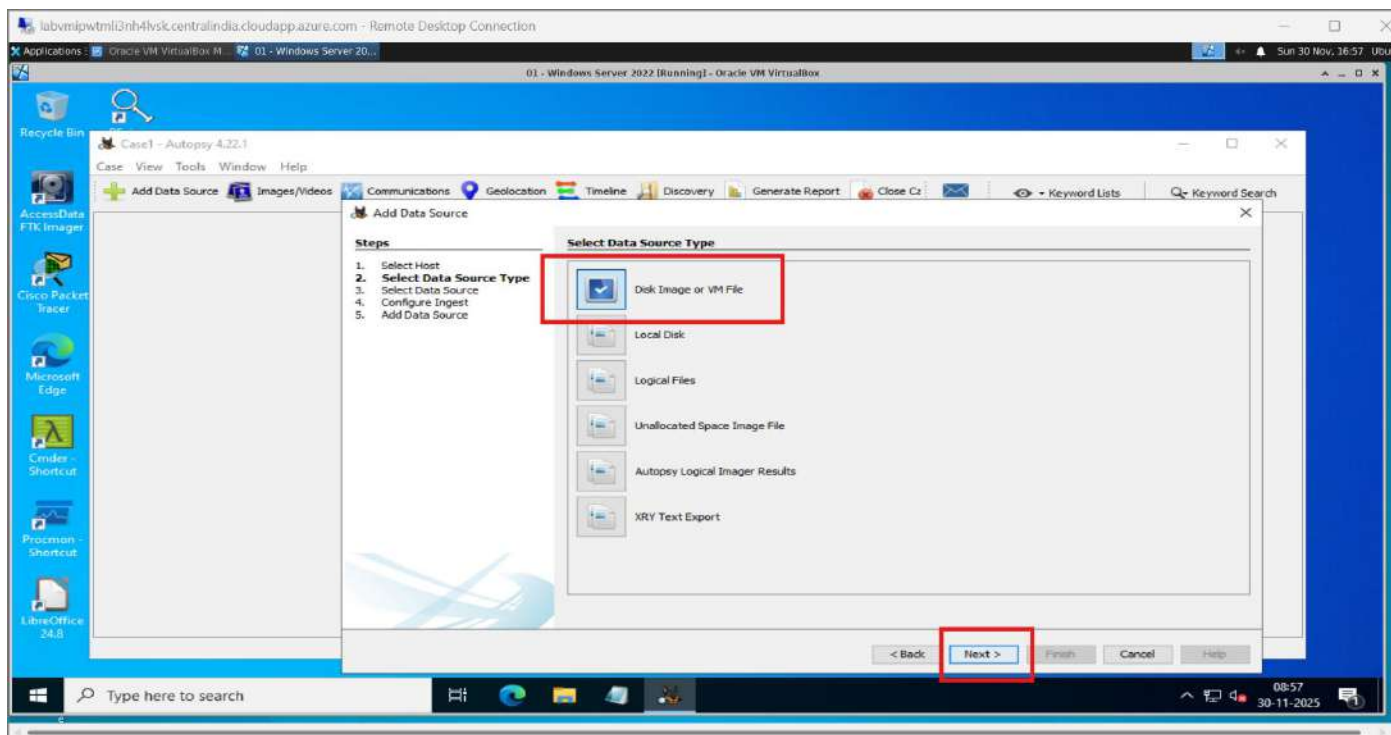




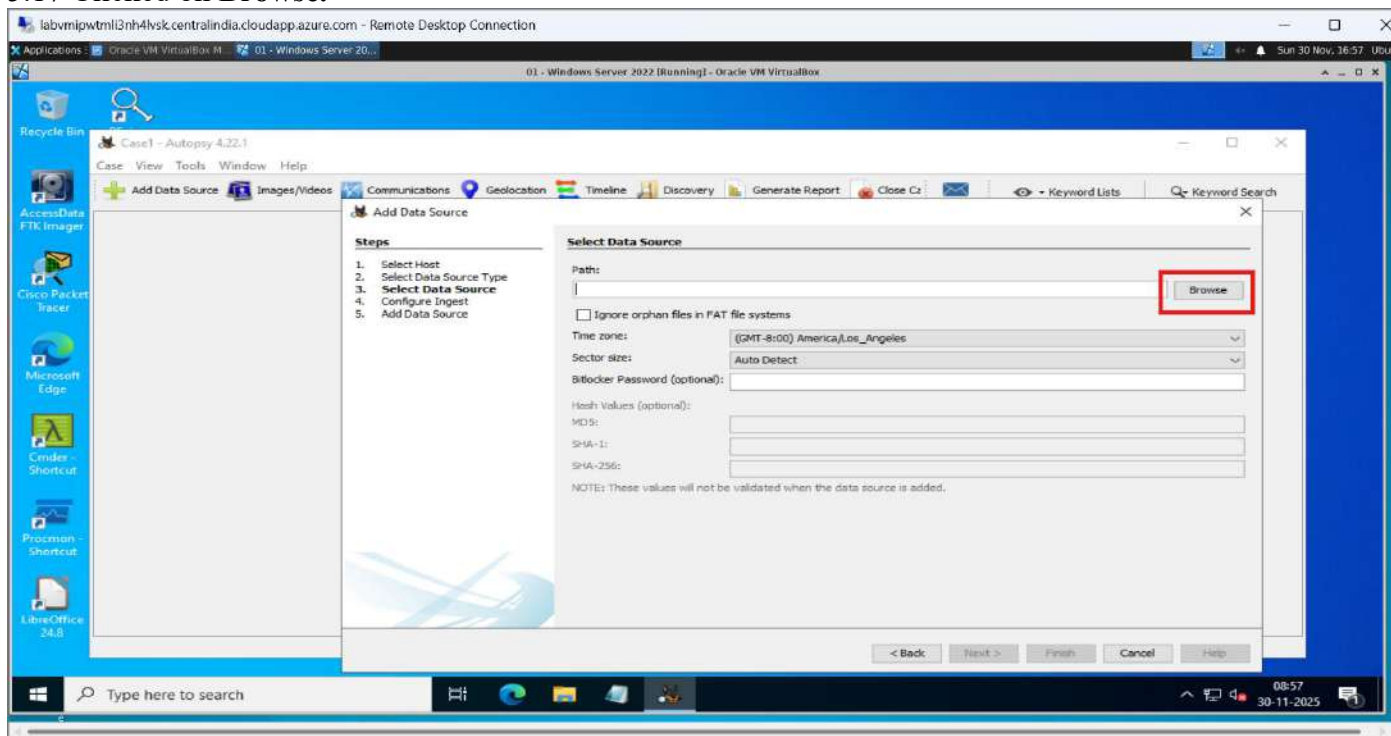
5.15 Selected the “Generate new host name based on data source name” and Clicked Next.



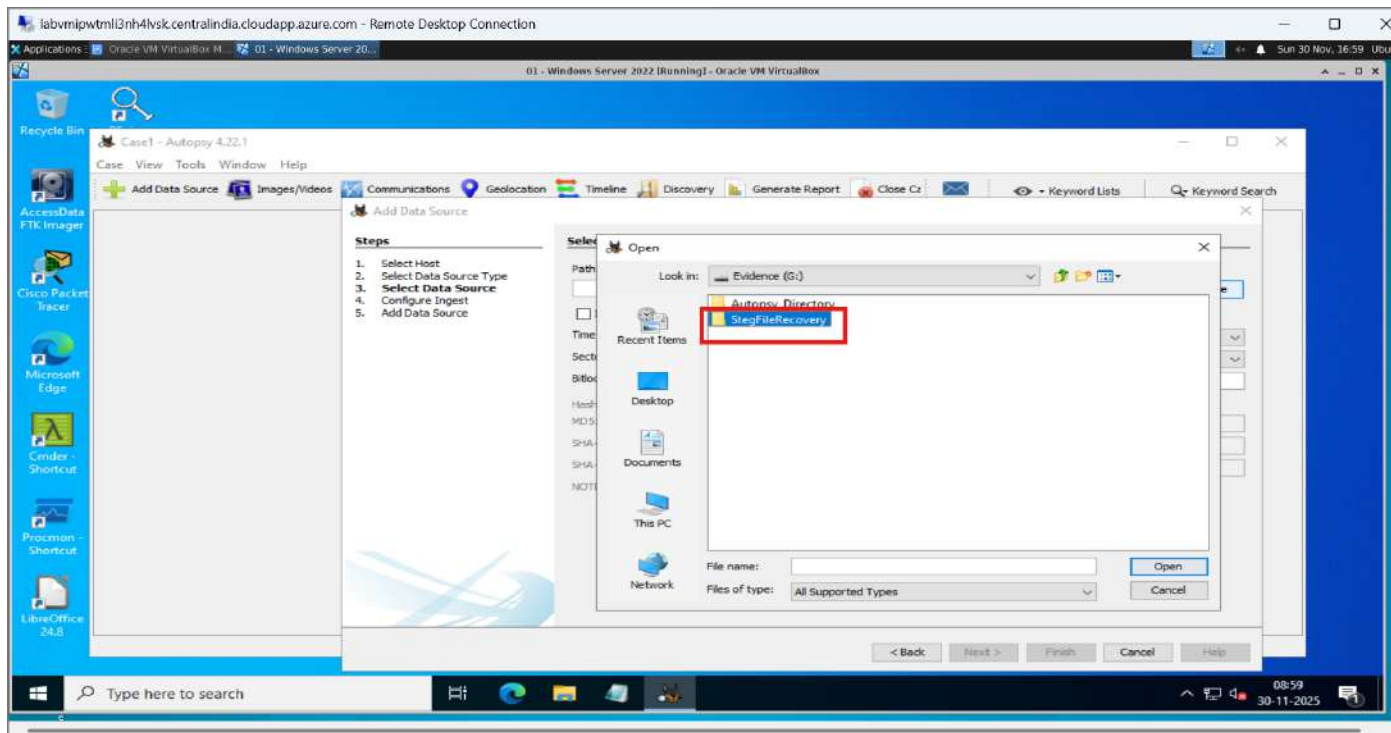
5.16 Selected the “Disk Image or VM File”, then Clicked on Next.



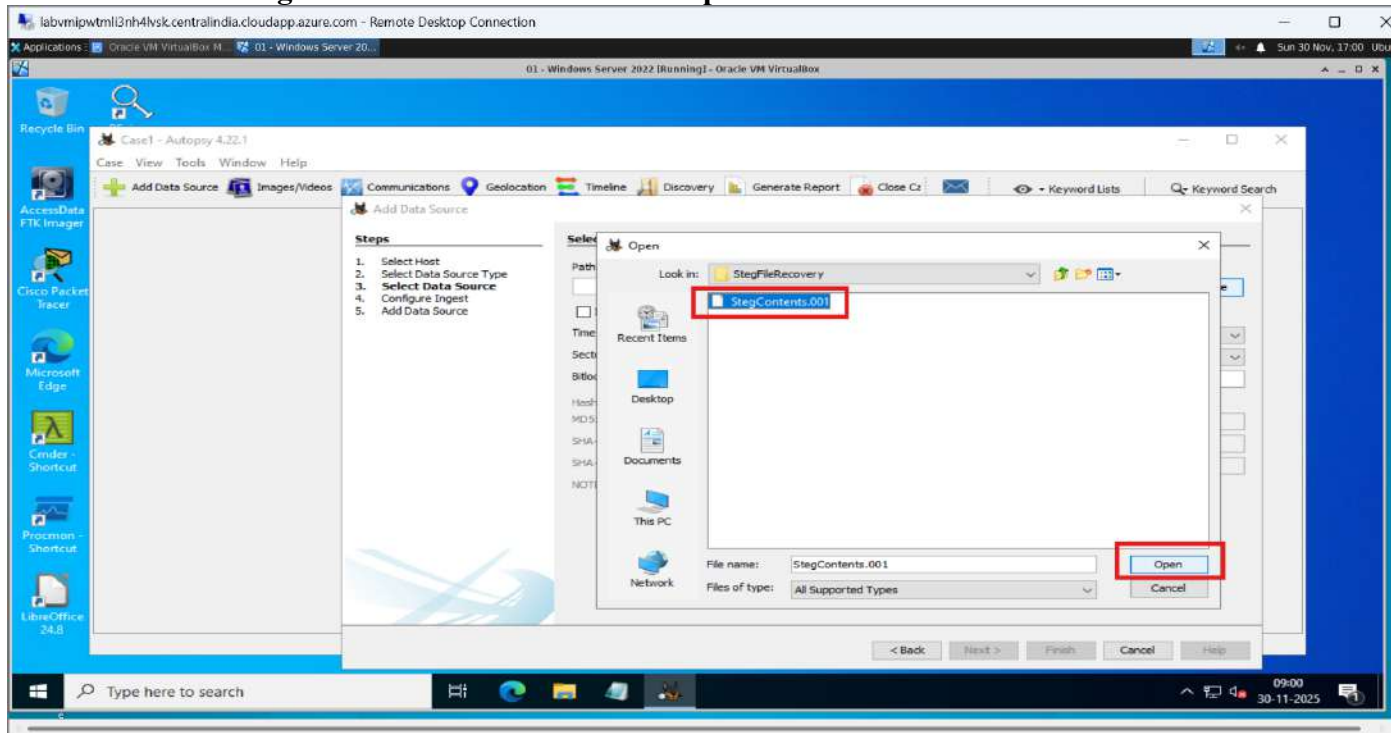
5.17 Clicked on **Browse**.



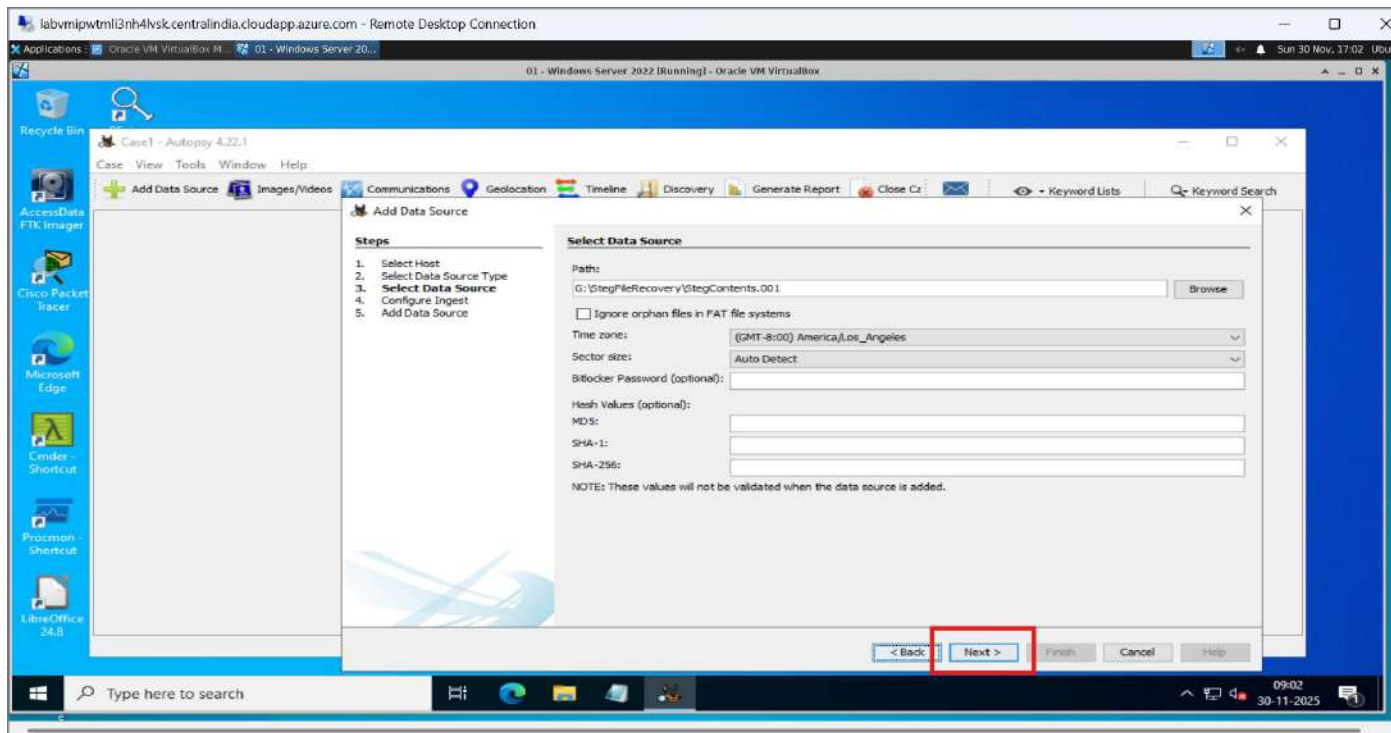
5.18 Navigate to **Evidence (G:) > StegFileRecovery**



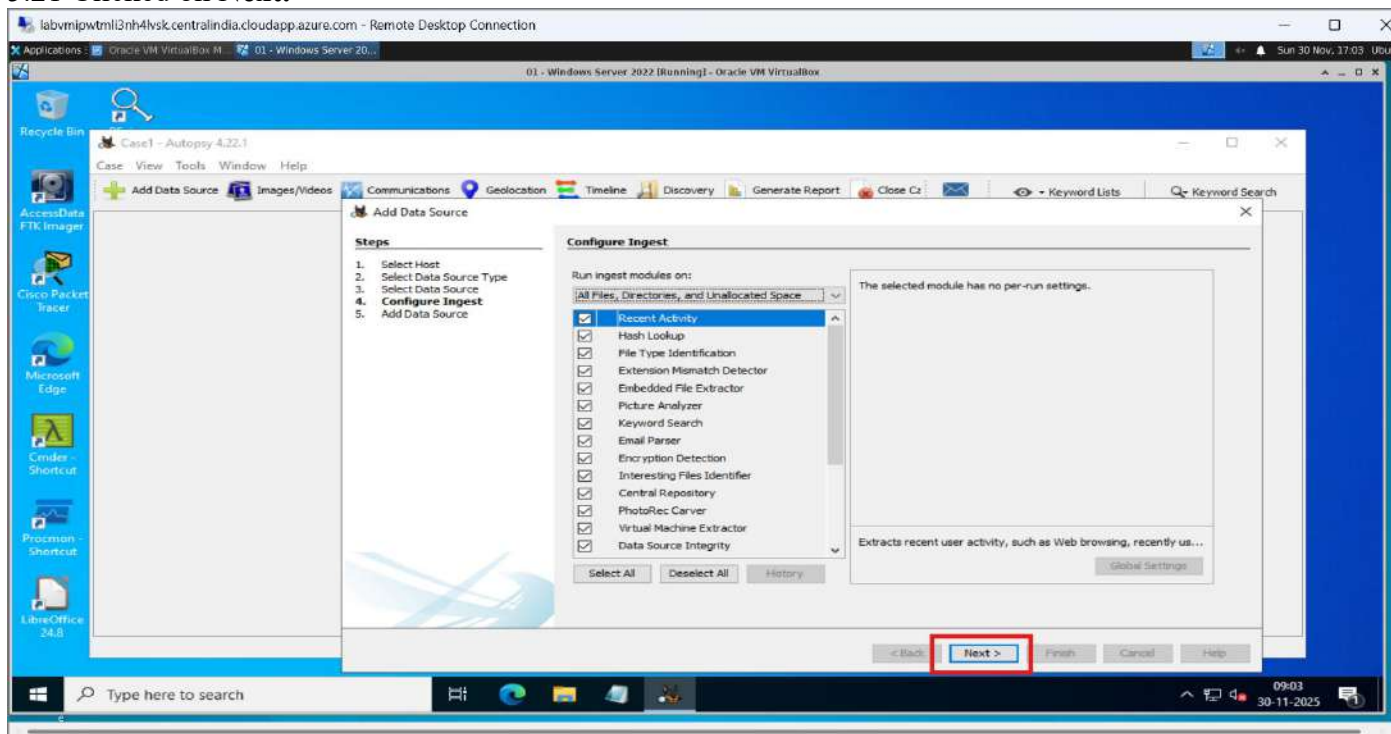
5.19 Selected the **StegContent.001** then Clicked on **Open**.



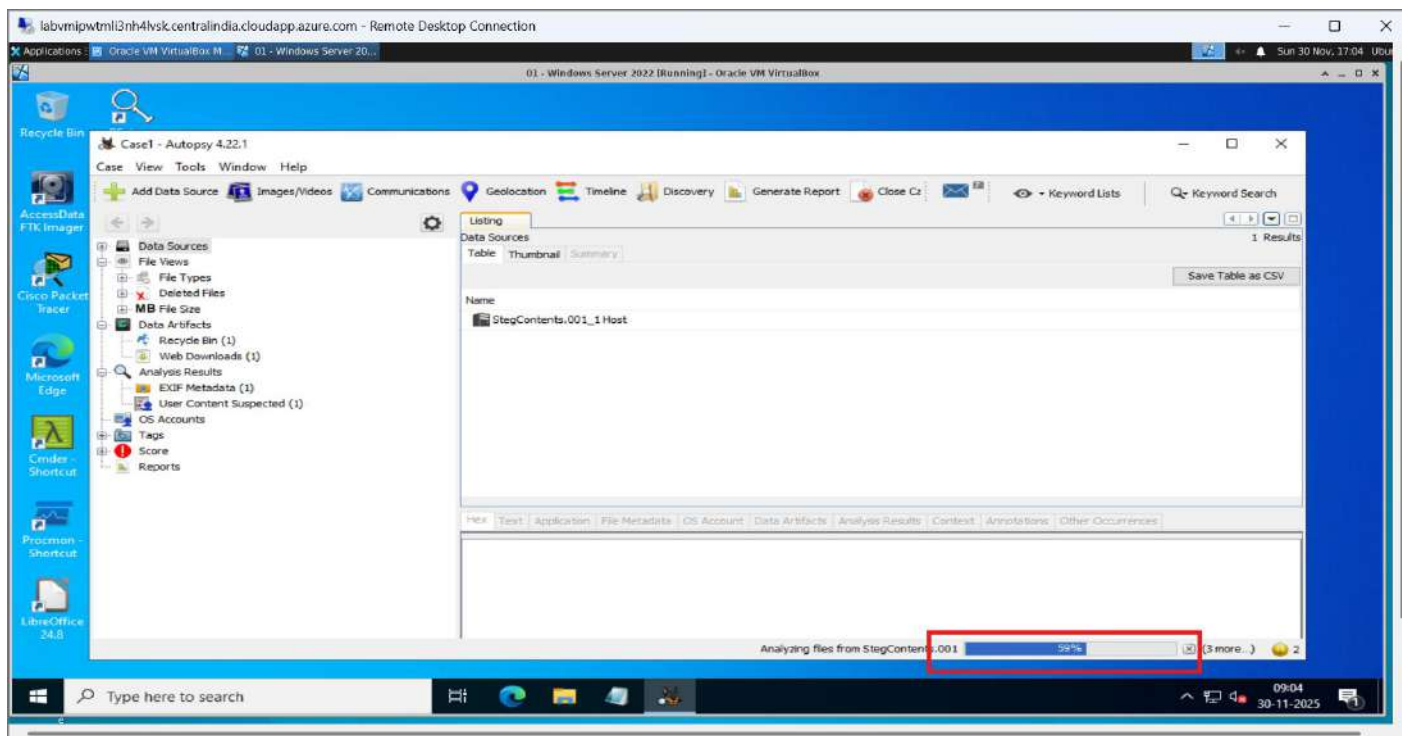
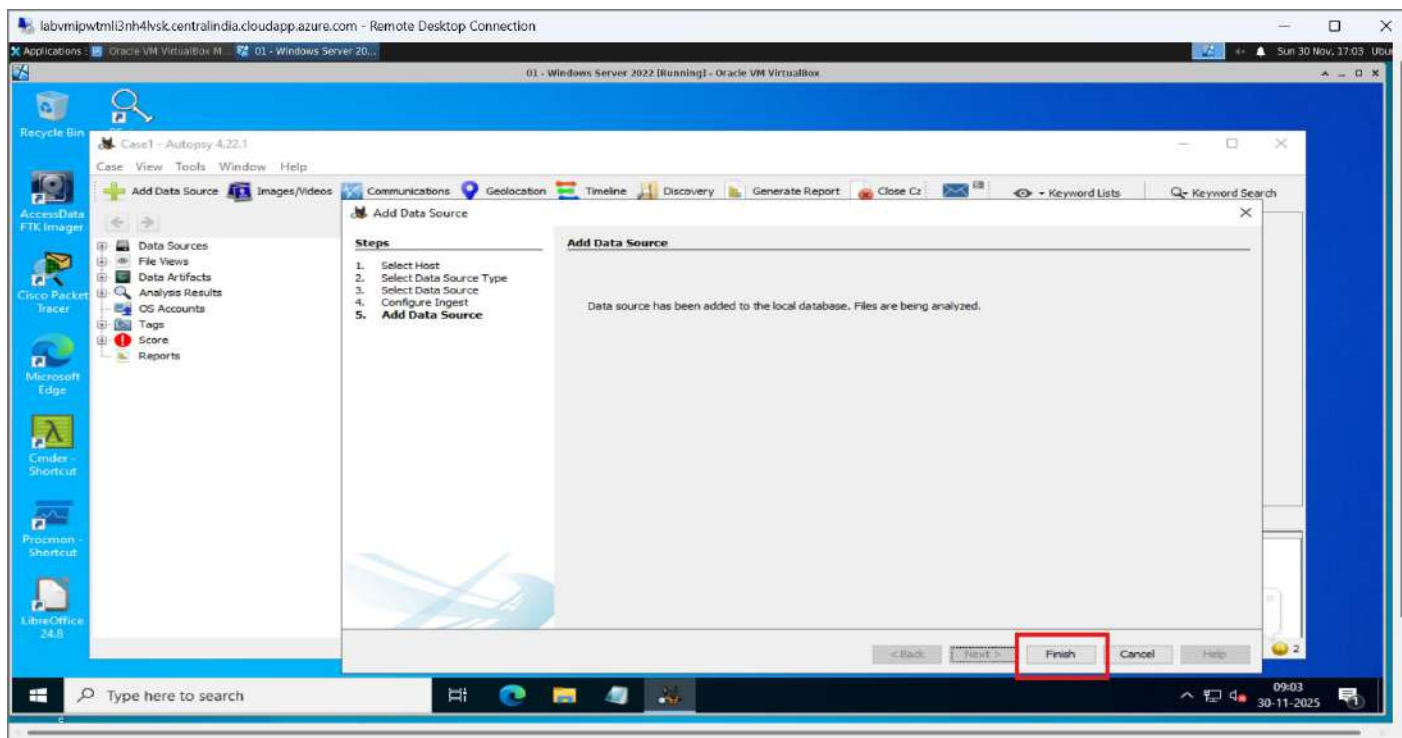
5.20 Clicked on **Next**.



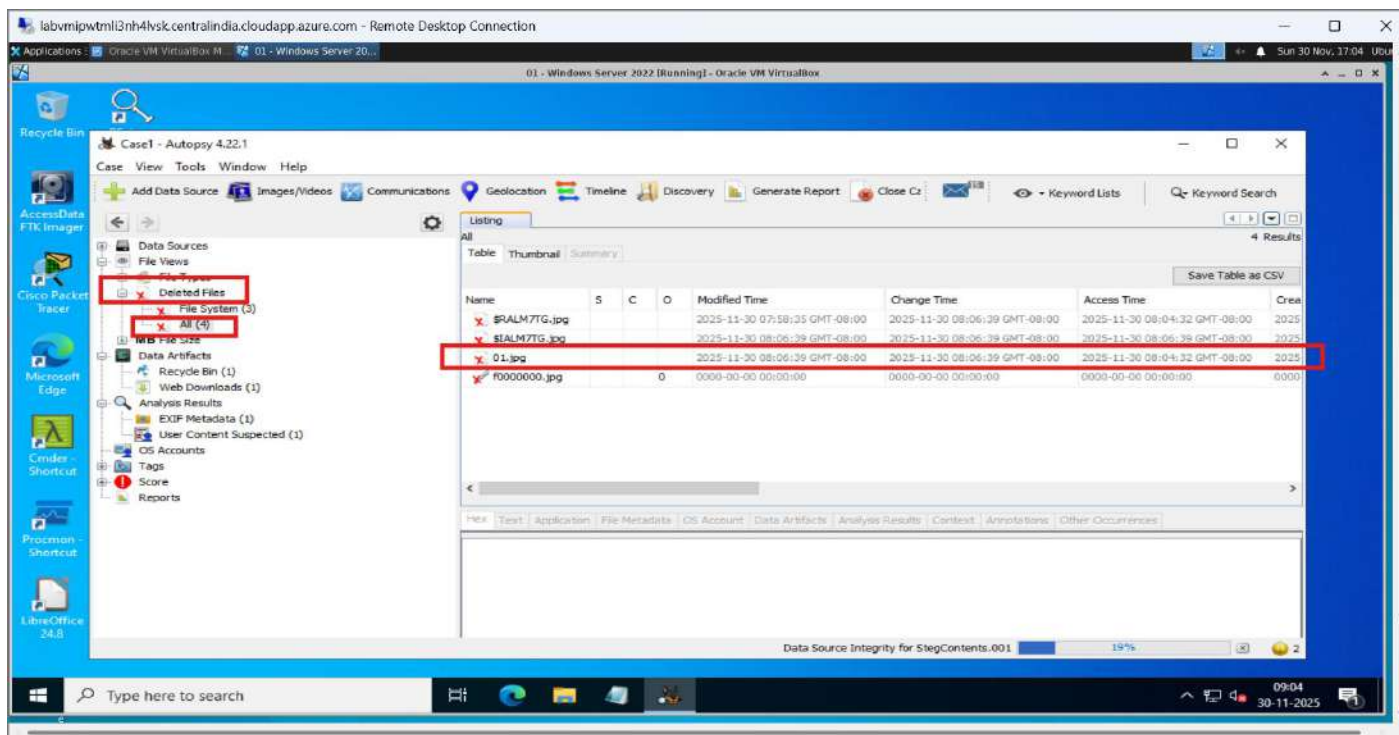
5.21 Clicked on Next.



5.22 Now clicked on Finish.

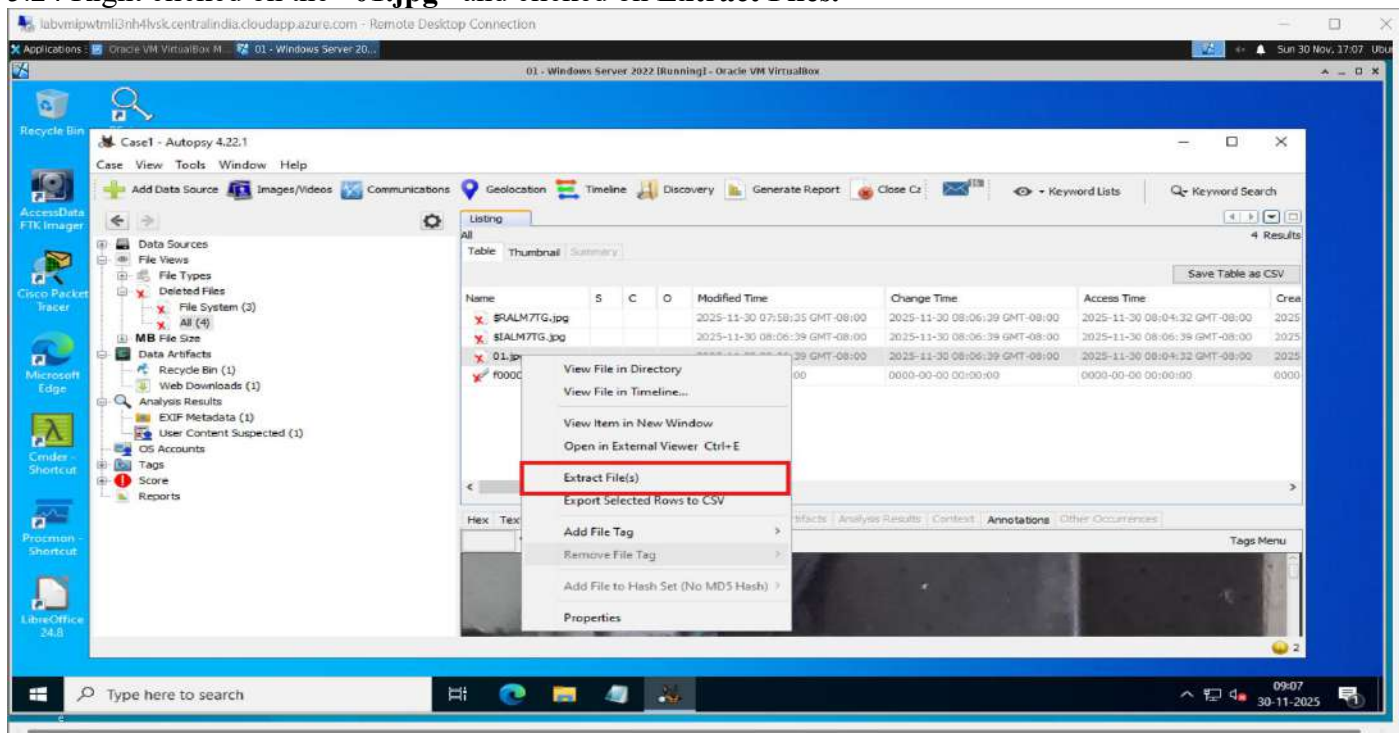


5.23 Expand the **Deleted Files** and clicked on **All**

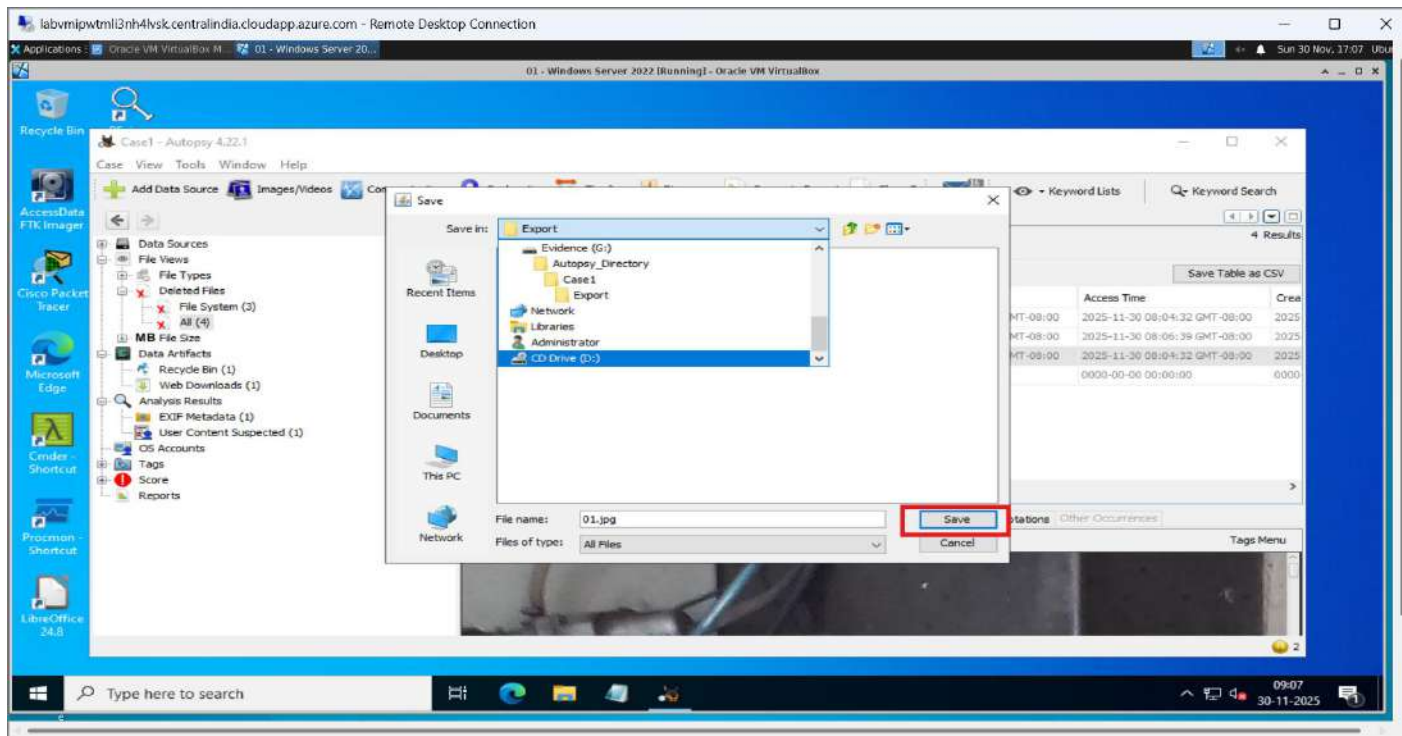


Here we can see the deleted file 01.jpg

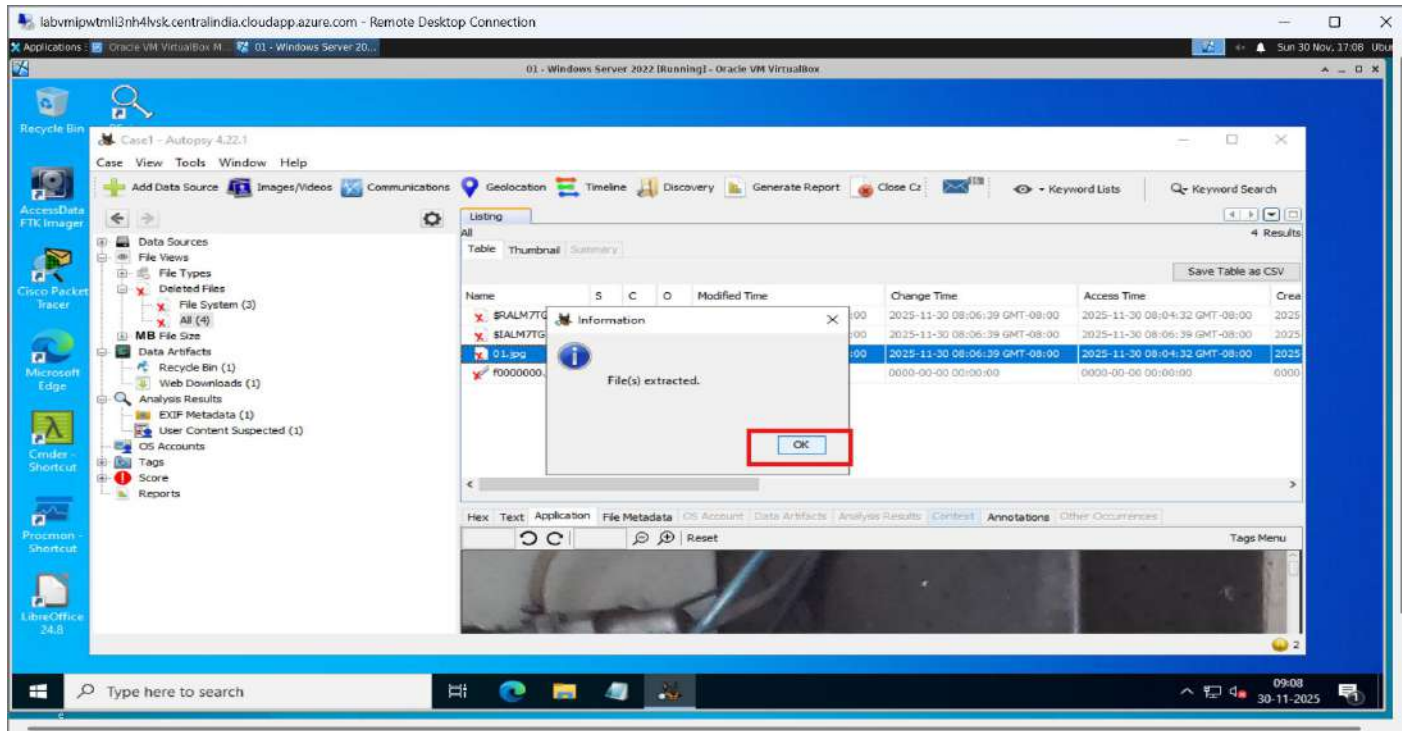
5.24 Right clicked on the “01.jpg” and clicked on **Extract Files**.



5.25 Clicked on save

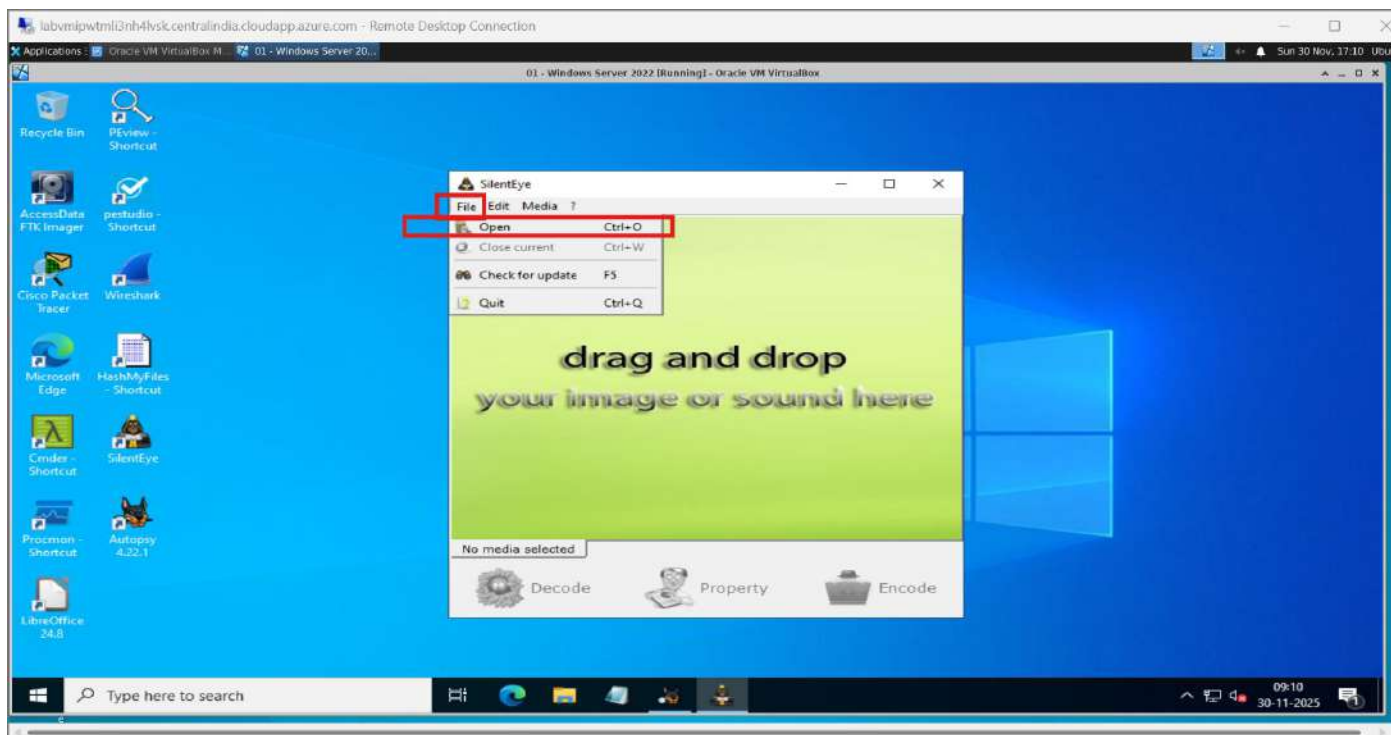


5.26 Clicked on ok

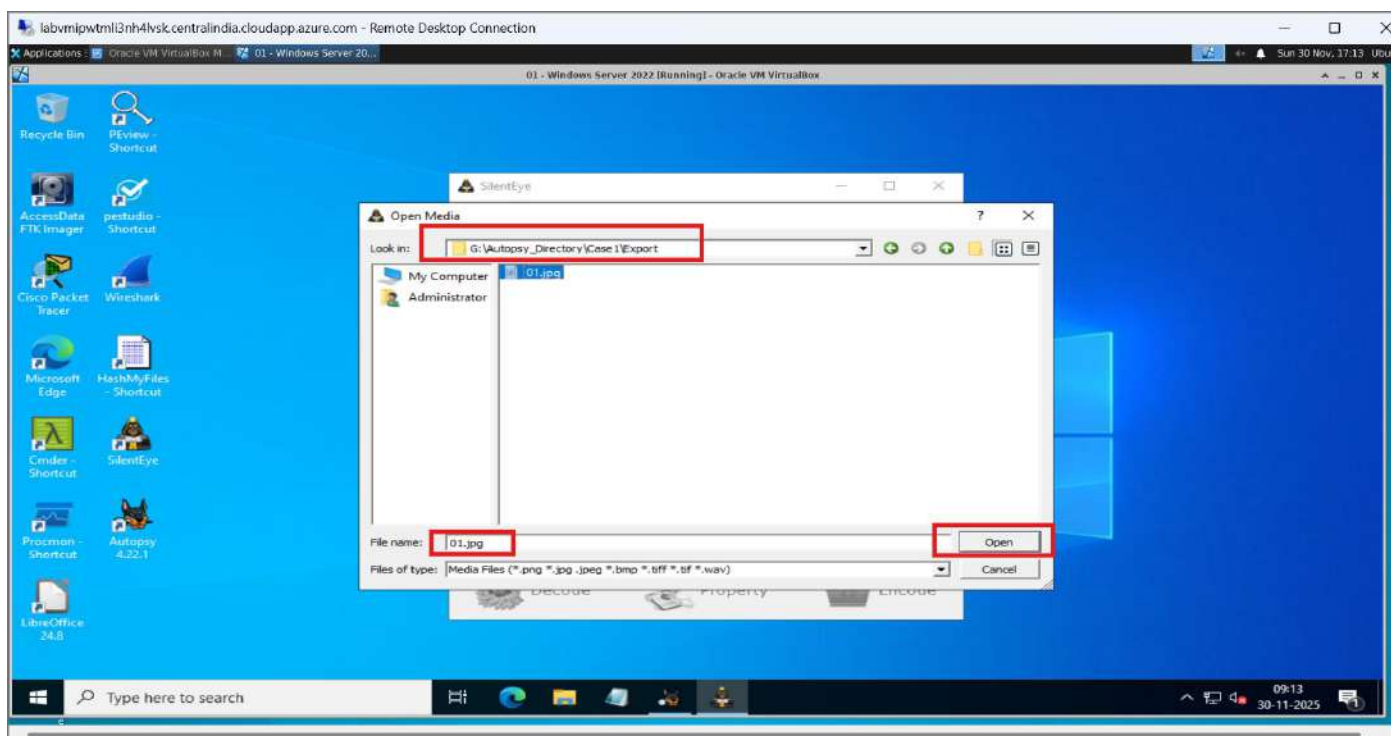


Task 6: Extract the Hidden Text from the Recovered Image:

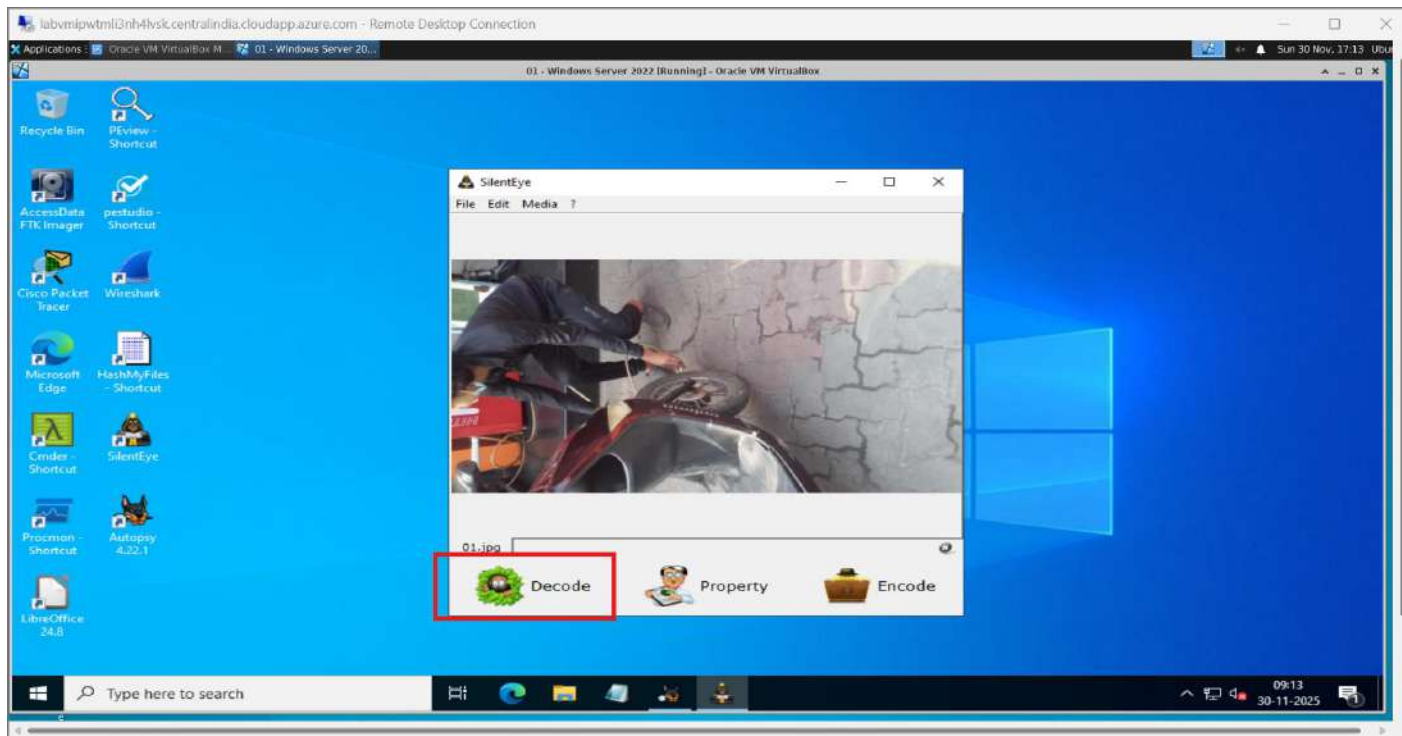
6.1 Opened the SilentEye, and clicked on File > Open.



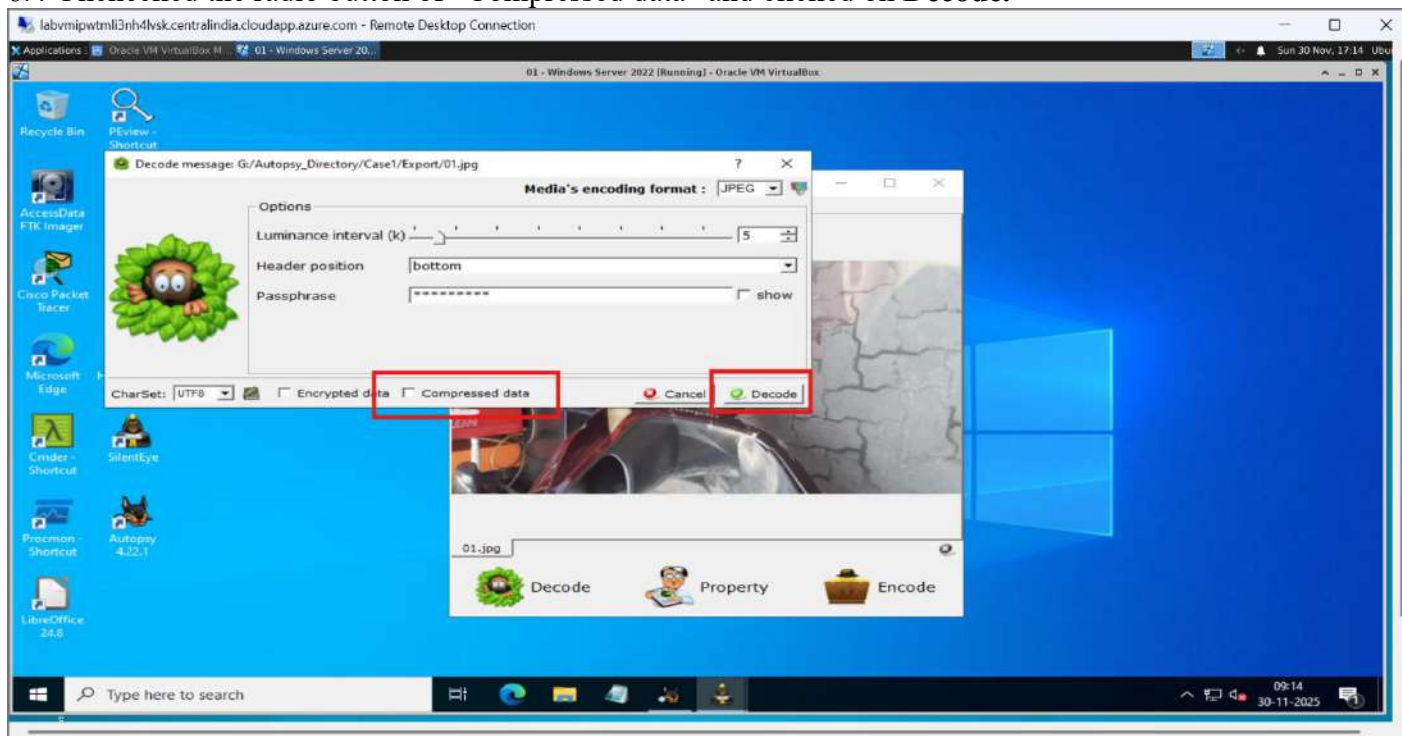
6.2 Selected the **01.jpg** file from the recovered folder and clicked on **Open**.



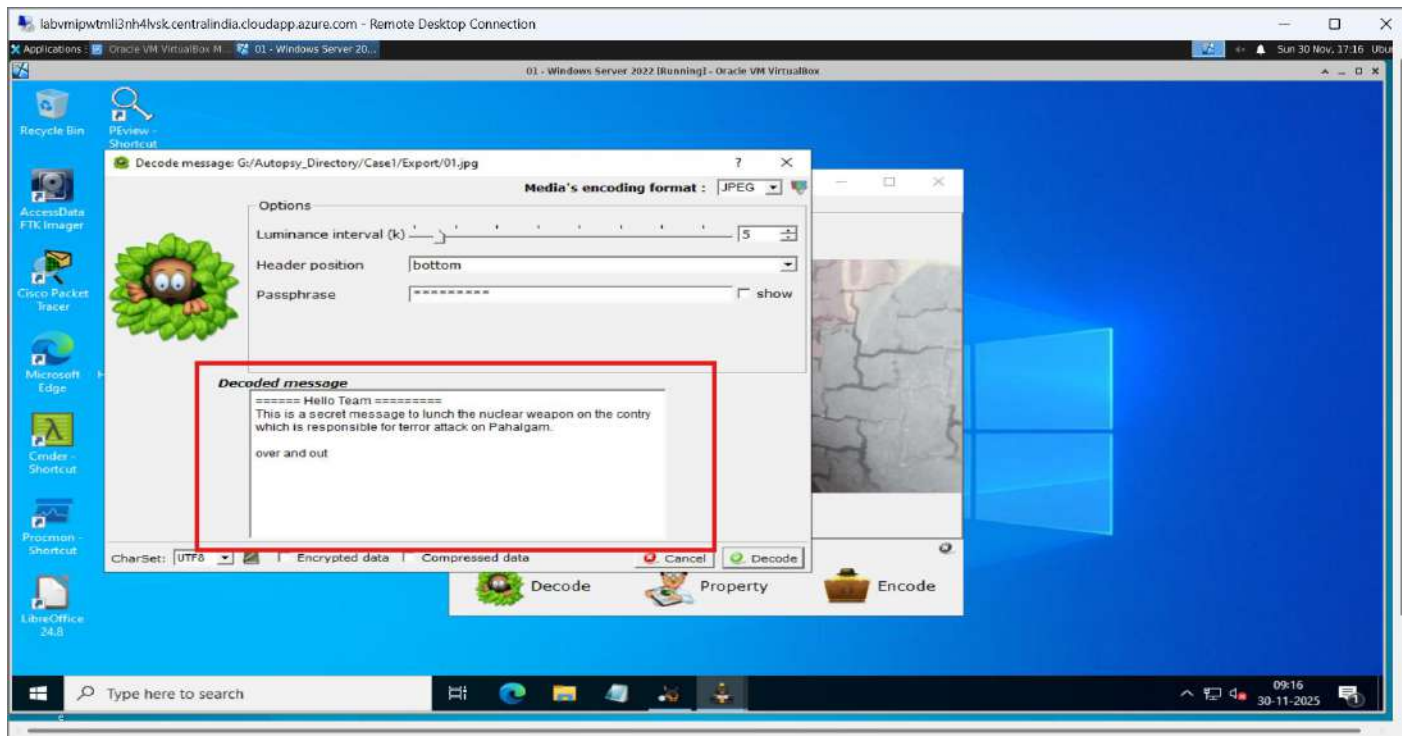
6.3 Clicked on **Decode**.



6.4 Unchecked the radio button of “Compressed data” and clicked on **Decode**.



6.5 The Message has been recovered.



Conclusion

This project effectively showcases how digital forensics and steganography intersect in modern cybersecurity investigations. By embedding text inside an image, deleting it, and later recovering it using forensic tools, the project demonstrates that:

- Hidden data can survive file deletion
- Forensic tools can recover both visible and hidden information
- Steganography leaves detectable artifacts that professionals can analyze

Overall, this project provides hands-on experience with real forensic methodologies and highlights the importance of secure data handling, digital evidence acquisition, and investigative techniques used in cybersecurity.
